

hetero_rules_run

September 14, 2026

1 Crossover Resolution Under Lag-Context Learning Rules

The crossover experiment of `transition_matrix.ipynb`, run over the learning rules of *Context-Dependent Sequential Memory* instead of a catalogue of transition matrices T .

1.1 The problem

Two sequences share a contiguous run of patterns. Cued from one branch, the trajectory reaches the run and has to *choose* – but the state there is identical whichever branch brought it, so under the plain $\mu \rightarrow \mu + 1$ chain the drive toward the two continuations is symmetric and the branch falls to noise and ω disorder.

1.2 What is different from `transition_matrix.ipynb`

That notebook’s gate $\sum_{\beta} T_{\alpha\beta}(m^{\beta})^d$ is a **sum of pure powers**: every overlap sits in its own term and no term multiplies two *different* overlaps. So a rule like “drive $\xi^{\mu+1}$ only when m_{μ} **and** $m_{\mu-1}$ are both large” has no matrix, and that notebook says so explicitly and does not attempt it.

The write-up’s context rules are exactly that. Its fields are fourth-order of bidegree $(2, 1)$ – three overlap factors – spent at **different lags**:

$$h_i^{\text{bigram}} = \sum_{\mu} \xi_i^{\mu+1} m_{\mu} |m_{\mu-1}|^2, \quad h_i^{\text{coh3}} = \sum_{\mu} \xi_i^{\mu+1} m_{\mu} m_{\mu-1} m_{\mu-2}.$$

`kuramoto_context.py` is the one-term generalization that covers all of them:

$$\dot{\theta}_i = -\sin \theta_i \sum_a \sum_{\mu} \sum_{\text{terms}} w \xi_i^{a, \mu+s} \prod_{l \in \text{lags}} m_i^{a, \mu-l}$$

A rule is a **table of terms**, one row each: a target offset s , d source lags, and a weight. $d = 3$ with lags $(0, 0, 0)$ is m_{μ}^3 , the existing dynamics; everything else the write-up asks for is a different row.

1.3 Contents

§	what	ω
1	The learning-rule catalogue – nine $d = 3$ rules, raw and gain-corrected	–

§	what	ω
2	Validation: forward reproduces the baseline exactly	$\sigma_\omega = 0.3$
3	Benchmark topologies and success metrics	$\sigma_\omega = 0.3$
4	The benchmark experiments – all four topologies, raw and gain	$\sigma_\omega = 0.3$
5	Knob 1 – the reach r of the gate	$\sigma_\omega = 0.3$
6	Knob 2 – the multilag weights	$\sigma_\omega = 0.3$
7	Sampled trial – time series by topology	$\sigma_\omega = 0.3$
8	Constant intrinsic frequency	$\sigma_\omega = 0, \omega \in \{0, 0.1, 0.3, 1, 3\}$
9	The full benchmark at fixed ω	$\sigma_\omega = 0, \omega \in \{0.1, 0.3, 1\}$
10	Seed sweep – 100 instances at a sampled corruption rate, run twice	$\sigma_\omega = 0.3$, then $\omega \equiv 0.3$
–	Findings	–

On ω . CONFIG sets `frequency_mean = 0`, `frequency_std = 0.3`: quenched disorder, $\omega_i \sim \mathcal{N}(0, 0.3)$, a different intrinsic frequency per oscillator. Every section marked $\sigma_\omega = 0.3$ runs that default; §8 and §9 are the only ones that remove it. Two things to know about the disorder rows:

- **The draw is never resampled.** Every `ContextNetwork` in this notebook is built with `seed=10`, and `omega` is drawn once per network from that seed, so one ω vector is reused across every trial, rule and geometry. §9's spread is over patterns and corruption rate, not over disorder realizations – on the ω axis every result here is $n = 1$.
- **phase_noise is inert.** CONFIG carries `phase_noise = 0.01`, but `mode = "ode"` and the integrator passes noise only in `"sde"` mode. Every run is a deterministic ODE.

Robustness sweeps – corruption ratio, network size, rule mixtures, boundary and frequency conditions – are in the companion notebook `robustness_sweep.ipynb`.

1.4 What this notebook measures

Rules – nine, all $d = 3$ so they cost the same: `self`, `forward`, `skip-two`, `self+forward`, `forward+skip2`, `bigram`, `multilag`, `coh bigram`, `coh trigram`. Each is run **raw** and **gain-corrected** (`kc.proportional_gain`) side by side.

Topologies – written **before-shared-after**: unshared patterns before the shared run, its length L , unshared after. All are **smooth** mutation chains, two branches each.

tag	spec	P	shared	L
2-1-2	(a) eq (7)	5	{2}	1
2-2-2	(c) eq (9)	6	{2, 3}	2
2-3-2	the failure case	7	{2, 3, 4}	3

tag	spec	P	shared	L
1-1-1	legacy	3	{1}	1

Metrics – four, per rule and per topology:

1. **Strict full-retrieval success rate:** the `correct` count $\div$ `cueing attempts`. An attempt is `correct` only if every unshared pattern of the cued branch is decoded **in stored order**. No partial credit. The two failure modes are reported separately – `crossed` (the other branch appeared first) and `incomplete` (the path ran out) – over the same denominator. Shared positions are excluded: both branches store the same array there, so `argmax` always awards the label to whichever was added first.
2. **Overlap quality:** for a *successful* attempt, the mean over the cued branch’s unshared patterns of each one’s *peak* overlap. Averaged over successful attempts only, so a failure lowers metric 1 and contributes nothing here. §10 adds its **variance across attempts**.
3. **Branch-correct fraction:** at the shared state, did the cued branch’s continuation reach the higher peak overlap? Counted over *every* attempt, so it separates “picked the wrong branch” from “picked the right one but never locked on”. Chance is 0.5.
4. **Margin:** the signed gap between those two peak overlaps.

1.5 Note — every rule below is compared raw and gain-corrected

The proportional gain correction – a per-term rescaling that equalizes drive strength across rules with different lag structure, math under *Why lag-heavy rules run weaker* – lives in `kc.proportional_gain(rule, corruption_rate)`, previously named `unit_gain`. Every experiment below runs **both** `RAW_CATALOGUE` and `GAIN_CATALOGUE`, so the effect of the correction is measured rather than assumed.

Integration time is not a substitute for it. *Does a longer duration rescue the raw rules?* sweeps `duration` from 50 to 80 on the hardest geometry and returns **bit-identical** outcome fractions, because the trajectories hit the simulator’s stationary early-stop well inside the shortest duration tested. Durations are therefore capped at 80.0 (the legacy 1-1-1 stays at 60.0), verified to reproduce the longer runs exactly, rule by rule; only the per-term weights differ between the raw and gain runs.

1.5.1 Note on implementation

Stored patterns are real ($\xi = \cos(\text{phase}) \in \{-1, +1\}$), so every overlap is real and $|m|^2 = m^2$. The write-up’s complex-overlap rules therefore collapse onto the real forms in the table below, and the “coherent” rules differ from the magnitude-gated ones by *sign sensitivity* rather than by phase.

A rule is **relative**, so unlike a transition matrix it does not have the end of the chain baked in and the inherited `boundary` argument still means what it always did: `"self"` (default) clamps both ends, `"cycle"` wraps both. Clamping at the *start* is what lets a cue at pattern 0 launch the sequence at all — a lag-1 gate there has no earlier overlap to read, and reading m_0 twice degrades the term to the plain forward rule.

```
[40]: %load_ext autoreload
      %autoreload 2
```

```

%matplotlib inline

from collections import Counter

import numpy as np
import matplotlib.pyplot as plt

import sys
from pathlib import Path

root_dir = Path.cwd().parent.parent
if str(root_dir) not in sys.path:
    sys.path.append(str(root_dir))

import kuramoto.kuramoto_library as krm
import kuramoto.hetero_learning.kuramoto_hetero as kc

```

The autoreload extension is already loaded. To reload it, use:

```
%reload_ext autoreload
```

2 Setup

```

[41]: # Same configuration as transition_matrix.ipynb, so every row below is directly
      ↳ comparable
      # to that notebook's numbers.
      CONFIG = dict(
          dt=0.05,
          T=30.0,
          frequency_std=0.3,
          phase_noise=0.01,
          d=3,
          corruption_rate=0.2,
          tolerance=0.5,
          mode="ode",
      )

      # 200, not 100. Branch choice -- the quantity these benchmarks exist to measure
      ↳ -- improves
      # sharply with N: on 1-1-1, `coh trigram` goes 70% -> 95% branch-correct and
      ↳ `bigram` 52% -> 85%
      # between N=100 and N=200. N=400 adds little over 200 (98% / 82%) for 4x the
      ↳ cost, so 200 is the
      # knee.

```

```

#
# CAVEAT, and it is the same trap that produced a false finding in
↳robustness_sweep SS2:
# `incomplete` RISES with  $N$  under a fixed duration cap (coh trigram 5% -> 22%),
↳because a larger
# network needs longer to traverse and  $T$  is capped at 80 here. So  $N$  buys the
↳decision and the cap
# spends part of it on completion. Read `branch_correct` rather than `correct`
↳when comparing
# across  $N$ , or raise the durations in GEOMETRIES alongside  $N$ .
N = 200
TRIALS = 20      # trials per rule; each runs 2 cueing attempts
MARGIN = 0.01
SEED_POOL = 10_000

# frequency_std must stay > 0: an exact, uncorrupted cue has theta in {0, pi},
↳so
# sin(theta_i) = 0 and the drive term vanishes identically -- with omega = 0 the
# trajectory is frozen at the cue and every rule scores "incomplete".
assert CONFIG["frequency_std"] > 0

```

3 1. The learning-rule catalogue

3.1 The rule catalogue

Section 2 of the write-up, minus the two rules that are not tables of relative indices (see the note below). Every rule is $d = 3$, i.e. fourth order, so they all cost the same.

name	eq	term (s , lags)	gate at pattern μ	what it says
self	(11)	$(0, (0, 0, 0))$	$\xi^\mu m_\mu^3$	Autoassociative stabilizer. No sequence at all; the control that should never traverse.
forward	(12)	$(1, (0, 0, 0))$	$\xi^{\mu+1} m_\mu^3$	Markov-1. The existing <code>xi_next</code> dynamics, reproduced exactly. Baseline.

name	eq	term (s, lags)	gate at pattern μ	what it says
skip-two	(13)	$(2, (0, 0, 0))$	$\xi^{\mu+2} m_\mu^3$	Still current-state-only: changes which pattern is driven, not what is read.
self+forward	(14)	mixture	—	Dwell plus advance. Tests whether inertia alone holds the branch.
forward+skip2	(14)	mixture	—	<code>transition_matrix.ipynb</code> 's <code>chain+skip</code> , as a rule.
bigram	(16)	$(1, (0, 1, 1))$	$\xi^{\mu+1} m_\mu m_{\mu-1} ^2$	The simplest context rule: gated by how much overlap with the <i>previous</i> pattern survives.
multilag	(17)	$(1, (0, k, k))$ summed	$\xi^{\mu+1} m_\mu \sum_k w_k m_{\mu-k} ^2$	Several recent overlaps gate the transition at once.
coh bigram	(18)	$(1, (0, 0, 1))$	$\xi^{\mu+1} m_\mu^2 m_{\mu-1}$	One factor moved from the gate to the drive: the lagged overlap enters linearly, so its <i>sign</i> matters.
coh trigram	(19)	$(1, (0, 1, 2))$	$\xi^{\mu+1} m_\mu m_{\mu-1} m_{\mu-2}$	Two-step context inside the same fourth-order budget. The write-up's headline rule.

3.1.1 Why lag-heavy rules run weaker

A rule that reads only m_μ has a gate of ≈ 1 whenever the state is on pattern μ . One that also reads $m_{\mu-l}$ is multiplied there by the residual overlap with a pattern l mutations back, about $(1 - 2\rho)^l$ – at $\rho = 0.2, 0.6$ per lag. So the coherent trigram's gate runs at $0.6 \cdot 0.36 \approx 0.22$ of the forward rule's, purely as an artifact of how many lags it reads rather than of whether that context is useful.

The correction. `kc.proportional_gain(rule, rho)` divides each term by $(1 - 2\rho)^{\sum \text{lags}}$ and the rule by its total weight, so every rule has total gate 1 at its own operating point. Named for what it does – a rescaling *proportional* to each term's own lag decay – since the correction depends on

both the rule and ρ . Raw and corrected run side by side in every experiment below, so the size of the effect is measured per rule and per geometry.

```
[42]: RAW_CATALOGUE = {
    "self":          kc.self_rule(),
    "forward":       kc.forward_rule(),
    "skip-two":      kc.skip_rule(nhop=2),
    "self+forward":  kc.mixture(kc.self_rule(weight=0.3), kc.
↳forward_rule(weight=1.0)),
    "forward+skip2": kc.mixture(kc.forward_rule(weight=1.0), kc.skip_rule(2,
↳weight=0.5)),
    "bigram":        kc.bigram_rule(),
    "multilag":      kc.multilag_rule((1.0, 1.0, 1.0)),
    "coh bigram":    kc.coherent_bigram_rule(),
    "coh trigram":   kc.coherent_trigram_rule(),
}

# The gain step, applied here at the configured rate so every experiment below
↳can compare
# a rule's raw and gain-corrected behavior directly.
GAIN_CATALOGUE = {name: kc.proportional_gain(rule, CONFIG["corruption_rate"])
                  for name, rule in RAW_CATALOGUE.items()}

print(f"{'rule':<15}{'target s':>9}{'lags':>12}{'raw w':>10}{'gain w':>10}")
print("-" * 56)
for name, rule in RAW_CATALOGUE.items():
    gain_rule = GAIN_CATALOGUE[name]
    for r, row in enumerate(rule):
        lags = ",".join(f"{int(l)}" for l in row[1:-1])
        print(f"{name if r == 0 else '':<15}{int(row[0]):>9}{('(' + lags + ')':
↳>12}"

                f"{row[-1]:>10.3f}{gain_rule[r, -1]:>10.3f}")
```

rule	target s	lags	raw w	gain w
self	0	(0,0,0)	1.000	1.000
forward	1	(0,0,0)	1.000	1.000
skip-two	2	(0,0,0)	1.000	1.000
self+forward	0	(0,0,0)	0.300	0.231
	1	(0,0,0)	1.000	0.769
forward+skip2	1	(0,0,0)	1.000	0.667
	2	(0,0,0)	0.500	0.333
bigram	1	(0,1,1)	1.000	2.778
multilag	1	(0,0,0)	1.000	0.333
	1	(0,1,1)	1.000	0.926
	1	(0,2,2)	1.000	2.572
coh bigram	1	(0,0,1)	1.000	1.667
coh trigram	1	(0,1,2)	1.000	4.630

3.1.2 Reading a rule table

Each row is one term: **target** s is which pattern it drives relative to μ , **lags** are which patterns it reads (counting *backwards*, so 0 is μ itself), and **weight** is its coefficient. **forward** is the single row (1, (0, 0, 0)) – drive $\xi^{\mu+1}$, read m_μ three times.

Both weight columns print: **raw** w is the builder weight, **gain** w the same term after `kc.proportional_gain`. **coh** trigram carries $4.63 = 1/0.6^3$ because its gate reads two lagged overlaps; a single lag-0 term like **forward**’s is untouched (gain = 1).

4 2. Validation: forward must reproduce the baseline exactly ($\sigma_\omega = 0.3$)

`forward_rule()` is the `xi_next` successor chain written as a rule, so `ContextNetwork` under it and the stock `KuramotoNetwork` should integrate the same trajectory. Any difference here means the extension changed the dynamics rather than generalizing them.

This is the same check `transition_matrix.ipynb` runs on `shift_transition(P)`, which came back at 4.4×10^{-15} .

```
[43]: check_multi = krm.generate_multi_sequence(N=N, seq_len=4,
      ↪corruption_rate=CONFIG["corruption_rate"],
      count=2, seed=7)

base_net = krm.KuramotoNetwork(N=N, seed=10, **CONFIG)
fwd_net = kc.ContextNetwork(rule=kc.forward_rule(), N=N, seed=10, **CONFIG)

ref = base_net.simulate(check_multi, cue=check_multi["A"], cue_idx=0)
new = fwd_net.simulate(check_multi, cue=check_multi["A"], cue_idx=0)

print("history shapes:", ref["theta_history"].shape, new["theta_history"].shape)
print("max |dtheta| :", np.abs(ref["theta_history"] - new["theta_history"]).
      ↪max())
```

```
history shapes: (578, 200) (578, 200)
max |dtheta| : 0.0
```

5 3. The benchmark topologies and the success metrics ($\sigma_\omega = 0.3$)

5.1 Building one crossover instance

`make_crossover_pair` and `classify_unshared` are `transition_matrix.ipynb`’s, unchanged: two sequences of `seq_len` patterns sharing a contiguous run at `shared_idx`, built as a mutation chain so consecutive patterns overlap at about 0.6, with both branches holding the same array object across the run so those overlaps are bit-identical.

Scoring is that notebook’s too – every **unshared** pattern of the cued sequence must come back, in order. Shared positions carry no information (**argmax** always awards the label to whichever branch was added first) and are skipped.

5.2 The geometries and why $\text{reach} \geq L$

The benchmarks of the write-up, plus the failure case and the legacy smallest crossover. For a shared run at indices $\ell \dots h$ of length L , the branch is decided while driving pattern $h + 1$, i.e. at $\mu = h$. A term reading lag l then reads pattern $h - l$, which is unshared only when $h - l < \ell$, so

$$\boxed{\text{reach} \geq L}$$

is what a rule needs to see anything that distinguishes the branches at all. The transition matrix needed $\text{nhop} \geq L + 1$ for the same reason; here the requirement lands on how far *back the gate reads* instead of how far *forward the drive jumps*.

tag	spec	P	shared indices	L	duration	reach needed
2-1-2	(a) eq (7)	5	$\{2\}$	1	80	1 — bigram, coh bigram
2-2-2	(c) eq (9)	6	$\{2, 3\}$	2	80	2 — only coh trigram
2-3-2	failure case	7	$\{2, 3, 4\}$	3	80	3 — nothing in the catalogue
1-1-1	legacy	3	$\{1\}$	1	60	1 — bigram, coh bigram

coh trigram, at reach 2, is over-provisioned for 2-1-2, exactly matched to 2-2-2, and still short for 2-3-2.

2-3-2 is the failure case. It is (c) with the shared segment grown from two patterns to three and everything else held fixed, so the only thing that changed is the length of the ambiguity. Every rule reads at most two patterns back, so at the branch point every gate factor is still inside the shared run where the branches are bit-identical.

5.3 The success metrics, in code

`classify_unshared` gives metric 1; `run_crossover_trial` adds metric 2 (`overlap_quality`, recorded only on successful attempts) and metrics 3-4 (`branch_correct`, `margin`, on every attempt). Every count is divided by the number of cueing attempts, so each panel is a fraction in $[0, 1]$.

```

[44]: def make_crossover_pair(N, corruption_rate, origin_seed, mutation_seeds,
                               seq_len, shared_idx):
    """The two sequences "A" and "B", `seq_len` patterns each, sharing a
    ↪ contiguous run at
    ↪ `shared_idx`. Two branches exactly -- every scorer below is written for a
    ↪ pair.

    Built as a mutation chain rather than a star around one origin, so
    ↪ consecutive patterns
    ↪ stay close enough for the dynamics to traverse. Both branches hold the same
    ↪ array object
    ↪ across the run, so those overlaps are bit-identical and the ambiguity is
    ↪ exact."""
    shared_idx = sorted(shared_idx)
    lo, hi = shared_idx[0], shared_idx[-1]
    if shared_idx != list(range(lo, hi + 1)):
        raise ValueError("shared_idx must be a contiguous run.")
    if not (0 < lo and hi < seq_len - 1):
        raise ValueError("The run needs at least one unshared pattern on each
    ↪ side.")

    seeds = iter(mutation_seeds)
    mutate = lambda v: krm.corrupt_phase(v, corruption_rate, np.random.
    ↪ default_rng(next(seeds)))

    run = [np.random.default_rng(origin_seed).choice([0.0, np.pi], size=N)]
    for _ in range(hi - lo):
        run.append(mutate(run[-1]))

    multi = krm.MultiSequence()
    for name in ("A", "B"):
        prefix, v = [], run[0]
        for _ in range(lo):
            v = mutate(v); prefix.append(v)
            # backwards off the front
        prefix.reverse()
        suffix, v = [], run[-1]
        for _ in range(seq_len - 1 - hi):
            v = mutate(v); suffix.append(v)
            # forwards off the back
        multi.add(krm.sequence_from_patterns(name, prefix + list(run) + suffix))
    return multi

def classify_unshared(retrieved, multi, name, shared_idx):
    """METRIC 1 -- outcome of one cueing attempt, on the strict criterion:
    ↪ every unshared
    ↪ pattern of the cued sequence must come back, in order.

```

```

correct      = all unshared positions decoded, in order, on the cued branch
crossed      = the other branch showed up at an unshared position first
incomplete = the path ran out before all of them appeared"""
other = "B" if name == "A" else "A"
shared = set(shared_idx)
pending = iter([k for k in range(len(multi[name])) if k not in shared])
target = next(pending, None)

for seq, idx in retrieved:
    if target is None:
        break
    if idx in shared:
        continue
    if seq == other and idx >= target:
        return "crossed"
    if seq == name and idx == target:
        target = next(pending, None)
return "correct" if target is None else "incomplete"

def overlap_quality(overlaps, labels, name, seq_len, shared_idx):
    """METRIC 2 -- mean, over the cued branch's UNSHARED patterns, of each
    ↪ pattern's peak
    overlap. Only meaningful for an attempt classified "correct"; callers must
    ↪ check that
    separately. Shared positions are excluded because both branches store the
    ↪ identical
    array there, so they carry no rule-discriminating information."""
    shared = set(shared_idx)
    own_unshared = [labels.index((name, k)) for k in range(seq_len) if k not in
    ↪ shared]
    return float(overlaps[:, own_unshared].max(axis=0).mean())

```

5.3.1 The branch-choice diagnostic

The write-up asks for one more measurement than the three-way outcome: “a branch-choice diagnostic at the shared state, comparing correct and wrong successor drive”.

For a shared run ending at index h , the two candidate continuations are (cued, $h+1$) and (other, $h+1$). `branch_correct` records whether the cued branch’s continuation reached the higher peak overlap, and `margin` by how much. Scored on **every** attempt including those the decoder calls `incomplete`, so it separates “picked the wrong branch” from “picked the right one but never locked onto it hard enough to clear *tolerance*”.

```

[45]: # The benchmark topologies, one place. Every experiment below indexes this by
    ↪ tag.
GEOMETRIES = {

```

```

    "2-1-2": dict(seq_len=5, shared_idx=(2,),      duration=80.0),      # (a) eq
    ↪(7)
    "2-2-2": dict(seq_len=6, shared_idx=(2, 3),    duration=80.0),      # (c) eq
    ↪(9)
    "2-3-2": dict(seq_len=7, shared_idx=(2, 3, 4), duration=80.0),      # the
    ↪failure case
    "1-1-1": dict(seq_len=3, shared_idx=(1,),      duration=60.0),      # legacy
}

# The write-up's benchmarks plus the failure case; the knobs and the seed sweep
    ↪run on this
# subset rather than on the legacy geometry as well.
SPEC = ["2-1-2", "2-2-2", "2-3-2"]
FOCUS = ["2-1-2", "2-2-2", "2-3-2"]

for tag, g in GEOMETRIES.items():
    lo, hi = min(g["shared_idx"]), max(g["shared_idx"])
    print(f"{tag:<8} P={g['seq_len']} shared={set(g['shared_idx'])} "
          f"L={hi - lo + 1} unshared before={lo}, after={g['seq_len'] - 1 -
    ↪hi} "
          f"duration={g['duration']:g}")

def run_crossover_trial(net, rng, seq_len, shared_idx, duration,
    ↪corruption_rate=None):
    """One crossover instance, cued once from each sequence's first pattern.

    corruption_rate: defaults to CONFIG's; the seed sweep passes a per-trial
    ↪draw."""
    rate = CONFIG["corruption_rate"] if corruption_rate is None else
    ↪corruption_rate
    origin_seed = int(rng.integers(SEED_POOL))
    # 12 is an upper bound over the geometries (2-3-2 consumes 10), held fixed
    ↪so the
    # draw sequence is identical at every geometry.
    mutation_seeds = rng.choice(SEED_POOL, size=12, replace=False).tolist()
    multi = make_crossover_pair(N, rate, origin_seed, mutation_seeds, seq_len,
    ↪shared_idx)
    labels = multi.labels()
    hi = max(shared_idx)

    out = {}
    for name in ("A", "B"):
        other = "B" if name == "A" else "A"
        result = net.simulate(multi, cue=multi[name], cue_idx=0, T=duration)
        retrieved = net.decode_sequence_online(result, multi, margin=MARGIN)

```

```

outcome = classify_unshared(retrieved, multi, name, shared_idx)

overlaps = net.overlaps(result["theta_history"], multi)
own_peak = overlaps[:, labels.index((name, hi + 1))].max()
other_peak = overlaps[:, labels.index((other, hi + 1))].max()

out[name] = dict(result=result, retrieved=retrieved, outcome=outcome,
                 branch_correct=bool(own_peak > other_peak),
                 margin=float(own_peak - other_peak),
                 overlap_quality=(overlap_quality(overlaps, labels,
↳name, seq_len, shared_idx)
                                     if outcome == "correct" else None),
                                     # the cued branch's own peak overlap at each position,
↳in stored
                                     # order -- the succession of retrieved peaks of
↳section 9
                                     peaks=[overlaps[:, labels.index((name, k))].max()
                                             for k in range(seq_len)])

return multi, out

def rule_sweep(catalogue, seq_len, shared_idx, duration, trials=TRIALS, seed=0,
               verbose=True, config_overrides=None, rate_sampler=None):
    """The sweep over whatever catalogue it is handed. Every rule gets a
↳freshly seeded
    `default_rng(seed)`, so all of them see the same crossover instances.

    Returns, per rule, all four metrics: the `correct`/`crossed`/`incomplete`
↳counter
    (metric 1), `mean_overlap` + `overlap_var` over the successful attempts
↳only (metric 2),
    `branch_correct` (metric 3) and `margin` (metric 4).

    config_overrides: dict merged over CONFIG for this call (e.g.
↳frequency_mean /
    frequency_std) -- used by the constant-omega section.
    rate_sampler: called as rate_sampler(rng) once per trial to draw that
↳trial's
    corruption rate; None keeps CONFIG's fixed rate -- used by the seed sweep.
↳"""

    net_config = {**CONFIG, **(config_overrides or {})}
    stats, first = {}, {}
    for rule, table in catalogue.items():
        net = kc.ContextNetwork(rule=table, N=N, seed=10, **net_config)
        rng = np.random.default_rng(seed)
        counter, branch_vals, margins, qualities = Counter(), [], [], []

```

```

peaks, rates = [], []
for trial in range(trials):
    rate = rate_sampler(rng) if rate_sampler is not None else None
    if rate is not None:
        rates.append(rate)
    multi, out = run_crossover_trial(net, rng, seq_len, shared_idx,
    ↪duration,
                                corruption_rate=rate)

    if trial == 0:
        first[rule] = (net, multi, out)
    for name in ("A", "B"):
        counter[out[name]["outcome"]] += 1
        branch_vals.append(float(out[name]["branch_correct"]))
        margins.append(out[name]["margin"])
        peaks.append(out[name]["peaks"])
        if out[name]["overlap_quality"] is not None:
            qualities.append(out[name]["overlap_quality"])
    attempts = sum(counter.values())
    # Branch choice is the quantity these benchmarks exist to measure, so
    ↪it is reported
    # with a spread like every other metric. It is a proportion over
    ↪Bernoulli attempts, so
    # its error bar is the binomial standard error  $\sqrt{p(1-p)/n}$  -- the s.
    ↪d. of the
    # attempts themselves is a function of p and carries no extra
    ↪information. `margin` is
    # the continuous quantity underneath it and does get a genuine s.d.
    p_branch = float(np.mean(branch_vals))
    stats[rule] = dict(counter=counter, attempts=attempts,
                        branch_correct=p_branch,
                        branch_se=float(np.sqrt(p_branch * (1 - p_branch) /
    ↪attempts)),
                        branch_vals=np.asarray(branch_vals),
                        margin=float(np.mean(margins)), margin_std=float(np.
    ↪std(margins)),
                        mean_overlap=float(np.mean(qualities)) if qualities
    ↪else float("nan"),
                        overlap_var=float(np.var(qualities)) if qualities
    ↪else float("nan"),
                        peaks=np.asarray(peaks), rates=rates)

    if verbose:
        q = stats[rule]["mean_overlap"]
        q_str = f"{q:+.3f}" if q == q else " n/a"
        print(f"{rule:<15} correct={counter['correct'] / attempts:>5.0%} "
              f"crossed={counter['crossed'] / attempts:>5.0%} "
              f"incomplete={counter['incomplete'] / attempts:>5.0%} ")

```

```

        f"overlap={q_str} "
        f"branch-correct={p_branch:>5.0%} +/"
↪-{stats[rule]['branch_se']:<5.3f}"
        f" margin={np.mean(margins):>+6.3f} +/-{np.std(margins):.
↪3f}")
    return stats, first

kinds = ["correct", "crossed", "incomplete"]

def zeroed(values):
    """nan -> 0.0. `mean_overlap` is nan when a rule had no successful attempt,
↪to average;
    plotting nan leaves a gap and the line or bar simply vanishes. A rule that
↪never
    retrieves anything should read as a flat zero, which is what it scored."""
    return [0.0 if v != v else float(v) for v in values]

def plot_rule_stats(stats, title, axes=None, show_overlap_var=False):
    """Three panels, one per reported metric family.

    Left -- metric 1, each outcome count divided by the number of cueing
↪attempts, so the
        bar is a fraction in [0, 1] and `correct` is read against a fixed
↪ceiling
        rather than against the other rules.
    Middle -- metric 2, overlap quality on the successful attempts only (error
↪bars =
        1 standard deviation across those attempts, when
↪`show_overlap_var`).
    Right -- metric 3, the branch-choice diagnostic, which the outcome bar
↪cannot show.
        Chance is 0.5."""
    names = list(stats)
    attempts = stats[names[0]]["attempts"]
    if axes is None:
        _, axes = plt.subplots(1, 3, figsize=(20, 4.4))
        axes = np.atleast_1d(axes) # a single-row subplots() call hands back a
↪bare Axes

    bottom = np.zeros(len(names))
    for kind in kinds:
        heights = np.array([stats[n]["counter"][kind] / attempts for n in
↪names])

```

```

        axes[0].bar(names, heights, bottom=bottom, label=kind)
        bottom += heights
    axes[0].set_ylabel("fraction of cueing attempts")
    axes[0].set_ylim(0, 1.0)
    axes[0].set_title(f"{title} -- strict full retrieval ({attempts} attempts_
↳per rule)")
    axes[0].legend(fontsize=8)

    quality = [stats[n]["mean_overlap"] for n in names]
    err = ([np.sqrt(stats[n]["overlap_var"]) if stats[n]["overlap_var"] ==_
↳stats[n]["overlap_var"] else 0.0
            for n in names] if show_overlap_var else None)
    bars = axes[1].bar(names, zeroed(quality), yerr=err, capsize=3, color="tab:
↳green")
    for bar, v in zip(bars, quality):
        if v != v:
            axes[1].text(bar.get_x() + bar.get_width() / 2, 0.02, "0 (none_
↳correct)",
                        ha="center", va="bottom", fontsize=7, rotation=90,
↳color="tab:red")
    axes[1].axhline(CONFIG["tolerance"], color="k", ls="--", lw=1,
↳label="tolerance")
    axes[1].set_ylabel("mean peak overlap (successful attempts only)")
    axes[1].set_ylim(0, 1.05)
    axes[1].set_title(f"{title} -- overlap quality")
    axes[1].legend(fontsize=8)

    axes[2].bar(names, [stats[n]["branch_correct"] for n in names], color="tab:
↳purple")
    axes[2].axhline(0.5, color="k", ls="--", lw=1, label="chance")
    axes[2].set_ylabel("fraction branch-correct")
    axes[2].set_ylim(0, 1.05)
    axes[2].set_title(f"{title} -- branch choice at the shared state")
    axes[2].legend(fontsize=8)

    for ax in axes:
        ax.set_xticks(range(len(names)))
        ax.set_xticklabels(names, rotation=20, ha="right", fontsize=9)
        ax.grid(True, axis="y", alpha=0.3)
    return axes

```

2-1-2	P=5	shared={2}	L=1	unshared before=2, after=2	duration=80
2-2-2	P=6	shared={2, 3}	L=2	unshared before=2, after=2	duration=80
2-3-2	P=7	shared={2, 3, 4}	L=3	unshared before=2, after=2	duration=80
1-1-1	P=3	shared={1}	L=1	unshared before=1, after=1	duration=60

5.3.2 Does a longer duration rescue the raw rules?

Whether simply giving the integration more time closes the gap the gain correction closes. Below: the two rules most affected – **bigram** (one lag) and **coh trigram** (two) – on the hardest geometry at **duration** = 50, 60, 70, 80 – the whole range the 80.0 cap allows.

The outcome fractions come back **bit-identical at every duration**: the trajectories have already hit the simulator’s stationary early-stop well inside the shortest, so the extra time is never used. Measured separately before the cap was imposed, 80.0 also reproduces 100.0, 120.0 and 140.0 exactly, rule by rule and geometry by geometry, so nothing is lost by stopping there. That is the signature of a sub-tolerance fixed point under weaker drive, not of slow convergence – only a change to the drive strength itself moves it.

```
[46]: DURATION_CHECK = {
      "bigram":      kc.bigram_rule(),
      "coh trigram": kc.coherent_trigram_rule(),
    }

print("2-3-2 -- does a longer duration change the raw outcome?")
for duration in (50.0, 60.0, 70.0, 80.0):
    print(f"-- duration = {duration:.0f} --")
    _ = rule_sweep(DURATION_CHECK, **{**GEOMETRIES["2-3-2"], "duration":
    ↪duration}, trials=8)
```

```
2-3-2 -- does a longer duration change the raw outcome?
-- duration = 50 --
bigram          correct= 19%  crossed= 50%  incomplete= 31%  overlap=+0.760
branch-correct= 38% +/-0.121  margin=-0.008 +/-0.070
coh trigram      correct= 19%  crossed= 38%  incomplete= 44%  overlap=+0.741
branch-correct= 50% +/-0.125  margin=-0.008 +/-0.070
-- duration = 60 --
bigram          correct= 19%  crossed= 50%  incomplete= 31%  overlap=+0.760
branch-correct= 38% +/-0.121  margin=-0.008 +/-0.070
coh trigram      correct= 19%  crossed= 38%  incomplete= 44%  overlap=+0.741
branch-correct= 50% +/-0.125  margin=-0.008 +/-0.070
-- duration = 70 --
bigram          correct= 19%  crossed= 50%  incomplete= 31%  overlap=+0.760
branch-correct= 38% +/-0.121  margin=-0.008 +/-0.070
coh trigram      correct= 19%  crossed= 38%  incomplete= 44%  overlap=+0.741
branch-correct= 50% +/-0.125  margin=-0.008 +/-0.070
-- duration = 80 --
bigram          correct= 19%  crossed= 50%  incomplete= 31%  overlap=+0.760
branch-correct= 38% +/-0.121  margin=-0.008 +/-0.070
coh trigram      correct= 19%  crossed= 38%  incomplete= 44%  overlap=+0.741
branch-correct= 50% +/-0.125  margin=-0.008 +/-0.070
```

6 4. The benchmark experiments ($\sigma_\omega = 0.3$)

The whole catalogue, **raw** and **gain-corrected**, on each topology. Panels: strict full-retrieval outcome, overlap quality, branch choice.

The order is the point: (a) needs reach 1, which every context rule has; (c) needs reach 2, which only coh trigram has; 2-3-2 needs reach 3, which nothing has.

```
[47]: all_stats, all_first = {}, {}

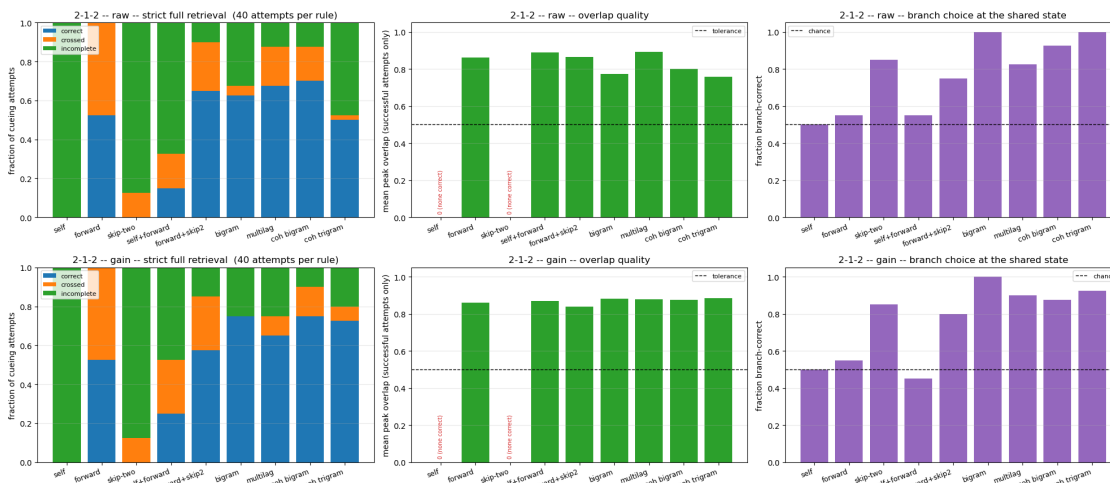
for tag in GEOMETRIES:
    geom = GEOMETRIES[tag]
    lo, hi = min(geom["shared_idx"]), max(geom["shared_idx"])
    print("=" * 100)
    print(f"{tag}    P={geom['seq_len']}, shared {set(geom['shared_idx'])}, L={hi - lo + 1}")
    print("-- raw --")
    stats_raw, first_raw = rule_sweep(RAW_CATALOGUE, **geom)
    print("-- gain --")
    stats_gain, _ = rule_sweep(GAIN_CATALOGUE, **geom)
    print()

    all_stats[tag] = stats_raw
    all_stats[tag + " gain"] = stats_gain
    all_first[tag] = first_raw

fig, axes = plt.subplots(2, 3, figsize=(20, 8.8))
plot_rule_stats(stats_raw, f"{tag} -- raw", axes=axes[0])
plot_rule_stats(stats_gain, f"{tag} -- gain", axes=axes[1])
plt.tight_layout(); plt.show()
```

```
=====
=====
2-1-2    P=5, shared {2}, L=1
-- raw --
self      correct= 0%  crossed= 0%  incomplete= 100%  overlap= n/a
branch-correct= 50% +/-0.079  margin=+0.005 +/-0.034
forward    correct= 52%  crossed= 48%  incomplete= 0%   overlap=+0.862
branch-correct= 55% +/-0.079  margin=+0.006 +/-0.090
skip-two   correct= 0%  crossed= 12%  incomplete= 88%  overlap= n/a
branch-correct= 85% +/-0.056  margin=+0.096 +/-0.064
self+forward correct= 15%  crossed= 18%  incomplete= 68%  overlap=+0.889
branch-correct= 55% +/-0.079  margin=-0.003 +/-0.068
forward+skip2 correct= 65%  crossed= 25%  incomplete= 10%  overlap=+0.865
branch-correct= 75% +/-0.068  margin=+0.060 +/-0.089
bigram     correct= 62%  crossed= 5%   incomplete= 32%  overlap=+0.774
branch-correct= 100% +/-0.000  margin=+0.124 +/-0.062
multilag   correct= 68%  crossed= 20%  incomplete= 12%  overlap=+0.891
branch-correct= 82% +/-0.060  margin=+0.099 +/-0.088
```

coh bigram	correct= 70%	crossed= 18%	incomplete= 12%	overlap=+0.801
branch-correct=	92% +/-0.042	margin=+0.095	+/-0.067	
coh trigram	correct= 50%	crossed= 2%	incomplete= 48%	overlap=+0.758
branch-correct=	100% +/-0.000	margin=+0.144	+/-0.053	
-- gain --				
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.005	+/-0.034	
forward	correct= 52%	crossed= 48%	incomplete= 0%	overlap=+0.862
branch-correct=	55% +/-0.079	margin=+0.006	+/-0.090	
skip-two	correct= 0%	crossed= 12%	incomplete= 88%	overlap= n/a
branch-correct=	85% +/-0.056	margin=+0.096	+/-0.064	
self+forward	correct= 25%	crossed= 28%	incomplete= 48%	overlap=+0.869
branch-correct=	45% +/-0.079	margin=-0.002	+/-0.081	
forward+skip2	correct= 57%	crossed= 28%	incomplete= 15%	overlap=+0.838
branch-correct=	80% +/-0.063	margin=+0.063	+/-0.078	
bigram	correct= 75%	crossed= 0%	incomplete= 25%	overlap=+0.881
branch-correct=	100% +/-0.000	margin=+0.141	+/-0.058	
multilag	correct= 65%	crossed= 10%	incomplete= 25%	overlap=+0.880
branch-correct=	90% +/-0.047	margin=+0.132	+/-0.081	
coh bigram	correct= 75%	crossed= 15%	incomplete= 10%	overlap=+0.876
branch-correct=	88% +/-0.052	margin=+0.091	+/-0.074	
coh trigram	correct= 72%	crossed= 8%	incomplete= 20%	overlap=+0.884
branch-correct=	92% +/-0.042	margin=+0.151	+/-0.078	



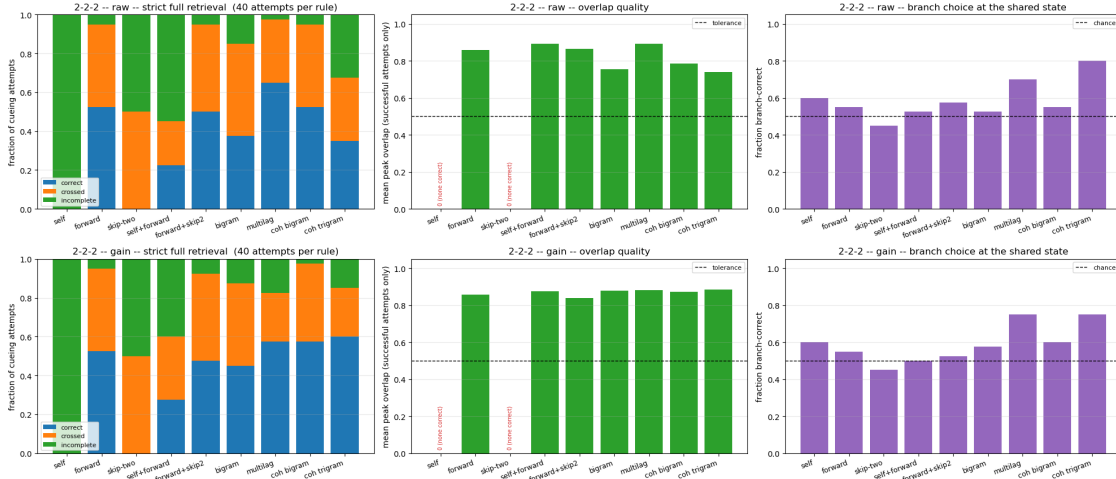
=====

2-2-2 P=6, shared {2, 3}, L=2

-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	60% +/-0.077	margin=+0.011	+/-0.074	

forward	correct= 52%	crossed= 42%	incomplete= 5%	overlap=+0.858
branch-correct=	55% +/-0.079	margin=+0.009 +/-0.077		
skip-two	correct= 0%	crossed= 50%	incomplete= 50%	overlap= n/a
branch-correct=	45% +/-0.079	margin=+0.008 +/-0.071		
self+forward	correct= 22%	crossed= 22%	incomplete= 55%	overlap=+0.892
branch-correct=	52% +/-0.079	margin=+0.003 +/-0.080		
forward+skip2	correct= 50%	crossed= 45%	incomplete= 5%	overlap=+0.865
branch-correct=	57% +/-0.078	margin=+0.006 +/-0.092		
bigram	correct= 38%	crossed= 48%	incomplete= 15%	overlap=+0.756
branch-correct=	52% +/-0.079	margin=+0.007 +/-0.073		
multilag	correct= 65%	crossed= 32%	incomplete= 2%	overlap=+0.891
branch-correct=	70% +/-0.072	margin=+0.044 +/-0.105		
coh bigram	correct= 52%	crossed= 42%	incomplete= 5%	overlap=+0.784
branch-correct=	55% +/-0.079	margin=+0.005 +/-0.079		
coh trigram	correct= 35%	crossed= 32%	incomplete= 32%	overlap=+0.739
branch-correct=	80% +/-0.063	margin=+0.054 +/-0.068		
-- gain --				
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	60% +/-0.077	margin=+0.011 +/-0.074		
forward	correct= 52%	crossed= 42%	incomplete= 5%	overlap=+0.858
branch-correct=	55% +/-0.079	margin=+0.009 +/-0.077		
skip-two	correct= 0%	crossed= 50%	incomplete= 50%	overlap= n/a
branch-correct=	45% +/-0.079	margin=+0.008 +/-0.071		
self+forward	correct= 28%	crossed= 32%	incomplete= 40%	overlap=+0.874
branch-correct=	50% +/-0.079	margin=+0.003 +/-0.077		
forward+skip2	correct= 48%	crossed= 45%	incomplete= 8%	overlap=+0.839
branch-correct=	52% +/-0.079	margin=+0.003 +/-0.079		
bigram	correct= 45%	crossed= 42%	incomplete= 12%	overlap=+0.878
branch-correct=	57% +/-0.078	margin=+0.006 +/-0.106		
multilag	correct= 57%	crossed= 25%	incomplete= 18%	overlap=+0.880
branch-correct=	75% +/-0.068	margin=+0.069 +/-0.109		
coh bigram	correct= 57%	crossed= 40%	incomplete= 2%	overlap=+0.873
branch-correct=	60% +/-0.077	margin=+0.005 +/-0.091		
coh trigram	correct= 60%	crossed= 25%	incomplete= 15%	overlap=+0.885
branch-correct=	75% +/-0.068	margin=+0.071 +/-0.112		



2-3-2 P=7, shared {2, 3, 4}, L=3

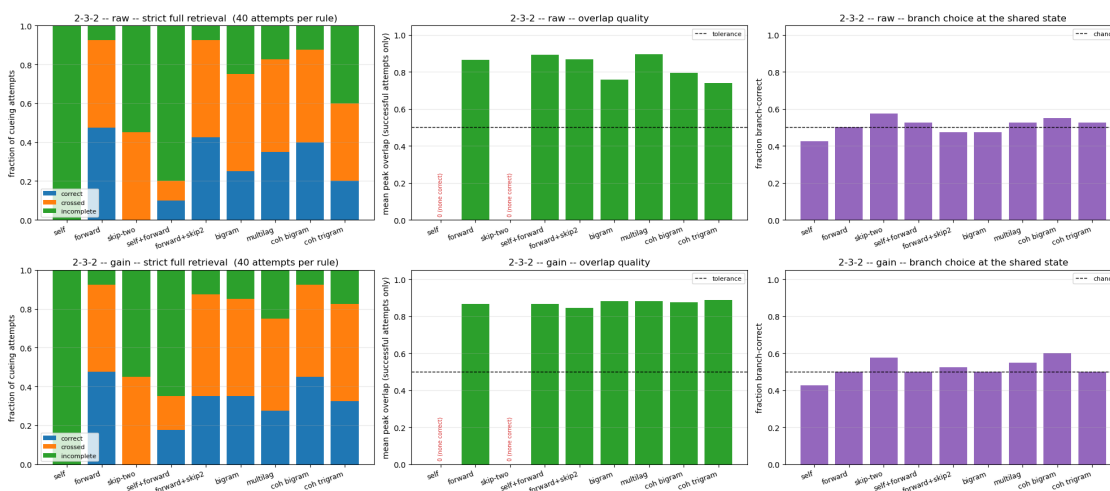
-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	42% +/-0.078	margin=-0.004 +/-0.072		
forward	correct= 48%	crossed= 45%	incomplete= 8%	overlap=+0.866
branch-correct=	50% +/-0.079	margin=+0.004 +/-0.088		
skip-two	correct= 0%	crossed= 45%	incomplete= 55%	overlap= n/a
branch-correct=	57% +/-0.078	margin=-0.002 +/-0.080		
self+forward	correct= 10%	crossed= 10%	incomplete= 80%	overlap=+0.893
branch-correct=	52% +/-0.079	margin=-0.001 +/-0.058		
forward+skip2	correct= 42%	crossed= 50%	incomplete= 8%	overlap=+0.867
branch-correct=	48% +/-0.079	margin=-0.001 +/-0.097		
bigram	correct= 25%	crossed= 50%	incomplete= 25%	overlap=+0.759
branch-correct=	48% +/-0.079	margin=-0.000 +/-0.059		
multilag	correct= 35%	crossed= 48%	incomplete= 18%	overlap=+0.895
branch-correct=	52% +/-0.079	margin=+0.003 +/-0.114		
coh bigram	correct= 40%	crossed= 48%	incomplete= 12%	overlap=+0.795
branch-correct=	55% +/-0.079	margin=+0.004 +/-0.063		
coh trigram	correct= 20%	crossed= 40%	incomplete= 40%	overlap=+0.738
branch-correct=	52% +/-0.079	margin=+0.002 +/-0.061		

-- gain --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	42% +/-0.078	margin=-0.004 +/-0.072		
forward	correct= 48%	crossed= 45%	incomplete= 8%	overlap=+0.866
branch-correct=	50% +/-0.079	margin=+0.004 +/-0.088		
skip-two	correct= 0%	crossed= 45%	incomplete= 55%	overlap= n/a
branch-correct=	57% +/-0.078	margin=-0.002 +/-0.080		
self+forward	correct= 18%	crossed= 18%	incomplete= 65%	overlap=+0.867
branch-correct=	50% +/-0.079	margin=-0.002 +/-0.063		

forward+skip2	correct= 35%	crossed= 52%	incomplete= 12%	overlap=+0.844
branch-correct=	52% +/-0.079	margin=+0.003 +/-0.074		
bigram	correct= 35%	crossed= 50%	incomplete= 15%	overlap=+0.882
branch-correct=	50% +/-0.079	margin=-0.002 +/-0.107		
multilag	correct= 28%	crossed= 48%	incomplete= 25%	overlap=+0.881
branch-correct=	55% +/-0.079	margin=+0.001 +/-0.107		
coh bigram	correct= 45%	crossed= 48%	incomplete= 8%	overlap=+0.875
branch-correct=	60% +/-0.077	margin=+0.004 +/-0.091		
coh trigram	correct= 32%	crossed= 50%	incomplete= 18%	overlap=+0.887
branch-correct=	50% +/-0.079	margin=-0.004 +/-0.120		



1-1-1 P=3, shared {1}, L=1

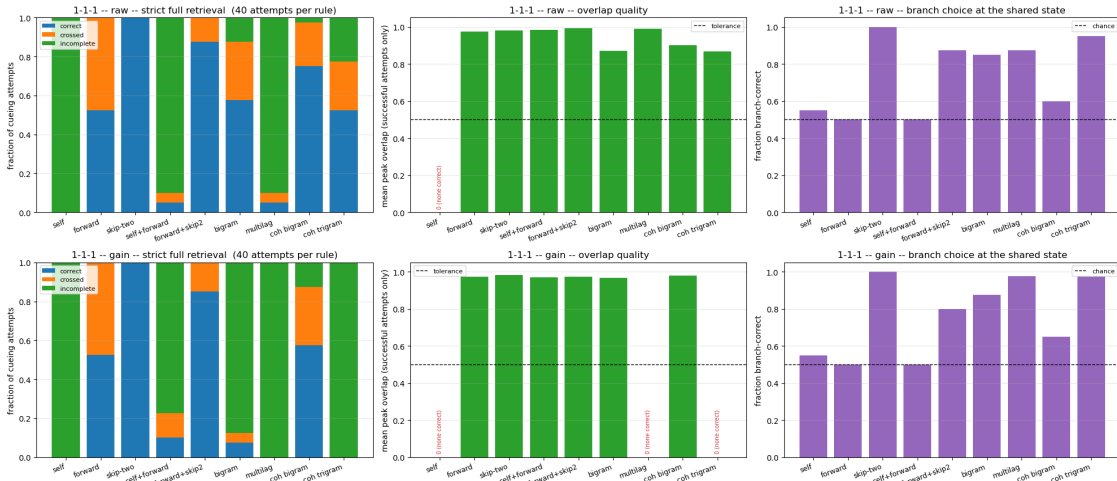
-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	55% +/-0.079	margin=+0.000 +/-0.010		
forward	correct= 52%	crossed= 48%	incomplete= 0%	overlap=+0.974
branch-correct=	50% +/-0.079	margin=+0.005 +/-0.311		
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+0.981
branch-correct=	100% +/-0.000	margin=+0.478 +/-0.046		
self+forward	correct= 5%	crossed= 5%	incomplete= 90%	overlap=+0.982
branch-correct=	50% +/-0.079	margin=+0.002 +/-0.100		
forward+skip2	correct= 88%	crossed= 12%	incomplete= 0%	overlap=+0.993
branch-correct=	88% +/-0.052	margin=+0.279 +/-0.217		
bigram	correct= 57%	crossed= 30%	incomplete= 12%	overlap=+0.871
branch-correct=	85% +/-0.056	margin=+0.058 +/-0.070		
multilag	correct= 5%	crossed= 5%	incomplete= 90%	overlap=+0.989
branch-correct=	88% +/-0.052	margin=+0.039 +/-0.100		
coh bigram	correct= 75%	crossed= 22%	incomplete= 2%	overlap=+0.902

```

branch-correct= 60% +/-0.077  margin=+0.076 +/-0.154
coh trigram      correct= 52%  crossed= 25%  incomplete= 22%  overlap=+0.867
branch-correct= 95% +/-0.034  margin=+0.109 +/-0.064
-- gain --
self             correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 55% +/-0.079  margin=+0.000 +/-0.010
forward          correct= 52%  crossed= 48%  incomplete=  0%  overlap=+0.974
branch-correct= 50% +/-0.079  margin=+0.005 +/-0.311
skip-two         correct= 100%  crossed=  0%  incomplete=  0%  overlap=+0.981
branch-correct= 100% +/-0.000  margin=+0.478 +/-0.046
self+forward     correct= 10%  crossed= 12%  incomplete= 78%  overlap=+0.970
branch-correct= 50% +/-0.079  margin=+0.002 +/-0.140
forward+skip2    correct= 85%  crossed= 15%  incomplete=  0%  overlap=+0.973
branch-correct= 80% +/-0.063  margin=+0.228 +/-0.256
bigram           correct=  8%  crossed=  5%  incomplete= 88%  overlap=+0.968
branch-correct= 88% +/-0.052  margin=+0.051 +/-0.099
multilag         correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 98% +/-0.025  margin=+0.052 +/-0.026
coh bigram       correct= 57%  crossed= 30%  incomplete= 12%  overlap=+0.979
branch-correct= 65% +/-0.075  margin=+0.114 +/-0.253
coh trigram      correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 98% +/-0.025  margin=+0.053 +/-0.022

```



6.0.1 Note on skip-two

skip-two scores 100% on both panels, the same artefact `transition_matrix.ipynb` documents for its skip `nhop=2` matrix: at $P = 3$ the rule drives $\mu \rightarrow \mu + 2$, so cued at $A[0]$ it targets $A[2]$ directly and the shared centre is never a target of the drive at all. It sidesteps the crossover rather than resolving it. The middle label still shows up in the decoded path because the centre sits *between* $A[0]$ and $A[2]$ in overlap space – a decoded label is not a driven pattern. At $P = 5$ it drops

back to the pack.

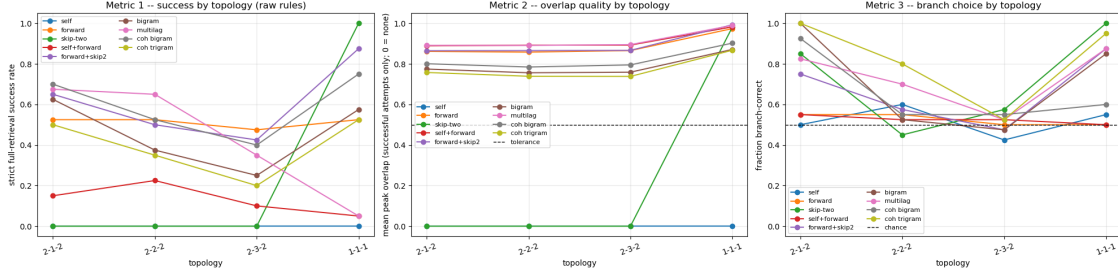
6.1 All topologies side by side

Raw rules only: metrics 1, 2 and 3 across the four topologies.

```
[48]: PANEL_TAGS = SPEC + ["1-1-1"]

fig, axes = plt.subplots(1, 3, figsize=(20, 5.0))
for rule in RAW_CATALOGUE:
    axes[0].plot(PANEL_TAGS, [all_stats[t][rule]["counter"]["correct"] /
    ↪all_stats[t][rule]["attempts"]
                                for t in PANEL_TAGS], "-o", label=rule)
    axes[1].plot(PANEL_TAGS, zeroed([all_stats[t][rule]["mean_overlap"] for t
    ↪in PANEL_TAGS]),
                "-o", label=rule)
    axes[2].plot(PANEL_TAGS, [all_stats[t][rule]["branch_correct"] for t in
    ↪PANEL_TAGS],
                "-o", label=rule)
axes[0].set_ylabel("strict full-retrieval success rate")
axes[0].set_title("Metric 1 -- success by topology (raw rules)")
axes[1].axhline(CONFIG["tolerance"], color="k", ls="--", lw=1,
    ↪label="tolerance")
axes[1].set_ylabel("mean peak overlap (successful attempts only; 0 = none)")
axes[1].set_title("Metric 2 -- overlap quality by topology")
axes[2].axhline(0.5, color="k", ls="--", lw=1, label="chance")
axes[2].set_ylabel("fraction branch-correct")
axes[2].set_title("Metric 3 -- branch choice by topology")
for ax in axes:
    ax.set_xlabel("topology"); ax.set_ylim(-0.05, 1.05)
    ax.tick_params(axis="x", rotation=20)
    ax.legend(fontsize=8, ncol=2); ax.grid(alpha=0.3)
plt.tight_layout(); plt.show()

print(f"{'rule':<15}" + "".join(f"{'t':>26}" for t in PANEL_TAGS))
print(f"{'':<15}" + "".join(f"{'correct / overlap / branch':>26}" for t in
    ↪PANEL_TAGS))
print("-" * (15 + 26 * len(PANEL_TAGS)))
for rule in RAW_CATALOGUE:
    row = ""
    for t in PANEL_TAGS:
        s = all_stats[t][rule]
        q = f"{s['mean_overlap']:.2f}" if s["mean_overlap"] ==
    ↪s["mean_overlap"] else " n/a"
        row += f"{s['counter']['correct'] / s['attempts']:>11.0%} / {q:>6} /
    ↪{s['branch_correct']:>6.0%}"
    print(f"{'rule':<15}{row}")
```



rule	2-1-2				2-2-2			
2-3-2	1-1-1							
	correct	overlap	branchcorrect	overlap	branchcorrect	correct	overlap	branch
self	0% /	n/a /	50%	0% /	n/a /	60%	0%	
/ n/a / 42%	0% /	n/a /	55%					
forward	52% /	0.86 /	55%	52% /	0.86 /	55%	48%	
/ 0.87 / 50%	52% /	0.97 /	50%					
skip-two	0% /	n/a /	85%	0% /	n/a /	45%	0%	
/ n/a / 57%	100% /	0.98 /	100%					
self+forward	15% /	0.89 /	55%	22% /	0.89 /	52%	10%	
/ 0.89 / 52%	5% /	0.98 /	50%					
forward+skip2	65% /	0.86 /	75%	50% /	0.87 /	57%	42%	
/ 0.87 / 48%	88% /	0.99 /	88%					
bigram	62% /	0.77 /	100%	38% /	0.76 /	52%	25%	
/ 0.76 / 48%	57% /	0.87 /	85%					
multilag	68% /	0.89 /	82%	65% /	0.89 /	70%	35%	
/ 0.90 / 52%	5% /	0.99 /	88%					
coh bigram	70% /	0.80 /	92%	52% /	0.78 /	55%	40%	
/ 0.79 / 55%	75% /	0.90 /	60%					
coh trigram	50% /	0.76 /	100%	35% /	0.74 /	80%	20%	
/ 0.74 / 52%	52% /	0.87 /	95%					

6.1.1 Reading the two panels together

The outcome bar and the branch-choice bar say different things, and on 1-1-1 they say *opposite* things: **forward** is at chance on branch choice and scores well on **correct**, while **bigram**, **multilag** and **coh trigram** are far above chance on branch choice and score badly on **correct**, the difference sitting in **incomplete**.

That splits the problem in two. **Picking the branch** every context rule does and the current-state-only rules do not: the gate at the shared state is $m_{\star}|m_{\mu-1}|^2$, and $m_{\mu-1}$ is larger for the branch the trajectory came from. **Committing to it** needs the driven pattern's own overlap to amplify itself; **forward**'s gate is m_{μ}^3 , cubic in the pattern it drives, while a context rule spends factors on lagged overlaps and has only m_{μ}^1 or m_{μ}^2 left, so it settles between the two continuations.

`coh bigram` does both, and is the only context rule keeping two factors on the current overlap ($m_\mu^2 m_{\mu-1}$) – the only one whose `incomplete` fraction stays near `forward`’s on all three geometries.

On sample size. 40 attempts per rule: a fraction carries a standard error of ~ 8 points and differences under ~ 15 points are not resolved. What survives is `forward` at chance versus the context rules at 85-98% on branch choice at $P = 3$, their matching collapse into `incomplete`, and the whole catalogue at chance on the shared run of three.

7 5. Knob 1 — the reach ($\sigma_\omega = 0.3$)

The catalogue fixes each rule’s lags. Both context families generalize to an arbitrary reach r with the same number of factors:

$$\text{bigram_rule}(\text{lag}=r) : \xi^{\mu+1} m_\mu |m_{\mu-r}|^2 \quad \text{coherent_trigram_rule}(\text{lag}=r) : \xi^{\mu+1} m_\mu m_{\mu-r+1} m_{\mu-r}$$

At $r = 1$ the coherent family degenerates to `coh bigram`; at $r = 2$ it is the write-up’s eq (19). The box above predicts a threshold at $r = L$. On 2-1-2 that is $r \geq 1$, so every reach tested qualifies and the curve should simply sit above chance; the informative panels are 2-2-2 ($L = 2$) and 2-3-2, where only $r \geq 3$ reads an unshared pattern at all.

```
[49]: REACH = [1, 2, 3, 4]

reach_stats = {}
for tag in FOCUS:
    geom = GEOMETRIES[tag]
    need = max(geom["shared_idx"]) - min(geom["shared_idx"]) + 1
    print(f"{tag}: P = {geom['seq_len']}, shared {set(geom['shared_idx'])} "
          f"branch information first available at reach = {need}")
    catalogue = {f"bigram r={r}": kc.bigram_rule(lag=r) for r in REACH}
    catalogue.update({f"coh r={r}": kc.coherent_trigram_rule(lag=r) for r in
    REACH})
    reach_stats[tag] = rule_sweep(catalogue, **geom)[0]
    print()

fig, axes = plt.subplots(1, len(FOCUS), figsize=(6.0 * len(FOCUS), 4.4))
for ax, tag in zip(np.atleast_1d(axes), FOCUS):
    stats = reach_stats[tag]
    geom = GEOMETRIES[tag]
    need = max(geom["shared_idx"]) - min(geom["shared_idx"]) + 1
    for prefix, style in [("bigram", "-o"), ("coh", "--s")]:
        ax.plot(REACH, [stats[f"{prefix} r={r}"]["branch_correct"] for r in
        REACH],
                style, lw=2, label=f"{prefix}")
        ax.axvline(need, color="tab:grey", ls=":", lw=1.5, label=f"reach = L =
        {need}")
```

```

ax.axhline(0.5, color="k", ls=":", lw=1, label="chance")
ax.set_xlabel("reach $r$ of the gate"); ax.set_xticks(REACH)
ax.set_ylabel("fraction branch-correct"); ax.set_ylim(-0.05, 1.05)
ax.set_title(tag); ax.legend(fontsize=9); ax.grid(alpha=0.3)
plt.tight_layout(); plt.show()

```

2-1-2: P = 5, shared {2} branch information first available at reach = 1

bigram r=1	correct= 62%	crossed= 5%	incomplete= 32%	overlap=+0.774
branch-correct=	100% +/-0.000	margin=+0.124	+/-0.062	
bigram r=2	correct= 38%	crossed= 2%	incomplete= 60%	overlap=+0.750
branch-correct=	100% +/-0.000	margin=+0.151	+/-0.049	
bigram r=3	correct= 15%	crossed= 2%	incomplete= 82%	overlap=+0.734
branch-correct=	100% +/-0.000	margin=+0.171	+/-0.044	
bigram r=4	correct= 2%	crossed= 2%	incomplete= 95%	overlap=+0.731
branch-correct=	100% +/-0.000	margin=+0.168	+/-0.044	
coh r=1	correct= 70%	crossed= 18%	incomplete= 12%	overlap=+0.801
branch-correct=	92% +/-0.042	margin=+0.095	+/-0.067	
coh r=2	correct= 50%	crossed= 2%	incomplete= 48%	overlap=+0.758
branch-correct=	100% +/-0.000	margin=+0.144	+/-0.053	
coh r=3	correct= 32%	crossed= 2%	incomplete= 65%	overlap=+0.740
branch-correct=	100% +/-0.000	margin=+0.165	+/-0.045	
coh r=4	correct= 5%	crossed= 2%	incomplete= 92%	overlap=+0.738
branch-correct=	100% +/-0.000	margin=+0.170	+/-0.044	

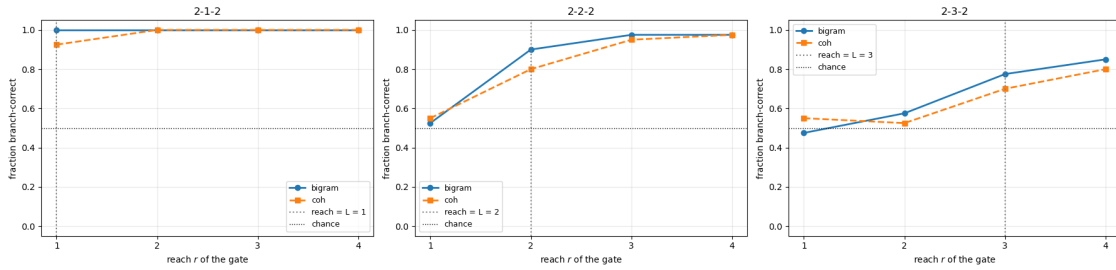
2-2-2: P = 6, shared {2, 3} branch information first available at reach = 2

bigram r=1	correct= 38%	crossed= 48%	incomplete= 15%	overlap=+0.756
branch-correct=	52% +/-0.079	margin=+0.007	+/-0.073	
bigram r=2	correct= 22%	crossed= 15%	incomplete= 62%	overlap=+0.734
branch-correct=	90% +/-0.047	margin=+0.079	+/-0.065	
bigram r=3	correct= 8%	crossed= 0%	incomplete= 92%	overlap=+0.718
branch-correct=	98% +/-0.025	margin=+0.111	+/-0.057	
bigram r=4	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	98% +/-0.025	margin=+0.122	+/-0.053	
coh r=1	correct= 52%	crossed= 42%	incomplete= 5%	overlap=+0.784
branch-correct=	55% +/-0.079	margin=+0.005	+/-0.079	
coh r=2	correct= 35%	crossed= 32%	incomplete= 32%	overlap=+0.739
branch-correct=	80% +/-0.063	margin=+0.054	+/-0.068	
coh r=3	correct= 12%	crossed= 0%	incomplete= 88%	overlap=+0.726
branch-correct=	95% +/-0.034	margin=+0.101	+/-0.060	
coh r=4	correct= 2%	crossed= 0%	incomplete= 98%	overlap=+0.719
branch-correct=	98% +/-0.025	margin=+0.120	+/-0.054	

2-3-2: P = 7, shared {2, 3, 4} branch information first available at reach = 3

bigram r=1	correct= 25%	crossed= 50%	incomplete= 25%	overlap=+0.759
branch-correct=	48% +/-0.079	margin=-0.000	+/-0.059	
bigram r=2	correct= 10%	crossed= 32%	incomplete= 57%	overlap=+0.726
branch-correct=	57% +/-0.078	margin=+0.003	+/-0.064	
bigram r=3	correct= 5%	crossed= 5%	incomplete= 90%	overlap=+0.710

branch-correct=	78% +/-0.066	margin=	+0.055 +/-0.070		
bigram r=4	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	85% +/-0.056	margin=	+0.070 +/-0.062		
coh r=1	correct= 40%	crossed= 48%	incomplete= 12%	overlap=	+0.795
branch-correct=	55% +/-0.079	margin=	+0.004 +/-0.063		
coh r=2	correct= 20%	crossed= 40%	incomplete= 40%	overlap=	+0.738
branch-correct=	52% +/-0.079	margin=	+0.002 +/-0.061		
coh r=3	correct= 5%	crossed= 12%	incomplete= 82%	overlap=	+0.707
branch-correct=	70% +/-0.072	margin=	+0.036 +/-0.066		
coh r=4	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	80% +/-0.063	margin=	+0.067 +/-0.064		



7.0.1 Why reach alone does not rescue 2-3-2

Reach $\geq L$ is necessary and not sufficient. `coh r=3` on 2-3-2 *does* read m_{a_0} , the only overlap that can distinguish the branches, and is still at chance.

What it reads has already decayed by the time it is read. The gate compares $m_{a_{\ell-1}}$ against $m_{b_{\ell-1}}$ – the traversed against the untraversed predecessor of the shared run. Both are the *same distance* from the shared patterns in mutation steps, so their static overlaps are identical and the entire signal is the dynamic residual of having passed through one of them. The cell below measures that residual on the baseline **forward** trajectory, cued from $A[0]$, reading the gap as the state peaks on each pattern in turn.

```
[50]: GAP_TRIALS = 10

fwd = kc.ContextNetwork(rule=kc.forward_rule(), N=N, seed=10, **CONFIG)
fig, ax = plt.subplots(figsize=(9, 4.6))

for tag, style in zip(FOCUS, ("-o", "--s", ":-^")):
    geom = GEOMETRIES[tag]
    P, shared, duration = geom["seq_len"], geom["shared_idx"], geom["duration"]
    lo, hi = min(shared), max(shared)
    L = hi - lo + 1
    gaps = np.zeros((GAP_TRIALS, P))
    for trial in range(GAP_TRIALS):
```

```

rng = np.random.default_rng(trial)
multi = make_crossover_pair(
    N, CONFIG["corruption_rate"], int(rng.integers(SEED_POOL)),
    rng.choice(SEED_POOL, size=12, replace=False).tolist(), P, shared)
labels = multi.labels()
result = fwd.simulate(multi, cue=multi["A"], cue_idx=0, T=duration)
ov = fwd.overlaps(result["theta_history"], multi)
gap = ov[:, labels.index(("A", lo - 1))] - ov[:, labels.index(("B", lo
↳ - 1))]

    for k in range(P):
        gaps[trial, k] = gap[int(np.argmax(ov[:, labels.index(("A", k))]))]

    print(f"{tag} (L={L})    mean gap  m(A[{lo-1}]) - m(B[{lo-1}])  at the peak
↳ of each pattern")
    print("      " + " ".join(f"A[{k}]{'*' if k in shared else ''}{'^' if k ==
↳ hi+1 else ''}")
                                f"{gaps[:, k].mean():>9.3f}    " for k in range(P)))
    print("      * = shared,  ^ = the branch point, where the gate still has to
↳ be reading it\n")

    line, = ax.plot(range(P), gaps.mean(0), style, lw=2, label=f"{tag} 
↳ (L={L})")
    ax.plot(hi + 1, gaps[:, hi + 1].mean(), "o", ms=14, mfc="none", mew=2,
↳ color=line.get_color())

ax.axhline(0.0, color="k", ls=":", lw=1)
ax.set_xlabel("state is peaking on pattern A[k]")
ax.set_ylabel(r"$m(A[\ell-1]) - m(B[\ell-1])$")
ax.set_title("Lifetime of the branch signal (ring = the branch point)")
ax.legend(fontsize=9); ax.grid(alpha=0.3)
plt.tight_layout(); plt.show()

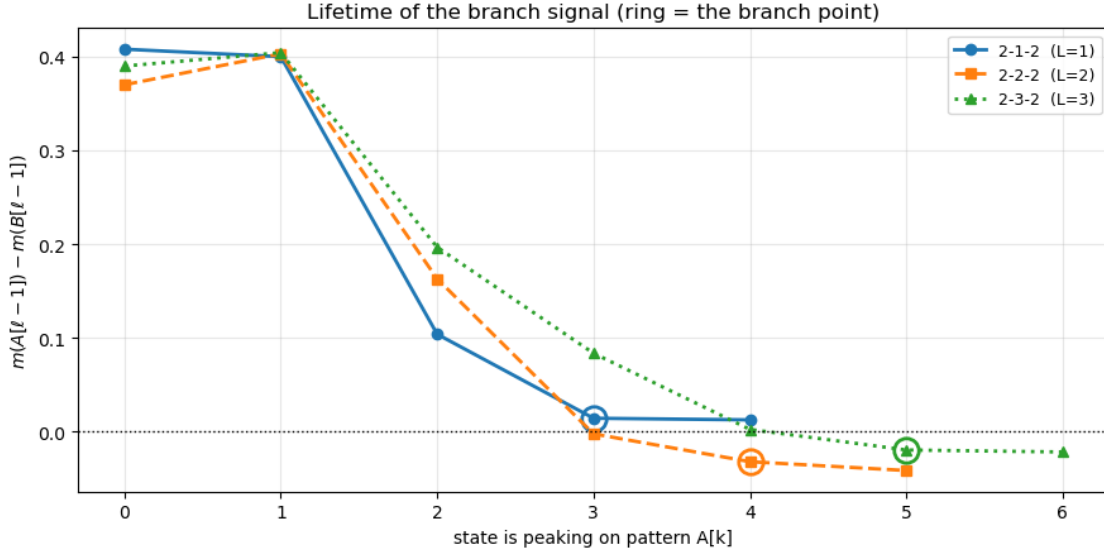
```

2-1-2 (L=1) mean gap m(A[1]) - m(B[1]) at the peak of each pattern
A[0] +0.408 A[1] +0.400 A[2]* +0.104 A[3]^ +0.014 A[4]
+0.012
* = shared, ^ = the branch point, where the gate still has to be reading
it

2-2-2 (L=2) mean gap m(A[1]) - m(B[1]) at the peak of each pattern
A[0] +0.370 A[1] +0.403 A[2]* +0.163 A[3]* -0.003 A[4]^
-0.032 A[5] -0.042
* = shared, ^ = the branch point, where the gate still has to be reading
it

2-3-2 (L=3) mean gap m(A[1]) - m(B[1]) at the peak of each pattern
A[0] +0.390 A[1] +0.404 A[2]* +0.197 A[3]* +0.083 A[4]*
+0.002 A[5]^ -0.020 A[6] -0.022

* = shared, ^ = the branch point, where the gate still has to be reading it



8 6. Knob 2 — the multilag weights ($\sigma_\omega = 0.3$)

multilag is the one rule in the catalogue with weights to sweep, eq (17):

$$h_i^{\text{multilag}} = \sum_{\mu} \xi_i^{\mu+1} m_{\mu} (w_0 |m_{\mu}|^2 + w_1 |m_{\mu-1}|^2 + w_2 |m_{\mu-2}|^2)$$

w_0 is the ungated forward term and w_1, w_2 the context gates, so $(1, 0, 0)$ is **forward** and $(0, 1, 0)$ is **bigram**. After `proportional_gain` the weights are *shares* summing to 1.

This is the trade-off the two panels above set up – w_0 buys commitment, w_1 and w_2 buy branch information – so the grid should show them pulling against each other.

```
[51]: GRID_W = [0.0, 0.5, 1.0, 2.0]      # weight on the lag-1 and lag-2 gates, with
      ↪ w0 = 1

grid_stats = {}
for tag in FOCUS:
    geom = GEOMETRIES[tag]
    catalogue = {(w1, w2): kc.multilag_rule((1.0, w1, w2))
                  for w1 in GRID_W for w2 in GRID_W}
    grid_stats[tag] = rule_sweep(catalogue, **geom, trials=15, verbose=False)[0]
```

```

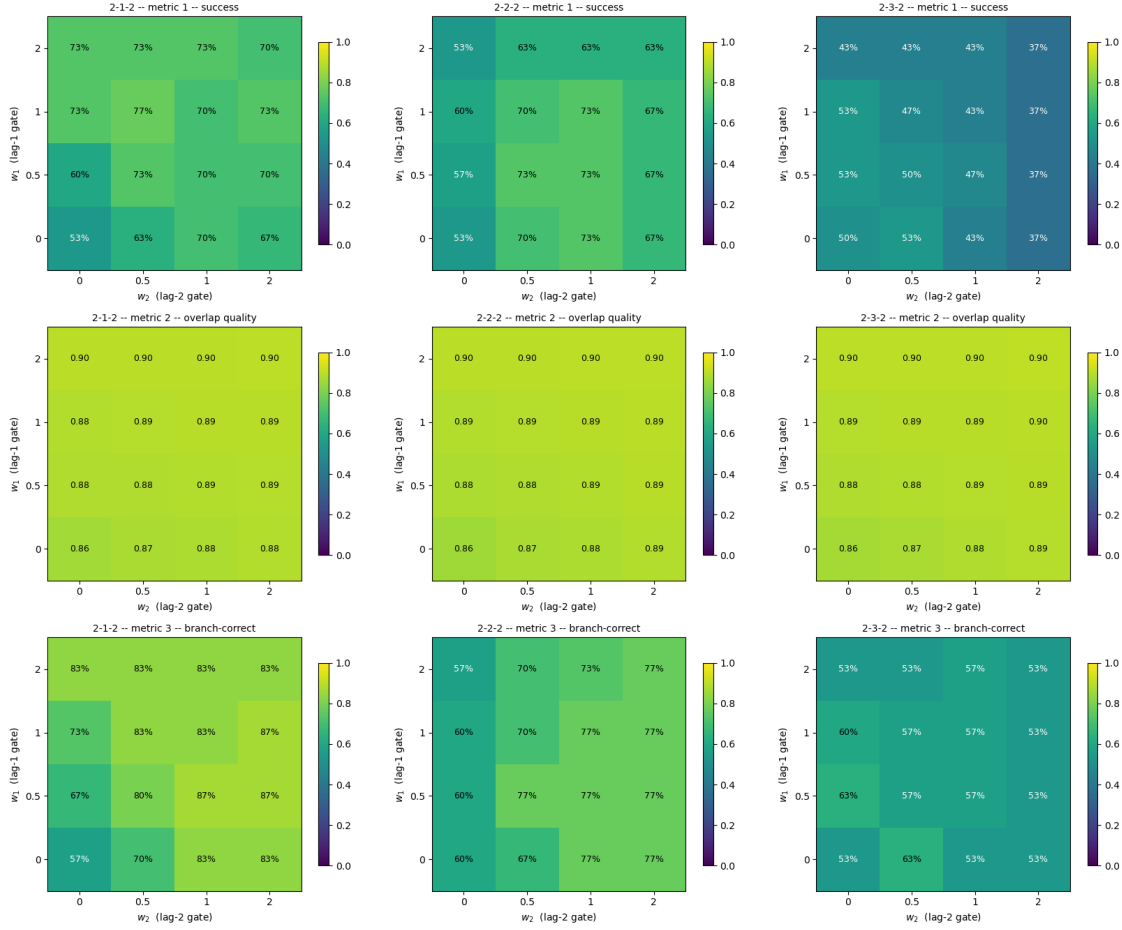
    print(f"{tag}    (correct / branch-correct)          " + "".join(f"w2={w2:<10g}"
↪for w2 in GRID_W))
    for w1 in GRID_W:
        row = ""
        for w2 in GRID_W:
            s = grid_stats[tag][(w1, w2)]
            row += f"{s['counter']['correct'] / s['attempts']:>7.0%} /
↪{s['branch_correct']:>5.0%}"
            print(f"    w1={w1:<5g}                    {row}")
        print()

fig, axes = plt.subplots(3, len(FOCUS), figsize=(5.6 * len(FOCUS), 13.5))
METRIC_ROWS = [("correct", "metric 1 -- success"),
                ("overlap", "metric 2 -- overlap quality"),
                ("branch", "metric 3 -- branch-correct")]
for col, tag in enumerate(FOCUS):
    stats = grid_stats[tag]
    for row, (key, label) in enumerate(METRIC_ROWS):
        def value(s):
            if key == "correct":
                return s["counter"]["correct"] / s["attempts"]
            if key == "overlap":
                return 0.0 if s["mean_overlap"] != s["mean_overlap"] else
↪s["mean_overlap"]
            return s["branch_correct"]
        grid = np.array([[value(stats[(w1, w2)]) for w2 in GRID_W] for w1 in
↪GRID_W])
        ax = axes[row, col]
        im = ax.imshow(grid, cmap="viridis", vmin=0.0, vmax=1.0, origin="lower")
        for i in range(len(GRID_W)):
            for j in range(len(GRID_W)):
                ax.text(j, i, f"{grid[i, j]:.0%}" if key != "overlap" else
↪f"{grid[i, j]:.2f}",
                        ha="center", va="center",
                        color="w" if grid[i, j] < 0.6 else "k", fontsize=9)
        ax.set_xticks(range(len(GRID_W)), [f"{w:g}" for w in GRID_W])
        ax.set_yticks(range(len(GRID_W)), [f"{w:g}" for w in GRID_W])
        ax.set_xlabel(r"$w_2$ (lag-2 gate)"); ax.set_ylabel(r"$w_1$ (lag-1
↪gate)")
        ax.set_title(f"{tag} -- {label}", fontsize=10)
        fig.colorbar(im, ax=ax, shrink=0.8)
plt.tight_layout(); plt.show()

```

2-1-2	(correct / branch-correct)	w2=0	w2=0.5	w2=1
w2=2				
w1=0		53% / 57%	63% / 70%	70% / 83%
67% / 83%				

w1=0.5	60% / 67%	73% / 80%	70% / 87%
70% / 87%			
w1=1	73% / 73%	77% / 83%	70% / 83%
73% / 87%			
w1=2	73% / 83%	73% / 83%	73% / 83%
70% / 83%			
2-2-2 (correct / branch-correct)	w2=0	w2=0.5	w2=1
w2=2			
w1=0	53% / 60%	70% / 67%	73% / 77%
67% / 77%			
w1=0.5	57% / 60%	73% / 77%	73% / 77%
67% / 77%			
w1=1	60% / 60%	70% / 70%	73% / 77%
67% / 77%			
w1=2	53% / 57%	63% / 70%	63% / 73%
63% / 77%			
2-3-2 (correct / branch-correct)	w2=0	w2=0.5	w2=1
w2=2			
w1=0	50% / 53%	53% / 63%	43% / 53%
37% / 53%			
w1=0.5	53% / 63%	50% / 57%	47% / 57%
37% / 53%			
w1=1	53% / 60%	47% / 57%	43% / 57%
37% / 53%			
w1=2	43% / 53%	43% / 53%	43% / 57%
37% / 53%			



9 7. Sampled Trial: Time series plots for varying topologies ($\sigma_\omega = 0.3$)

One trial, kept from each topology's sweep so there is something concrete behind the averages. A **trial** is one freshly built pair of branching sequences plus the two recall attempts on it, cued from $A[0]$ and from $B[0]$; §4 runs 20 of them per rule and the bar charts average over all 40 attempts. This is the first of those 20, saved verbatim.

Three figures, one per topology in **SPEC**, each four rules deep and two columns wide (cued A, cued B). Every panel plots the state's overlap with all stored patterns over time. Every rule sees the *same* sampled trial – each restarts its generator from the same seed, and ω is fixed besides – so the rows differ by rule and nothing else.

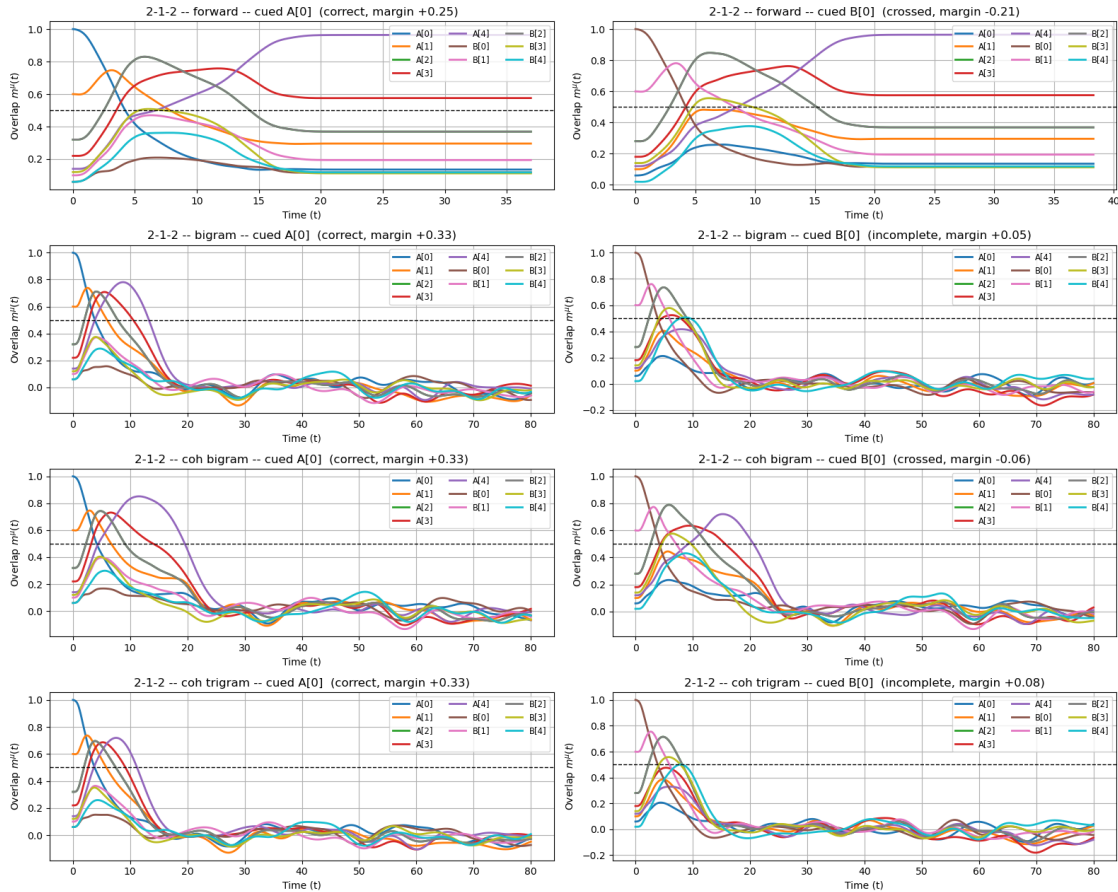
Read the tail of each trace. Under **forward** the two continuations rise together out of the shared run, which is the ambiguity itself made visible. Under **bigram** and **coh trigram** they separate – and then neither clears **tolerance**, which is the **incomplete** column of the bar chart. On 2-3-2 they do not separate at all: the shared run is longer than any gate's reach.

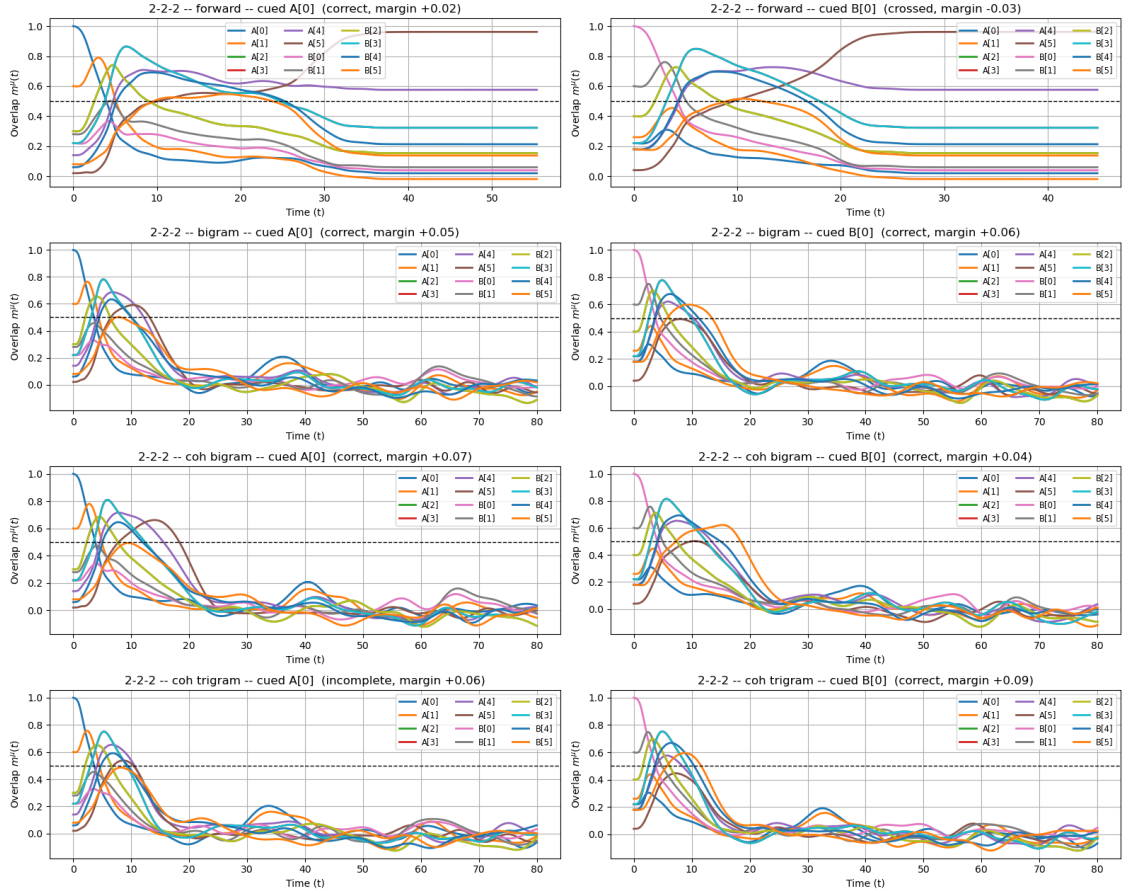
```

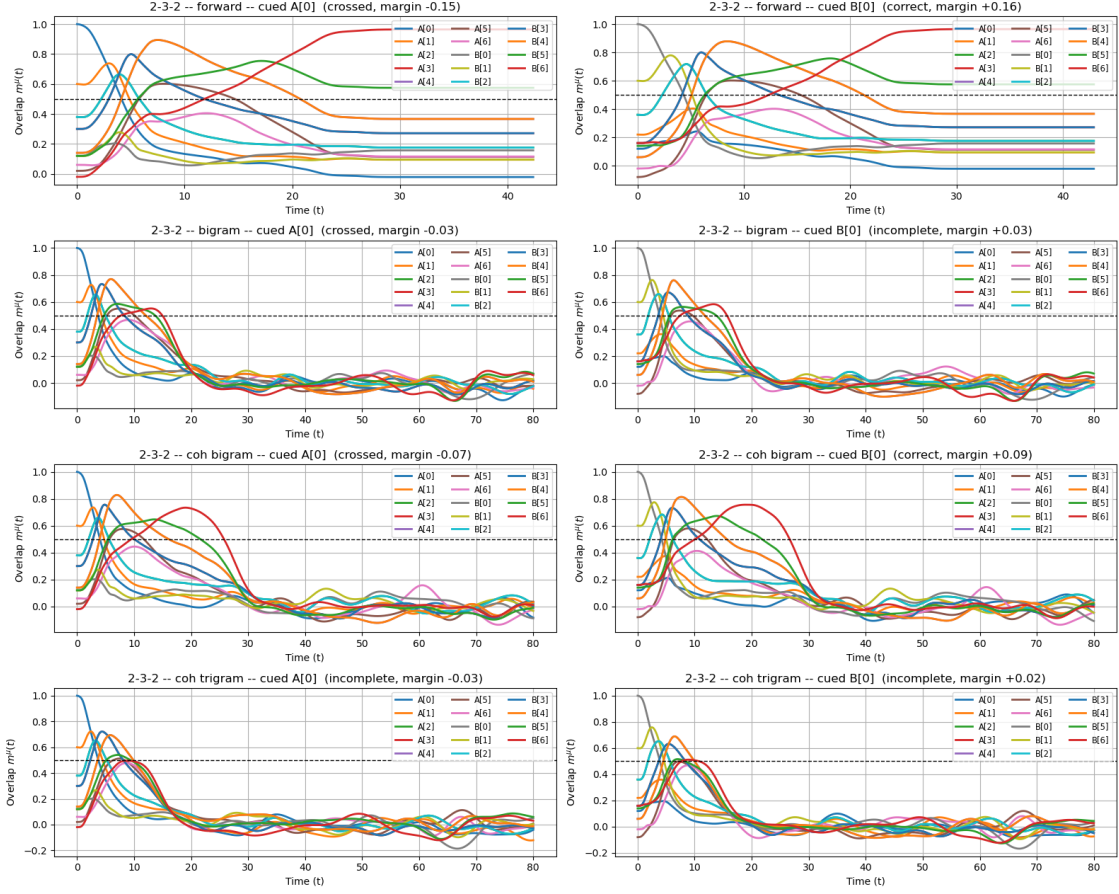
[52]: PICKED = ["forward", "bigram", "coh bigram", "coh trigram"]
TRACE_TAGS = SPEC

for tag in TRACE_TAGS:
    first = all_first[tag]
    fig, axes = plt.subplots(len(PICKED), 2, figsize=(16, 3.2 * len(PICKED)))
    for row, rule in enumerate(PICKED):
        net, multi, out = first[rule]
        for col, name in enumerate(("A", "B")):
            net.plot(out[name]["result"], multi, ax=axes[row, col],
                    title=f"{tag} -- {rule} -- cued {name}[0] ",
                    ↪({out[name]['outcome']}, "
                        f"margin {out[name]['margin']:+.2f}"))
            axes[row, col].axhline(CONFIG["tolerance"], color="k", ls="--", ↪
            ↪lw=1)
    plt.tight_layout(); plt.show()

```







10 8. Constant intrinsic frequency ($\sigma_\omega = 0$, constant $\omega \in \{0, 0.1, 0.3, 1, 3\}$)

Everything above runs at `frequency_std = 0.3 – disorder`, a different ω_i per oscillator. Here the disorder is removed and every oscillator gets the **same** constant ω , swept over several non-zero values: a uniform drift rather than heterogeneity.

Why it is a separate axis: at an exact cue every phase is in $\{0, \pi\}$, so $\sin \theta_i = 0$ and the coupling vanishes; with $\omega \equiv 0$ too, the state is an exact fixed point and nothing moves (hence the `frequency_std > 0` assertion at the top). A constant $\omega \neq 0$ is the minimal thing that breaks that degeneracy without reintroducing disorder. $\omega \equiv 0$ is kept as the frozen control – it should score 0% correct for every rule; `robustness_sweep.ipynb §6` runs it properly, from a displaced start.

Run on 2-1-2, where the context rules do have enough reach, so a change is attributable to ω rather than to an unsolvable geometry.

10.0.1 What ω is actually doing: symmetry breaking, not pacing

In a standard Kuramoto model a *uniform* ω is removable – transform to a rotating frame, and it drops out, because the coupling depends only on phase differences. **That does not hold here.** The drive is

$$\dot{\theta}_i = \omega_i - \sin \theta_i V_i(\cos \theta),$$

anchored to the stored patterns on the $\{0, \pi\}$ lattice rather than difference-only, so it is not rotation-covariant and a uniform ω genuinely rotates the state *relative to the memories*. ω is therefore a real physical parameter, not a gauge choice.

Its job is to **break the fixed-point degeneracy**. At an exact cue $\sin \theta_i = 0$, so the coupling cannot move the state at all; $\omega \neq 0$ is the only thing that takes it off the lattice, after which the coupling does the work of the transition. This predicts an asymmetric response, and that is what the sweep finds:

- **$\omega = 0$ is singular, not slow.** Exactly zero is an exact fixed point – 0% correct for every rule. Any $\varepsilon > 0$ is a different regime. The discontinuity at the origin is real.
- **There is a ceiling, not a peak.** Performance is flat across the low end and decays above it; nothing is gained by increasing ω . Beyond the plateau the drift simply competes with the coupling that is trying to hold each pattern.
- **The floor is a rate limit.** Measured on the single-sequence chain of `robustness_sweep.ipynb`: below $\omega \approx 0.01$ the escape becomes slow enough that a fixed duration runs out – at $T = 54$, **forward** scores 0% at $\omega = 0.001$ but 100% at $T = 300$. So ω breaks the symmetry *and* sets the escape rate; the rate only binds at the very bottom.

That leaves a usable band of roughly $\omega \in [0.02, 0.2]$ at the durations used here, and **OMEGA_OPT = 0.05** is taken as the operating point. For reference, on the single-sequence chain at $P = 10$ that setting beats the $\sigma_\omega = 0.3$ default on every rule – **bigram** 37% -> 85% strict, **coh bigram** 60% -> 100%, **coh trigram** 23% -> 55%.

One caveat carried forward: **multilag** is far more ω -tolerant than the rest (it still scores at $\omega = 1$ where **bigram** and **coh bigram** are at zero), and there is a narrow non-monotone dip near $\omega \approx 0.45$ where **forward** collapses and then recovers. Both are visible in the sweep below; neither is explained here.

```
[53]: # Resolved finely below 0.3: the interesting structure is the low plateau and
      ↪where it
      # ends, not the decade above it. 0.0 is the frozen control (exact fixed point,
      ↪see above).
      OMEGA_CONST = [0.0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.3, 0.45, 0.6, 1.0, 3.0]
      OMEGA_OPT = 0.05      # the operating point measured below; §9 and §10 run here
      OMEGA_GEOMETRY = "2-1-2"

      omega_stats = {}
      for omega in OMEGA_CONST:
          print(f"-- omega = {omega:g} (constant, frequency_std = 0), 
          ↪{OMEGA_GEOMETRY} --")
```

```

    omega_stats[omega] = rule_sweep(
        RAW_CATALOGUE, **GEOMETRIES[OMEGA_GEOMETRY],
        config_overrides={"frequency_mean": omega, "frequency_std": 0.0})[0]
    print()

fig, axes = plt.subplots(1, 3, figsize=(20, 5.0))
for rule in RAW_CATALOGUE:
    axes[0].plot(OMEGA_CONST, [omega_stats[w][rule]["counter"]["correct"] /
    ↪omega_stats[w][rule]["attempts"]
                        for w in OMEGA_CONST], "-o", label=rule)
    axes[1].plot(OMEGA_CONST, zeroed([omega_stats[w][rule]["mean_overlap"] for
    ↪w in OMEGA_CONST]),
                "-o", label=rule)
    axes[2].plot(OMEGA_CONST, [omega_stats[w][rule]["branch_correct"] for w in
    ↪OMEGA_CONST],
                "-o", label=rule)
axes[0].set_ylabel("strict full-retrieval success rate")
axes[0].set_title(f"{OMEGA_GEOMETRY} -- metric 1 vs constant  $\omega$ 
    ↪( $\sigma_\omega = 0$ )")
axes[1].axhline(CONFIG["tolerance"], color="k", ls="--", lw=1,
    ↪label="tolerance")
axes[1].set_ylabel("mean peak overlap (successful attempts only; 0 = none)")
axes[1].set_title(f"{OMEGA_GEOMETRY} -- metric 2 vs constant  $\omega$ ")
axes[2].axhline(0.5, color="k", ls="--", lw=1, label="chance")
axes[2].set_ylabel("fraction branch-correct")
axes[2].set_title(f"{OMEGA_GEOMETRY} -- metric 3 vs constant  $\omega$ ")
for ax in axes:
    ax.set_xlabel(r"constant  $\omega$  (same for every oscillator)")
    ax.set_xscale("symlog", linthresh=0.01)
    ax.axvline(OMEGA_OPT, color="k", ls=":", lw=1)
    ax.set_xticks(OMEGA_CONST, [f"{w:g}" for w in OMEGA_CONST])
    ax.set_ylim(-0.05, 1.05); ax.legend(fontsize=8, ncol=2); ax.grid(alpha=0.3)
plt.tight_layout(); plt.show()

print(f"strict full-retrieval success rate, {OMEGA_GEOMETRY}, frequency_std =
    ↪0")
print(f"{'rule':<15}" + " ".join(f"w={w:<8g}" for w in OMEGA_CONST) +
    ↪f"{'(disorder 0.3)':>16}")
for rule in RAW_CATALOGUE:
    base = all_stats[OMEGA_GEOMETRY][rule]["counter"]["correct"] /
    ↪all_stats[OMEGA_GEOMETRY][rule]["attempts"]
    row = " ".join(f"{omega_stats[w][rule]['counter']['correct'] /
    ↪omega_stats[w][rule]['attempts']:<10.0%}"
                    for w in OMEGA_CONST)
    print(f"{'rule':<15}-{row}-{base:>16.0%}")

```

-- omega = 0 (constant, frequency_std = 0), 2-1-2 --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
skip-two	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
forward+skip2	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
multilag	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
coh bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
coh trigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=+0.011	+/-0.053	
-- omega = 0.01 (constant, frequency_std = 0), 2-1-2 --				
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	55% +/-0.079	margin=+0.011	+/-0.053	
forward	correct= 2%	crossed= 5%	incomplete= 92%	overlap=+0.934
branch-correct=	60% +/-0.077	margin=+0.003	+/-0.043	
skip-two	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.187	+/-0.046	
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=-0.002	+/-0.025	
forward+skip2	correct= 32%	crossed= 12%	incomplete= 55%	overlap=+0.925
branch-correct=	75% +/-0.068	margin=+0.068	+/-0.108	
bigram	correct= 62%	crossed= 10%	incomplete= 28%	overlap=+0.922
branch-correct=	90% +/-0.047	margin=+0.144	+/-0.105	
multilag	correct= 45%	crossed= 10%	incomplete= 45%	overlap=+0.927
branch-correct=	90% +/-0.047	margin=+0.106	+/-0.103	
coh bigram	correct= 42%	crossed= 5%	incomplete= 52%	overlap=+0.928
branch-correct=	92% +/-0.042	margin=+0.111	+/-0.086	
coh trigram	correct= 52%	crossed= 2%	incomplete= 45%	overlap=+0.914
branch-correct=	98% +/-0.025	margin=+0.184	+/-0.087	
-- omega = 0.02 (constant, frequency_std = 0), 2-1-2 --				
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	55% +/-0.079	margin=+0.011	+/-0.053	
forward	correct= 2%	crossed= 5%	incomplete= 92%	overlap=+0.930
branch-correct=	57% +/-0.078	margin=+0.003	+/-0.043	
skip-two	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.178	+/-0.046	
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	48% +/-0.079	margin=-0.001	+/-0.026	

forward+skip2	correct= 32%	crossed= 15%	incomplete= 52%	overlap=+0.923
branch-correct=	72% +/-0.071	margin=+0.064	+/-0.108	
bigram	correct= 65%	crossed= 18%	incomplete= 18%	overlap=+0.919
branch-correct=	85% +/-0.056	margin=+0.130	+/-0.123	
multilag	correct= 42%	crossed= 10%	incomplete= 48%	overlap=+0.925
branch-correct=	90% +/-0.047	margin=+0.104	+/-0.101	
coh bigram	correct= 55%	crossed= 15%	incomplete= 30%	overlap=+0.926
branch-correct=	90% +/-0.047	margin=+0.106	+/-0.098	
coh trigram	correct= 60%	crossed= 5%	incomplete= 35%	overlap=+0.910
branch-correct=	95% +/-0.034	margin=+0.183	+/-0.091	

-- omega = 0.05 (constant, frequency_std = 0), 2-1-2 --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	55% +/-0.079	margin=+0.011	+/-0.053	
forward	correct= 10%	crossed= 12%	incomplete= 78%	overlap=+0.937
branch-correct=	52% +/-0.079	margin=+0.003	+/-0.087	
skip-two	correct= 0%	crossed= 2%	incomplete= 98%	overlap= n/a
branch-correct=	98% +/-0.025	margin=+0.161	+/-0.051	
self+forward	correct= 2%	crossed= 2%	incomplete= 95%	overlap=+0.896
branch-correct=	48% +/-0.079	margin=+0.000	+/-0.039	
forward+skip2	correct= 38%	crossed= 28%	incomplete= 35%	overlap=+0.917
branch-correct=	65% +/-0.075	margin=+0.048	+/-0.121	
bigram	correct= 68%	crossed= 12%	incomplete= 20%	overlap=+0.910
branch-correct=	88% +/-0.052	margin=+0.129	+/-0.106	
multilag	correct= 45%	crossed= 18%	incomplete= 38%	overlap=+0.921
branch-correct=	82% +/-0.060	margin=+0.090	+/-0.111	
coh bigram	correct= 57%	crossed= 28%	incomplete= 15%	overlap=+0.919
branch-correct=	75% +/-0.068	margin=+0.085	+/-0.113	
coh trigram	correct= 70%	crossed= 8%	incomplete= 22%	overlap=+0.899
branch-correct=	92% +/-0.042	margin=+0.167	+/-0.089	

-- omega = 0.1 (constant, frequency_std = 0), 2-1-2 --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	55% +/-0.079	margin=+0.011	+/-0.053	
forward	correct= 38%	crossed= 40%	incomplete= 22%	overlap=+0.927
branch-correct=	50% +/-0.079	margin=+0.002	+/-0.135	
skip-two	correct= 0%	crossed= 2%	incomplete= 98%	overlap= n/a
branch-correct=	98% +/-0.025	margin=+0.148	+/-0.049	
self+forward	correct= 2%	crossed= 2%	incomplete= 95%	overlap=+0.930
branch-correct=	48% +/-0.079	margin=+0.001	+/-0.039	
forward+skip2	correct= 45%	crossed= 38%	incomplete= 18%	overlap=+0.907
branch-correct=	60% +/-0.077	margin=+0.041	+/-0.124	
bigram	correct= 72%	crossed= 10%	incomplete= 18%	overlap=+0.895
branch-correct=	90% +/-0.047	margin=+0.129	+/-0.087	
multilag	correct= 55%	crossed= 28%	incomplete= 18%	overlap=+0.917
branch-correct=	75% +/-0.068	margin=+0.089	+/-0.121	
coh bigram	correct= 65%	crossed= 28%	incomplete= 8%	overlap=+0.909
branch-correct=	75% +/-0.068	margin=+0.086	+/-0.105	

coh trigram	correct= 72%	crossed= 10%	incomplete= 18%	overlap=+0.878
branch-correct=	92% +/-0.042	margin=+0.151	+/-0.078	

```
-- omega = 0.2 (constant, frequency_std = 0), 2-1-2 --
```

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	52% +/-0.079	margin=+0.010	+/-0.052	
forward	correct= 50%	crossed= 48%	incomplete= 2%	overlap=+0.905
branch-correct=	52% +/-0.079	margin=+0.005	+/-0.113	
skip-two	correct= 0%	crossed= 2%	incomplete= 98%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.127	+/-0.046	
self+forward	correct= 5%	crossed= 8%	incomplete= 88%	overlap=+0.924
branch-correct=	45% +/-0.079	margin=-0.000	+/-0.055	
forward+skip2	correct= 60%	crossed= 32%	incomplete= 8%	overlap=+0.894
branch-correct=	65% +/-0.075	margin=+0.054	+/-0.104	
bigram	correct= 80%	crossed= 8%	incomplete= 12%	overlap=+0.878
branch-correct=	100% +/-0.000	margin=+0.124	+/-0.056	
multilag	correct= 57%	crossed= 25%	incomplete= 18%	overlap=+0.910
branch-correct=	78% +/-0.066	margin=+0.091	+/-0.107	
coh bigram	correct= 85%	crossed= 10%	incomplete= 5%	overlap=+0.885
branch-correct=	92% +/-0.042	margin=+0.098	+/-0.061	
coh trigram	correct= 78%	crossed= 8%	incomplete= 15%	overlap=+0.877
branch-correct=	98% +/-0.025	margin=+0.151	+/-0.059	

```
-- omega = 0.3 (constant, frequency_std = 0), 2-1-2 --
```

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	60% +/-0.077	margin=+0.001	+/-0.014	
forward	correct= 48%	crossed= 52%	incomplete= 0%	overlap=+0.875
branch-correct=	57% +/-0.078	margin=+0.003	+/-0.062	
skip-two	correct= 0%	crossed= 2%	incomplete= 98%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.103	+/-0.043	
self+forward	correct= 25%	crossed= 25%	incomplete= 50%	overlap=+0.908
branch-correct=	52% +/-0.079	margin=+0.000	+/-0.094	
forward+skip2	correct= 65%	crossed= 28%	incomplete= 8%	overlap=+0.878
branch-correct=	75% +/-0.068	margin=+0.054	+/-0.080	
bigram	correct= 85%	crossed= 0%	incomplete= 15%	overlap=+0.874
branch-correct=	100% +/-0.000	margin=+0.130	+/-0.048	
multilag	correct= 60%	crossed= 25%	incomplete= 15%	overlap=+0.903
branch-correct=	78% +/-0.066	margin=+0.089	+/-0.101	
coh bigram	correct= 90%	crossed= 5%	incomplete= 5%	overlap=+0.877
branch-correct=	98% +/-0.025	margin=+0.091	+/-0.048	
coh trigram	correct= 80%	crossed= 2%	incomplete= 18%	overlap=+0.866
branch-correct=	95% +/-0.034	margin=+0.152	+/-0.066	

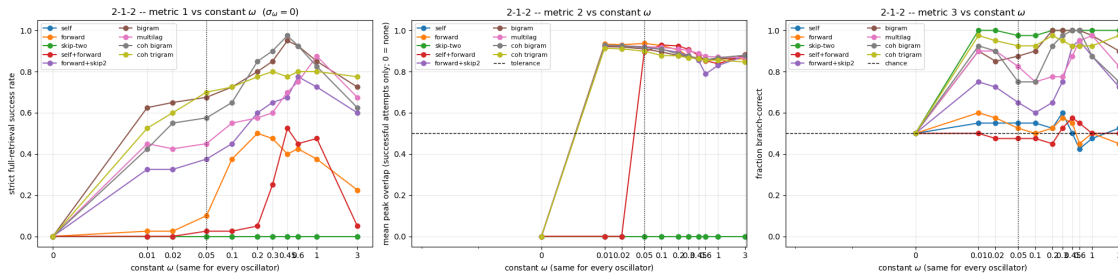
```
-- omega = 0.45 (constant, frequency_std = 0), 2-1-2 --
```

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.079	margin=-0.000	+/-0.003	
forward	correct= 40%	crossed= 60%	incomplete= 0%	overlap=+0.859
branch-correct=	55% +/-0.079	margin=+0.003	+/-0.039	

skip-two	correct= 0%	crossed= 2%	incomplete= 98%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.097 +/-0.030		
self+forward	correct= 52%	crossed= 48%	incomplete= 0%	overlap=+0.879
branch-correct=	57% +/-0.078	margin=+0.003 +/-0.101		
forward+skip2	correct= 68%	crossed= 22%	incomplete= 10%	overlap=+0.855
branch-correct=	92% +/-0.042	margin=+0.057 +/-0.047		
bigram	correct= 95%	crossed= 0%	incomplete= 5%	overlap=+0.866
branch-correct=	100% +/-0.000	margin=+0.171 +/-0.033		
multilag	correct= 70%	crossed= 15%	incomplete= 15%	overlap=+0.888
branch-correct=	88% +/-0.052	margin=+0.097 +/-0.073		
coh bigram	correct= 98%	crossed= 0%	incomplete= 2%	overlap=+0.861
branch-correct=	100% +/-0.000	margin=+0.101 +/-0.037		
coh trigram	correct= 78%	crossed= 5%	incomplete= 18%	overlap=+0.864
branch-correct=	92% +/-0.042	margin=+0.189 +/-0.081		
-- omega = 0.6 (constant, frequency_std = 0), 2-1-2 --				
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	42% +/-0.078	margin=-0.001 +/-0.011		
forward	correct= 42%	crossed= 57%	incomplete= 0%	overlap=+0.857
branch-correct=	45% +/-0.079	margin=-0.006 +/-0.119		
skip-two	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.179 +/-0.038		
self+forward	correct= 45%	crossed= 55%	incomplete= 0%	overlap=+0.853
branch-correct=	55% +/-0.079	margin=-0.001 +/-0.037		
forward+skip2	correct= 78%	crossed= 8%	incomplete= 15%	overlap=+0.788
branch-correct=	95% +/-0.034	margin=+0.056 +/-0.038		
bigram	correct= 92%	crossed= 0%	incomplete= 8%	overlap=+0.857
branch-correct=	100% +/-0.000	margin=+0.187 +/-0.055		
multilag	correct= 75%	crossed= 10%	incomplete= 15%	overlap=+0.873
branch-correct=	95% +/-0.034	margin=+0.097 +/-0.059		
coh bigram	correct= 92%	crossed= 0%	incomplete= 8%	overlap=+0.857
branch-correct=	100% +/-0.000	margin=+0.149 +/-0.064		
coh trigram	correct= 80%	crossed= 8%	incomplete= 12%	overlap=+0.855
branch-correct=	92% +/-0.042	margin=+0.176 +/-0.087		
-- omega = 1 (constant, frequency_std = 0), 2-1-2 --				
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	48% +/-0.079	margin=+0.003 +/-0.041		
forward	correct= 38%	crossed= 62%	incomplete= 0%	overlap=+0.862
branch-correct=	50% +/-0.079	margin=-0.001 +/-0.090		
skip-two	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.135 +/-0.038		
self+forward	correct= 48%	crossed= 52%	incomplete= 0%	overlap=+0.838
branch-correct=	50% +/-0.079	margin=-0.002 +/-0.091		
forward+skip2	correct= 72%	crossed= 18%	incomplete= 10%	overlap=+0.830
branch-correct=	88% +/-0.052	margin=+0.091 +/-0.089		
bigram	correct= 85%	crossed= 2%	incomplete= 12%	overlap=+0.860
branch-correct=	98% +/-0.025	margin=+0.158 +/-0.066		

multilag correct= 88% crossed= 2% incomplete= 10% overlap=+0.871
 branch-correct= 98% +/-0.025 margin=+0.099 +/-0.043
 coh bigram correct= 82% crossed= 12% incomplete= 5% overlap=+0.867
 branch-correct= 88% +/-0.052 margin=+0.098 +/-0.082
 coh trigram correct= 80% crossed= 8% incomplete= 12% overlap=+0.856
 branch-correct= 92% +/-0.042 margin=+0.173 +/-0.086

-- omega = 3 (constant, frequency_std = 0), 2-1-2 --
 self correct= 0% crossed= 0% incomplete= 100% overlap= n/a
 branch-correct= 52% +/-0.079 margin=+0.010 +/-0.053
 forward correct= 22% crossed= 48% incomplete= 30% overlap=+0.861
 branch-correct= 45% +/-0.079 margin=-0.002 +/-0.078
 skip-two correct= 0% crossed= 12% incomplete= 88% overlap= n/a
 branch-correct= 100% +/-0.000 margin=+0.125 +/-0.053
 self+forward correct= 5% crossed= 10% incomplete= 85% overlap=+0.882
 branch-correct= 50% +/-0.079 margin=-0.000 +/-0.035
 forward+skip2 correct= 60% crossed= 30% incomplete= 10% overlap=+0.876
 branch-correct= 72% +/-0.071 margin=+0.066 +/-0.094
 bigram correct= 72% crossed= 12% incomplete= 15% overlap=+0.873
 branch-correct= 90% +/-0.047 margin=+0.124 +/-0.085
 multilag correct= 68% crossed= 20% incomplete= 12% overlap=+0.868
 branch-correct= 82% +/-0.060 margin=+0.103 +/-0.088
 coh bigram correct= 62% crossed= 30% incomplete= 8% overlap=+0.879
 branch-correct= 75% +/-0.068 margin=+0.067 +/-0.088
 coh trigram correct= 78% crossed= 5% incomplete= 18% overlap=+0.847
 branch-correct= 98% +/-0.025 margin=+0.167 +/-0.073



strict full-retrieval success rate, 2-1-2, frequency_std = 0

rule	w=0	w=0.01	w=0.02	w=0.05	w=0.1	w=0.2	w=0.3
w=0.45	w=0.6	w=1	w=3	(disorder 0.3)			
self	0%	0%	0%	0%	0%	0%	0%
0%	0%	0%	0%		0%		
forward	0%	2%	2%	10%	38%	50%	48%
40%	42%	38%	22%		52%		
skip-two	0%	0%	0%	0%	0%	0%	0%
0%	0%	0%	0%		0%		

self+forward	0%	0%	0%	2%	2%	5%	25%
52%	45%	48%	5%		15%		
forward+skip2	0%	32%	32%	38%	45%	60%	65%
68%	78%	72%	60%		65%		
bigram	0%	62%	65%	68%	72%	80%	85%
95%	92%	85%	72%		62%		
multilag	0%	45%	42%	45%	55%	57%	60%
70%	75%	88%	68%		68%		
coh bigram	0%	42%	55%	57%	65%	85%	90%
98%	92%	82%	62%		70%		
coh trigram	0%	52%	60%	70%	72%	78%	80%
78%	80%	80%	78%		50%		

10.0.2 What the trajectories actually look like

Overlap with every stored pattern against time, on one crossover instance – the same instance in every panel – cued from $A[0]$, at three constant ω with $\sigma_\omega = 0$.

Read the tail of each row: whether the two continuations separate coming out of the shared centre, or rise together, and whether either clears `tolerance`. That is the `correct` / `crossed` / `incomplete` split of the bar charts above, one trajectory at a time.

$\omega \equiv 0$ is not plotted here because there is nothing to plot: under an exact cue every phase is in $\{0, \pi\}$, so $\sin \theta = 0$, and with no ω either the state is an exact fixed point – the integration hits the stationary early-stop within a few timesteps and the overlaps never move. That is the 0% column of the sweep above. To *see* $\omega \equiv 0$ move, it has to be started off the lattice: `robustness_sweep.ipynb` §6(a) plots exactly that, from a displaced cue.

```
[54]: OMEGA_TRACE_RULES = ["forward", "bigram", "coh trigram"]
      OMEGA_TRACE      = [0.1, 0.3, 1.0]      # constant; 0.0 is a fixed point, see_
      ↪above
      OMEGA_TRACE_SEED = 0

      geom = GEOMETRIES[OMEGA_GEOMETRY]
      fig, axes = plt.subplots(len(OMEGA_TRACE_RULES), len(OMEGA_TRACE),
                              figsize=(6.0 * len(OMEGA_TRACE), 3.2 *_
      ↪len(OMEGA_TRACE_RULES)),
                              sharex=True, sharey=True)

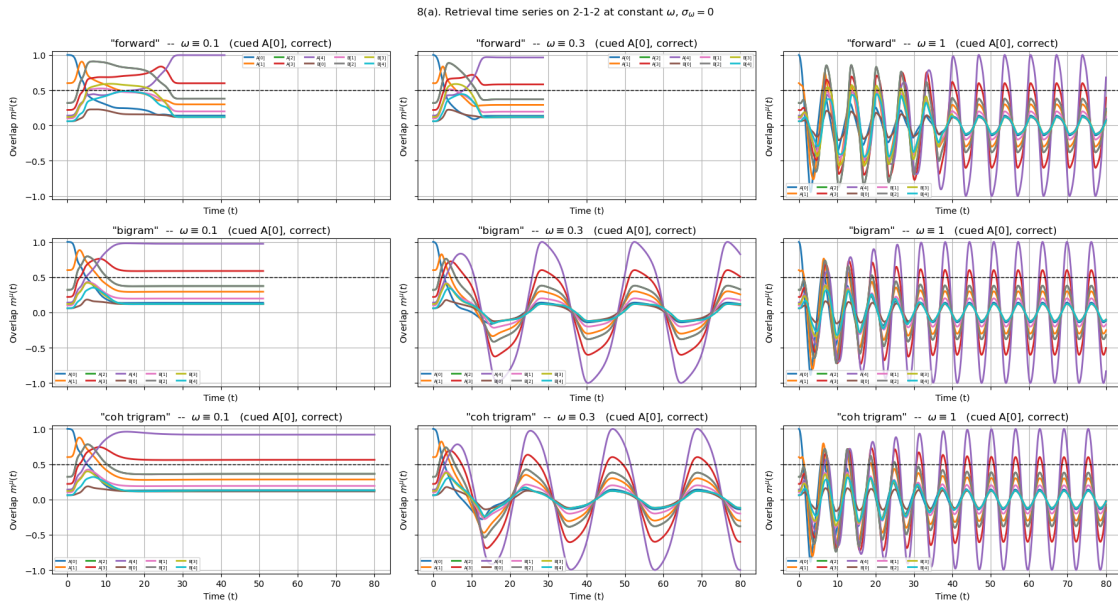
      for row, rule in enumerate(OMEGA_TRACE_RULES):
          for col, omega in enumerate(OMEGA_TRACE):
              net = kc.ContextNetwork(rule=RAW_CATALOGUE[rule], N=N, seed=10,
              **{**CONFIG, "frequency_mean": omega,_
      ↪"frequency_std": 0.0})
              # same seed in every panel, so it is one instance seen under nine_
      ↪conditions
              multi, out = run_crossover_trial(net, np.random._
      ↪default_rng(OMEGA_TRACE_SEED), **geom)
```

```

ax = axes[row, col]
net.plot(out["A"]["result"], multi, ax=ax,
         title=f'"{rule}" --  $\omega \equiv \omega$  (cued A[0], correct)'
         f'(\omega, {out["A"]["outcome"]})')
ax.axhline(CONFIG["tolerance"], color="k", ls="--", lw=1)
ax.set_ylim(-1.05, 1.05)
ax.legend(fontsize=5, ncol=5)

fig.suptitle(f'8(a). Retrieval time series on {OMEGA_GEOMETRY} at constant  $\omega$ 
 $\rightarrow \omega$ , "
            r" $\sigma_\omega = 0$ ", y=1.001)
plt.tight_layout(); plt.show()

```



11 9. The full benchmark at fixed ω ($\sigma_\omega = 0$, $\omega \in \{0.1, 0.3, 1\}$)

§4 runs at `frequency_std = 0.3`, a *different* intrinsic frequency per oscillator; §8 swept a constant ω on 2-1-2 only. This runs the **whole §4 suite** – the three SPEC topologies, raw and gain – with the disorder removed and every oscillator carrying the same non-zero ω .

The question it settles: are §4's conclusions – context rules above chance on branch choice where reach $\geq L$, at chance on 2-3-2 – a property of the *rules*, or an artefact of the frequency disorder driving the trajectory off each pattern? At $\sigma_\omega = 0$ the only thing moving the state off a stored pattern is a drift shared by every oscillator, so nothing about the escape is rule- or neuron-specific.

$\omega \equiv 0$ is not usable here (exact cue, $\sin \theta_i = 0$, no ω – an exact fixed point); §8 keeps it as the frozen control, `robustness_sweep.ipynb` §6 runs it displaced.

This now runs at OMEGA_OPT first, the operating point §8 measures, and over **every** topology in GEOMETRIES rather than the SPEC subset – 0.1, 0.3 and 1.0 are kept either side of it so the ω -dependence of the ordering stays visible. The final block prints all four metrics – outcome split, overlap quality with its spread, branch-correct and margin – for every rule, every topology, raw and gain, at $\omega = \omega_{\text{OMEGA_OPT}}$.

```
[55]: # OMEGA_OPT first: the operating point §8 measures. The rest are kept so the
# omega-dependence of the ordering is still visible either side of it.
FIXED_OMEGA = [OMEGA_OPT, 0.1, 0.3, 1.0]          # constant, frequency_std = 0
FIXED_OMEGA_GEOMETRIES = list(GEOMETRIES)        # every topology, not just SPEC

fixed_omega_stats = {}
for omega in FIXED_OMEGA:
    for tag in FIXED_OMEGA_GEOMETRIES:
        overrides = {"frequency_mean": omega, "frequency_std": 0.0}
        print("=" * 100)
        print(f"omega = {omega:g} (constant, frequency_std = 0)   {tag}")
        print("-- raw --")
        stats_raw, _ = rule_sweep(RAW_CATALOGUE, **GEOMETRIES[tag],
        ↪config_overrides=overrides)
        print("-- gain --")
        stats_gain, _ = rule_sweep(GAIN_CATALOGUE, **GEOMETRIES[tag],
        ↪config_overrides=overrides)
        fixed_omega_stats[(omega, tag, "raw")] = stats_raw
        fixed_omega_stats[(omega, tag, "gain")] = stats_gain
        print()

# --- every metric, every topology, at OMEGA_OPT
↪-----
print("=" * 108)
print(f"ALL METRICS AT THE OPERATING POINT  omega = {OMEGA_OPT:g} constant
↪(frequency_std = 0)")
print("=" * 108)
for tag in FIXED_OMEGA_GEOMETRIES:
    print(f"-- {tag} --")
    print(f"  {'rule':<15}{'cat':<6}{'correct':>9}{'crossed':>9}{'incompl':>9}"
          f"{'overlap':>18}{'branch':>9}{'margin':>9}")
    for cat in ("raw", "gain"):
        stats = fixed_omega_stats[(OMEGA_OPT, tag, cat)]
        for rule, s in stats.items():
            c, n = s["counter"], s["attempts"]
            ov = (f"{s['mean_overlap']:+.3f}+/-{s['overlap_var'] ** 0.5:.3f}"
                  if s["mean_overlap"] == s["mean_overlap"] else "n/a")
            print(f"  {rule:<15}{cat:<6}{c['correct'] / n:>9.0%}{c['crossed'] /
            ↪n:>9.0%}")
```

```

f"{c['incomplete'] / n:>9.0%}{ov:>18}{s['branch_correct']:>9.
↪0%}"

f"{s['margin']:>+9.3f}")

print()

```

```

=====
=====
omega = 0.05 (constant, frequency_std = 0)    2-1-2
-- raw --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 55% +/-0.079  margin=+0.011 +/-0.053
forward       correct= 10%  crossed= 12%  incomplete= 78%  overlap=+0.937
branch-correct= 52% +/-0.079  margin=+0.003 +/-0.087
skip-two      correct=  0%  crossed=  2%  incomplete= 98%  overlap=  n/a
branch-correct= 98% +/-0.025  margin=+0.161 +/-0.051
self+forward  correct=  2%  crossed=  2%  incomplete= 95%  overlap=+0.896
branch-correct= 48% +/-0.079  margin=+0.000 +/-0.039
forward+skip2 correct= 38%  crossed= 28%  incomplete= 35%  overlap=+0.917
branch-correct= 65% +/-0.075  margin=+0.048 +/-0.121
bigram        correct= 68%  crossed= 12%  incomplete= 20%  overlap=+0.910
branch-correct= 88% +/-0.052  margin=+0.129 +/-0.106
multilag      correct= 45%  crossed= 18%  incomplete= 38%  overlap=+0.921
branch-correct= 82% +/-0.060  margin=+0.090 +/-0.111
coh bigram    correct= 57%  crossed= 28%  incomplete= 15%  overlap=+0.919
branch-correct= 75% +/-0.068  margin=+0.085 +/-0.113
coh trigram   correct= 70%  crossed=  8%  incomplete= 22%  overlap=+0.899
branch-correct= 92% +/-0.042  margin=+0.167 +/-0.089
-- gain --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 55% +/-0.079  margin=+0.011 +/-0.053
forward       correct= 10%  crossed= 12%  incomplete= 78%  overlap=+0.937
branch-correct= 52% +/-0.079  margin=+0.003 +/-0.087
skip-two      correct=  0%  crossed=  2%  incomplete= 98%  overlap=  n/a
branch-correct= 98% +/-0.025  margin=+0.161 +/-0.051
self+forward  correct=  2%  crossed=  2%  incomplete= 95%  overlap=+0.895
branch-correct= 48% +/-0.079  margin=+0.000 +/-0.039
forward+skip2 correct= 38%  crossed= 28%  incomplete= 35%  overlap=+0.913
branch-correct= 68% +/-0.074  margin=+0.047 +/-0.117
bigram        correct= 68%  crossed= 20%  incomplete= 12%  overlap=+0.920
branch-correct= 82% +/-0.060  margin=+0.131 +/-0.127
multilag      correct= 48%  crossed= 12%  incomplete= 40%  overlap=+0.915
branch-correct= 88% +/-0.052  margin=+0.143 +/-0.112
coh bigram    correct= 57%  crossed= 22%  incomplete= 20%  overlap=+0.923
branch-correct= 80% +/-0.063  margin=+0.093 +/-0.114
coh trigram   correct= 57%  crossed= 10%  incomplete= 32%  overlap=+0.915
branch-correct= 90% +/-0.047  margin=+0.174 +/-0.105
=====
=====

```

```

=====
omega = 0.05 (constant, frequency_std = 0)    2-2-2
-- raw --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 62% +/-0.077  margin=+0.022 +/-0.083
forward        correct= 18%  crossed= 20%  incomplete= 62%  overlap=+0.935
branch-correct= 50% +/-0.079  margin=-0.000 +/-0.117
skip-two       correct=  0%  crossed= 50%  incomplete= 50%  overlap=  n/a
branch-correct= 50% +/-0.079  margin=+0.005 +/-0.131
self+forward    correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 50% +/-0.079  margin=-0.000 +/-0.035
forward+skip2   correct= 38%  crossed= 38%  incomplete= 25%  overlap=+0.915
branch-correct= 52% +/-0.079  margin=+0.013 +/-0.136
bigram         correct= 45%  crossed= 50%  incomplete=  5%  overlap=+0.911
branch-correct= 50% +/-0.079  margin=+0.005 +/-0.141
multilag       correct= 42%  crossed= 40%  incomplete= 18%  overlap=+0.923
branch-correct= 57% +/-0.078  margin=+0.020 +/-0.152
coh bigram     correct= 45%  crossed= 40%  incomplete= 15%  overlap=+0.921
branch-correct= 55% +/-0.079  margin=+0.015 +/-0.138
coh trigram    correct= 62%  crossed= 28%  incomplete= 10%  overlap=+0.904
branch-correct= 72% +/-0.071  margin=+0.081 +/-0.136
-- gain --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 62% +/-0.077  margin=+0.022 +/-0.083
forward        correct= 18%  crossed= 20%  incomplete= 62%  overlap=+0.935
branch-correct= 50% +/-0.079  margin=-0.000 +/-0.117
skip-two       correct=  0%  crossed= 50%  incomplete= 50%  overlap=  n/a
branch-correct= 50% +/-0.079  margin=+0.005 +/-0.131
self+forward    correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 50% +/-0.079  margin=-0.000 +/-0.034
forward+skip2   correct= 42%  crossed= 40%  incomplete= 18%  overlap=+0.911
branch-correct= 52% +/-0.079  margin=+0.012 +/-0.136
bigram         correct= 48%  crossed= 48%  incomplete=  5%  overlap=+0.921
branch-correct= 52% +/-0.079  margin=+0.013 +/-0.164
multilag       correct= 62%  crossed= 30%  incomplete=  8%  overlap=+0.920
branch-correct= 70% +/-0.072  margin=+0.094 +/-0.167
coh bigram     correct= 40%  crossed= 42%  incomplete= 18%  overlap=+0.925
branch-correct= 50% +/-0.079  margin=+0.008 +/-0.145
coh trigram    correct= 68%  crossed= 22%  incomplete= 10%  overlap=+0.921
branch-correct= 78% +/-0.066  margin=+0.117 +/-0.153

=====
=====
omega = 0.05 (constant, frequency_std = 0)    2-3-2
-- raw --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 42% +/-0.078  margin=-0.020 +/-0.092
forward        correct= 15%  crossed= 18%  incomplete= 68%  overlap=+0.921

```

branch-correct=	50% +/-0.079	margin=-0.001 +/-0.100			
skip-two	correct= 0%	crossed= 57%	incomplete= 42%	overlap=	n/a
branch-correct=	40% +/-0.077	margin=-0.019 +/-0.135			
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	50% +/-0.079	margin=+0.001 +/-0.027			
forward+skip2	correct= 20%	crossed= 40%	incomplete= 40%	overlap=	+0.913
branch-correct=	50% +/-0.079	margin=-0.008 +/-0.137			
bigram	correct= 40%	crossed= 48%	incomplete= 12%	overlap=	+0.911
branch-correct=	52% +/-0.079	margin=-0.001 +/-0.149			
multilag	correct= 20%	crossed= 32%	incomplete= 48%	overlap=	+0.921
branch-correct=	52% +/-0.079	margin=+0.005 +/-0.147			
coh bigram	correct= 32%	crossed= 48%	incomplete= 20%	overlap=	+0.921
branch-correct=	48% +/-0.079	margin=-0.005 +/-0.149			
coh trigram	correct= 32%	crossed= 50%	incomplete= 18%	overlap=	+0.903
branch-correct=	48% +/-0.079	margin=-0.006 +/-0.147			
-- gain --					
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	42% +/-0.078	margin=-0.020 +/-0.092			
forward	correct= 15%	crossed= 18%	incomplete= 68%	overlap=	+0.921
branch-correct=	50% +/-0.079	margin=-0.001 +/-0.100			
skip-two	correct= 0%	crossed= 57%	incomplete= 42%	overlap=	n/a
branch-correct=	40% +/-0.077	margin=-0.019 +/-0.135			
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	50% +/-0.079	margin=+0.000 +/-0.030			
forward+skip2	correct= 18%	crossed= 40%	incomplete= 42%	overlap=	+0.909
branch-correct=	48% +/-0.079	margin=-0.010 +/-0.130			
bigram	correct= 35%	crossed= 50%	incomplete= 15%	overlap=	+0.923
branch-correct=	50% +/-0.079	margin=-0.009 +/-0.171			
multilag	correct= 28%	crossed= 45%	incomplete= 28%	overlap=	+0.918
branch-correct=	50% +/-0.079	margin=-0.008 +/-0.170			
coh bigram	correct= 25%	crossed= 42%	incomplete= 32%	overlap=	+0.923
branch-correct=	48% +/-0.079	margin=-0.007 +/-0.149			
coh trigram	correct= 32%	crossed= 48%	incomplete= 20%	overlap=	+0.919
branch-correct=	48% +/-0.079	margin=-0.009 +/-0.177			

=====

=====

omega = 0.05 (constant, frequency_std = 0) 1-1-1

-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	57% +/-0.078	margin=+0.011 +/-0.057			
forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	60% +/-0.077	margin=+0.004 +/-0.016			
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=	+1.000
branch-correct=	100% +/-0.000	margin=+0.377 +/-0.041			
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	48% +/-0.079	margin=+0.000 +/-0.005			
forward+skip2	correct= 42%	crossed= 0%	incomplete= 57%	overlap=	+1.000

branch-correct=	95% +/-0.034	margin=+0.184	+/-0.171		
bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	100% +/-0.000	margin=+0.058	+/-0.019		
multilag	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	100% +/-0.000	margin=+0.061	+/-0.020		
coh bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	100% +/-0.000	margin=+0.044	+/-0.020		
coh trigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	100% +/-0.000	margin=+0.061	+/-0.020		
-- gain --					
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	57% +/-0.078	margin=+0.011	+/-0.057		
forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	60% +/-0.077	margin=+0.004	+/-0.016		
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=	+1.000
branch-correct=	100% +/-0.000	margin=+0.377	+/-0.041		
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	48% +/-0.079	margin=+0.000	+/-0.005		
forward+skip2	correct= 48%	crossed= 0%	incomplete= 52%	overlap=	+1.000
branch-correct=	95% +/-0.034	margin=+0.197	+/-0.172		
bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	100% +/-0.000	margin=+0.063	+/-0.020		
multilag	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	100% +/-0.000	margin=+0.065	+/-0.020		
coh bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	98% +/-0.025	margin=+0.048	+/-0.022		
coh trigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	100% +/-0.000	margin=+0.066	+/-0.020		

=====

=====

omega = 0.1 (constant, frequency_std = 0) 2-1-2

-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	55% +/-0.079	margin=+0.011	+/-0.053		
forward	correct= 38%	crossed= 40%	incomplete= 22%	overlap=	+0.927
branch-correct=	50% +/-0.079	margin=+0.002	+/-0.135		
skip-two	correct= 0%	crossed= 2%	incomplete= 98%	overlap=	n/a
branch-correct=	98% +/-0.025	margin=+0.148	+/-0.049		
self+forward	correct= 2%	crossed= 2%	incomplete= 95%	overlap=	+0.930
branch-correct=	48% +/-0.079	margin=+0.001	+/-0.039		
forward+skip2	correct= 45%	crossed= 38%	incomplete= 18%	overlap=	+0.907
branch-correct=	60% +/-0.077	margin=+0.041	+/-0.124		
bigram	correct= 72%	crossed= 10%	incomplete= 18%	overlap=	+0.895
branch-correct=	90% +/-0.047	margin=+0.129	+/-0.087		
multilag	correct= 55%	crossed= 28%	incomplete= 18%	overlap=	+0.917
branch-correct=	75% +/-0.068	margin=+0.089	+/-0.121		
coh bigram	correct= 65%	crossed= 28%	incomplete= 8%	overlap=	+0.909

```

branch-correct= 75% +/-0.068 margin=+0.086 +/-0.105
coh trigram    correct= 72% crossed= 10% incomplete= 18% overlap=+0.878
branch-correct= 92% +/-0.042 margin=+0.151 +/-0.078
-- gain --
self           correct= 0% crossed= 0% incomplete= 100% overlap= n/a
branch-correct= 55% +/-0.079 margin=+0.011 +/-0.053
forward        correct= 38% crossed= 40% incomplete= 22% overlap=+0.927
branch-correct= 50% +/-0.079 margin=+0.002 +/-0.135
skip-two       correct= 0% crossed= 2% incomplete= 98% overlap= n/a
branch-correct= 98% +/-0.025 margin=+0.148 +/-0.049
self+forward   correct= 2% crossed= 5% incomplete= 92% overlap=+0.926
branch-correct= 50% +/-0.079 margin=-0.000 +/-0.038
forward+skip2  correct= 55% crossed= 38% incomplete= 8% overlap=+0.900
branch-correct= 62% +/-0.077 margin=+0.047 +/-0.119
bigram         correct= 68% crossed= 20% incomplete= 12% overlap=+0.915
branch-correct= 82% +/-0.060 margin=+0.126 +/-0.118
multilag       correct= 55% crossed= 18% incomplete= 28% overlap=+0.910
branch-correct= 82% +/-0.060 margin=+0.136 +/-0.113
coh bigram     correct= 60% crossed= 28% incomplete= 12% overlap=+0.917
branch-correct= 72% +/-0.071 margin=+0.086 +/-0.111
coh trigram    correct= 62% crossed= 8% incomplete= 30% overlap=+0.911
branch-correct= 92% +/-0.042 margin=+0.177 +/-0.095

```

```

=====
=====

```

```

omega = 0.1 (constant, frequency_std = 0) 2-2-2

```

```

-- raw --
self           correct= 0% crossed= 0% incomplete= 100% overlap= n/a
branch-correct= 62% +/-0.077 margin=+0.022 +/-0.083
forward        correct= 40% crossed= 40% incomplete= 20% overlap=+0.925
branch-correct= 48% +/-0.079 margin=+0.000 +/-0.140
skip-two       correct= 0% crossed= 50% incomplete= 50% overlap= n/a
branch-correct= 48% +/-0.079 margin=+0.006 +/-0.107
self+forward   correct= 0% crossed= 0% incomplete= 100% overlap= n/a
branch-correct= 50% +/-0.079 margin=+0.000 +/-0.033
forward+skip2  correct= 45% crossed= 40% incomplete= 15% overlap=+0.909
branch-correct= 52% +/-0.079 margin=+0.015 +/-0.134
bigram         correct= 42% crossed= 50% incomplete= 8% overlap=+0.896
branch-correct= 52% +/-0.079 margin=+0.008 +/-0.113
multilag       correct= 50% crossed= 40% incomplete= 10% overlap=+0.919
branch-correct= 57% +/-0.078 margin=+0.032 +/-0.151
coh bigram     correct= 50% crossed= 45% incomplete= 5% overlap=+0.910
branch-correct= 55% +/-0.079 margin=+0.012 +/-0.124
coh trigram    correct= 60% crossed= 28% incomplete= 12% overlap=+0.878
branch-correct= 75% +/-0.068 margin=+0.065 +/-0.103
-- gain --
self           correct= 0% crossed= 0% incomplete= 100% overlap= n/a
branch-correct= 62% +/-0.077 margin=+0.022 +/-0.083

```

forward	correct= 40%	crossed= 40%	incomplete= 20%	overlap=+0.925
branch-correct=	48% +/-0.079	margin=+0.000 +/-0.140		
skip-two	correct= 0%	crossed= 50%	incomplete= 50%	overlap= n/a
branch-correct=	48% +/-0.079	margin=+0.006 +/-0.107		
self+forward	correct= 2%	crossed= 2%	incomplete= 95%	overlap=+0.911
branch-correct=	50% +/-0.079	margin=-0.000 +/-0.046		
forward+skip2	correct= 48%	crossed= 45%	incomplete= 8%	overlap=+0.902
branch-correct=	55% +/-0.079	margin=+0.011 +/-0.125		
bigram	correct= 45%	crossed= 50%	incomplete= 5%	overlap=+0.915
branch-correct=	50% +/-0.079	margin=+0.004 +/-0.151		
multilag	correct= 62%	crossed= 30%	incomplete= 8%	overlap=+0.916
branch-correct=	70% +/-0.072	margin=+0.088 +/-0.159		
coh bigram	correct= 48%	crossed= 42%	incomplete= 10%	overlap=+0.919
branch-correct=	55% +/-0.079	margin=+0.014 +/-0.139		
coh trigram	correct= 62%	crossed= 28%	incomplete= 10%	overlap=+0.916
branch-correct=	72% +/-0.071	margin=+0.094 +/-0.154		

```

=====
=====
omega = 0.1 (constant, frequency_std = 0)    2-3-2
-- raw --
self      correct=  0%  crossed=  0%  incomplete= 100%  overlap= n/a
branch-correct= 42% +/-0.078  margin=-0.020 +/-0.092
forward    correct= 35%  crossed= 38%  incomplete= 28%  overlap=+0.917
branch-correct= 50% +/-0.079  margin=+0.000 +/-0.137
skip-two   correct=  0%  crossed= 62%  incomplete= 38%  overlap= n/a
branch-correct= 38% +/-0.077  margin=-0.015 +/-0.111
self+forward correct=  2%  crossed=  2%  incomplete= 95%  overlap=+0.936
branch-correct= 50% +/-0.079  margin=+0.001 +/-0.047
forward+skip2 correct= 30%  crossed= 45%  incomplete= 25%  overlap=+0.910
branch-correct= 52% +/-0.079  margin=-0.001 +/-0.143
bigram     correct= 42%  crossed= 45%  incomplete= 12%  overlap=+0.892
branch-correct= 55% +/-0.079  margin=+0.006 +/-0.119
multilag   correct= 32%  crossed= 42%  incomplete= 25%  overlap=+0.921
branch-correct= 55% +/-0.079  margin=+0.006 +/-0.162
coh bigram  correct= 38%  crossed= 48%  incomplete= 15%  overlap=+0.910
branch-correct= 50% +/-0.079  margin=+0.001 +/-0.131
coh trigram correct= 30%  crossed= 48%  incomplete= 22%  overlap=+0.874
branch-correct= 55% +/-0.079  margin=+0.003 +/-0.105
-- gain --
self      correct=  0%  crossed=  0%  incomplete= 100%  overlap= n/a
branch-correct= 42% +/-0.078  margin=-0.020 +/-0.092
forward    correct= 35%  crossed= 38%  incomplete= 28%  overlap=+0.917
branch-correct= 50% +/-0.079  margin=+0.000 +/-0.137
skip-two   correct=  0%  crossed= 62%  incomplete= 38%  overlap= n/a
branch-correct= 38% +/-0.077  margin=-0.015 +/-0.111
self+forward correct=  2%  crossed=  2%  incomplete= 95%  overlap=+0.932
branch-correct= 52% +/-0.079  margin=+0.001 +/-0.044

```

forward+skip2	correct= 35%	crossed= 48%	incomplete= 18%	overlap=+0.903
branch-correct=	52% +/-0.079	margin=+0.001 +/-0.141		
bigram	correct= 40%	crossed= 48%	incomplete= 12%	overlap=+0.916
branch-correct=	52% +/-0.079	margin=-0.001 +/-0.158		
multilag	correct= 32%	crossed= 48%	incomplete= 20%	overlap=+0.913
branch-correct=	50% +/-0.079	margin=-0.004 +/-0.168		
coh bigram	correct= 35%	crossed= 50%	incomplete= 15%	overlap=+0.920
branch-correct=	48% +/-0.079	margin=-0.005 +/-0.151		
coh trigram	correct= 35%	crossed= 50%	incomplete= 15%	overlap=+0.916
branch-correct=	48% +/-0.079	margin=-0.009 +/-0.172		

=====

=====

omega = 0.1 (constant, frequency_std = 0) 1-1-1

-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	57% +/-0.078	margin=+0.011 +/-0.057		
forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	65% +/-0.075	margin=+0.003 +/-0.013		
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+0.999
branch-correct=	100% +/-0.000	margin=+0.403 +/-0.042		
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	38% +/-0.077	margin=+0.000 +/-0.004		
forward+skip2	correct= 55%	crossed= 0%	incomplete= 45%	overlap=+0.999
branch-correct=	95% +/-0.034	margin=+0.219 +/-0.171		
bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.053 +/-0.018		
multilag	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.056 +/-0.020		
coh bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.038 +/-0.017		
coh trigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.057 +/-0.019		

-- gain --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	57% +/-0.078	margin=+0.011 +/-0.057		
forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	65% +/-0.075	margin=+0.003 +/-0.013		
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+0.999
branch-correct=	100% +/-0.000	margin=+0.403 +/-0.042		
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	35% +/-0.075	margin=+0.000 +/-0.003		
forward+skip2	correct= 85%	crossed= 5%	incomplete= 10%	overlap=+0.999
branch-correct=	90% +/-0.047	margin=+0.294 +/-0.179		
bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.060 +/-0.020		
multilag	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.062 +/-0.020		

```

coh bigram      correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 100% +/-0.000  margin=+0.042 +/-0.019
coh trigram     correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 100% +/-0.000  margin=+0.063 +/-0.020

=====
=====
omega = 0.3 (constant, frequency_std = 0)  2-1-2
-- raw --
self           correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 60% +/-0.077  margin=+0.001 +/-0.014
forward        correct= 48%  crossed= 52%  incomplete=  0%  overlap=+0.875
branch-correct= 57% +/-0.078  margin=+0.003 +/-0.062
skip-two       correct=  0%  crossed=  2%  incomplete= 98%  overlap=  n/a
branch-correct= 100% +/-0.000  margin=+0.103 +/-0.043
self+forward   correct= 25%  crossed= 25%  incomplete= 50%  overlap=+0.908
branch-correct= 52% +/-0.079  margin=+0.000 +/-0.094
forward+skip2  correct= 65%  crossed= 28%  incomplete=  8%  overlap=+0.878
branch-correct= 75% +/-0.068  margin=+0.054 +/-0.080
bigram         correct= 85%  crossed=  0%  incomplete= 15%  overlap=+0.874
branch-correct= 100% +/-0.000  margin=+0.130 +/-0.048
multilag       correct= 60%  crossed= 25%  incomplete= 15%  overlap=+0.903
branch-correct= 78% +/-0.066  margin=+0.089 +/-0.101
coh bigram     correct= 90%  crossed=  5%  incomplete=  5%  overlap=+0.877
branch-correct= 98% +/-0.025  margin=+0.091 +/-0.048
coh trigram    correct= 80%  crossed=  2%  incomplete= 18%  overlap=+0.866
branch-correct= 95% +/-0.034  margin=+0.152 +/-0.066
-- gain --
self           correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 60% +/-0.077  margin=+0.001 +/-0.014
forward        correct= 48%  crossed= 52%  incomplete=  0%  overlap=+0.875
branch-correct= 57% +/-0.078  margin=+0.003 +/-0.062
skip-two       correct=  0%  crossed=  2%  incomplete= 98%  overlap=  n/a
branch-correct= 100% +/-0.000  margin=+0.103 +/-0.043
self+forward   correct= 45%  crossed= 42%  incomplete= 12%  overlap=+0.895
branch-correct= 52% +/-0.079  margin=-0.002 +/-0.118
forward+skip2  correct= 65%  crossed= 22%  incomplete= 12%  overlap=+0.855
branch-correct= 92% +/-0.042  margin=+0.057 +/-0.048
bigram         correct= 72%  crossed=  8%  incomplete= 20%  overlap=+0.894
branch-correct= 92% +/-0.042  margin=+0.130 +/-0.081
multilag       correct= 65%  crossed= 12%  incomplete= 22%  overlap=+0.890
branch-correct= 88% +/-0.052  margin=+0.133 +/-0.088
coh bigram     correct= 82%  crossed= 12%  incomplete=  5%  overlap=+0.891
branch-correct= 90% +/-0.047  margin=+0.098 +/-0.068
coh trigram    correct= 70%  crossed=  8%  incomplete= 22%  overlap=+0.895
branch-correct= 92% +/-0.042  margin=+0.160 +/-0.084

=====
=====

```

```

=====
omega = 0.3 (constant, frequency_std = 0)    2-2-2
-- raw --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 57% +/-0.078  margin=+0.011 +/-0.064
forward       correct= 42%  crossed= 57%  incomplete=  0%  overlap=+0.877
branch-correct= 42% +/-0.078  margin=-0.013 +/-0.077
skip-two      correct=  0%  crossed= 42%  incomplete= 57%  overlap=  n/a
branch-correct= 45% +/-0.079  margin=+0.009 +/-0.052
self+forward  correct= 25%  crossed= 28%  incomplete= 48%  overlap=+0.912
branch-correct= 48% +/-0.079  margin=-0.005 +/-0.115
forward+skip2 correct= 45%  crossed= 48%  incomplete=  8%  overlap=+0.880
branch-correct= 52% +/-0.079  margin=+0.005 +/-0.091
bigram        correct= 48%  crossed= 45%  incomplete=  8%  overlap=+0.870
branch-correct= 60% +/-0.077  margin=+0.015 +/-0.063
multilag      correct= 52%  crossed= 45%  incomplete=  2%  overlap=+0.904
branch-correct= 55% +/-0.079  margin=+0.021 +/-0.126
coh bigram    correct= 38%  crossed= 57%  incomplete=  5%  overlap=+0.866
branch-correct= 50% +/-0.079  margin=+0.002 +/-0.054
coh trigram   correct= 62%  crossed= 35%  incomplete=  2%  overlap=+0.877
branch-correct= 70% +/-0.072  margin=+0.070 +/-0.111
-- gain --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 57% +/-0.078  margin=+0.011 +/-0.064
forward       correct= 42%  crossed= 57%  incomplete=  0%  overlap=+0.877
branch-correct= 42% +/-0.078  margin=-0.013 +/-0.077
skip-two      correct=  0%  crossed= 42%  incomplete= 57%  overlap=  n/a
branch-correct= 45% +/-0.079  margin=+0.009 +/-0.052
self+forward  correct= 40%  crossed= 48%  incomplete= 12%  overlap=+0.892
branch-correct= 50% +/-0.079  margin=-0.008 +/-0.114
forward+skip2 correct= 48%  crossed= 48%  incomplete=  5%  overlap=+0.848
branch-correct= 52% +/-0.079  margin=-0.000 +/-0.057
bigram        correct= 42%  crossed= 48%  incomplete= 10%  overlap=+0.894
branch-correct= 52% +/-0.079  margin=+0.011 +/-0.107
multilag      correct= 57%  crossed= 32%  incomplete= 10%  overlap=+0.892
branch-correct= 68% +/-0.074  margin=+0.060 +/-0.124
coh bigram    correct= 45%  crossed= 50%  incomplete=  5%  overlap=+0.889
branch-correct= 52% +/-0.079  margin=+0.004 +/-0.085
coh trigram   correct= 60%  crossed= 30%  incomplete= 10%  overlap=+0.899
branch-correct= 70% +/-0.072  margin=+0.070 +/-0.127

=====
=====
omega = 0.3 (constant, frequency_std = 0)    2-3-2
-- raw --
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 48% +/-0.079  margin=-0.006 +/-0.070
forward       correct= 55%  crossed= 38%  incomplete=  8%  overlap=+0.876

```

branch-correct=	57% +/-0.078	margin=+0.006 +/-0.090			
skip-two	correct= 0%	crossed= 55%	incomplete= 45%	overlap=	n/a
branch-correct=	45% +/-0.079	margin=-0.000 +/-0.050			
self+forward	correct= 18%	crossed= 18%	incomplete= 65%	overlap=	+0.903
branch-correct=	50% +/-0.079	margin=-0.000 +/-0.089			
forward+skip2	correct= 42%	crossed= 45%	incomplete= 12%	overlap=	+0.877
branch-correct=	55% +/-0.079	margin=+0.008 +/-0.102			
bigram	correct= 28%	crossed= 48%	incomplete= 25%	overlap=	+0.868
branch-correct=	60% +/-0.077	margin=+0.007 +/-0.060			
multilag	correct= 32%	crossed= 48%	incomplete= 20%	overlap=	+0.904
branch-correct=	52% +/-0.079	margin=+0.001 +/-0.130			
coh bigram	correct= 45%	crossed= 48%	incomplete= 8%	overlap=	+0.869
branch-correct=	57% +/-0.078	margin=+0.004 +/-0.054			
coh trigram	correct= 30%	crossed= 55%	incomplete= 15%	overlap=	+0.872
branch-correct=	50% +/-0.079	margin=+0.000 +/-0.124			
-- gain --					
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	48% +/-0.079	margin=-0.006 +/-0.070			
forward	correct= 55%	crossed= 38%	incomplete= 8%	overlap=	+0.876
branch-correct=	57% +/-0.078	margin=+0.006 +/-0.090			
skip-two	correct= 0%	crossed= 55%	incomplete= 45%	overlap=	n/a
branch-correct=	45% +/-0.079	margin=-0.000 +/-0.050			
self+forward	correct= 30%	crossed= 40%	incomplete= 30%	overlap=	+0.889
branch-correct=	50% +/-0.079	margin=-0.007 +/-0.104			
forward+skip2	correct= 45%	crossed= 42%	incomplete= 12%	overlap=	+0.849
branch-correct=	57% +/-0.078	margin=+0.006 +/-0.064			
bigram	correct= 38%	crossed= 45%	incomplete= 18%	overlap=	+0.892
branch-correct=	52% +/-0.079	margin=+0.003 +/-0.114			
multilag	correct= 30%	crossed= 52%	incomplete= 18%	overlap=	+0.890
branch-correct=	45% +/-0.079	margin=-0.008 +/-0.122			
coh bigram	correct= 40%	crossed= 48%	incomplete= 12%	overlap=	+0.886
branch-correct=	50% +/-0.079	margin=+0.004 +/-0.095			
coh trigram	correct= 32%	crossed= 50%	incomplete= 18%	overlap=	+0.897
branch-correct=	50% +/-0.079	margin=-0.000 +/-0.133			

=====

=====

omega = 0.3 (constant, frequency_std = 0) 1-1-1

-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	48% +/-0.079	margin=+0.001 +/-0.007			
forward	correct= 42%	crossed= 57%	incomplete= 0%	overlap=	+0.900
branch-correct=	48% +/-0.079	margin=-0.038 +/-0.208			
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=	+0.986
branch-correct=	100% +/-0.000	margin=+0.445 +/-0.039			
self+forward	correct= 0%	crossed= 0%	incomplete= 100%	overlap=	n/a
branch-correct=	30% +/-0.072	margin=-0.000 +/-0.001			
forward+skip2	correct= 100%	crossed= 0%	incomplete= 0%	overlap=	+0.993

branch-correct=	100% +/-0.000	margin=+0.353	+/-0.033		
bigram	correct= 70%	crossed= 30%	incomplete= 0%	overlap=+0.917	
branch-correct=	45% +/-0.079	margin=+0.008	+/-0.250		
multilag	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a	
branch-correct=	100% +/-0.000	margin=+0.047	+/-0.017		
coh bigram	correct= 88%	crossed= 12%	incomplete= 0%	overlap=+0.897	
branch-correct=	82% +/-0.060	margin=+0.079	+/-0.075		
coh trigram	correct= 40%	crossed= 60%	incomplete= 0%	overlap=+0.866	
branch-correct=	62% +/-0.077	margin=-0.008	+/-0.090		
-- gain --					
self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a	
branch-correct=	48% +/-0.079	margin=+0.001	+/-0.007		
forward	correct= 42%	crossed= 57%	incomplete= 0%	overlap=+0.900	
branch-correct=	48% +/-0.079	margin=-0.038	+/-0.208		
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+0.986	
branch-correct=	100% +/-0.000	margin=+0.445	+/-0.039		
self+forward	correct= 0%	crossed= 5%	incomplete= 95%	overlap= n/a	
branch-correct=	38% +/-0.077	margin=-0.001	+/-0.011		
forward+skip2	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+0.982	
branch-correct=	100% +/-0.000	margin=+0.337	+/-0.032		
bigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a	
branch-correct=	100% +/-0.000	margin=+0.051	+/-0.018		
multilag	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a	
branch-correct=	100% +/-0.000	margin=+0.056	+/-0.019		
coh bigram	correct= 48%	crossed= 48%	incomplete= 5%	overlap=+0.989	
branch-correct=	50% +/-0.079	margin=+0.024	+/-0.282		
coh trigram	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a	
branch-correct=	100% +/-0.000	margin=+0.058	+/-0.020		

=====

=====

omega = 1 (constant, frequency_std = 0) 2-1-2

-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	48% +/-0.079	margin=+0.003	+/-0.041	
forward	correct= 38%	crossed= 62%	incomplete= 0%	overlap=+0.862
branch-correct=	50% +/-0.079	margin=-0.001	+/-0.090	
skip-two	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	100% +/-0.000	margin=+0.135	+/-0.038	
self+forward	correct= 48%	crossed= 52%	incomplete= 0%	overlap=+0.838
branch-correct=	50% +/-0.079	margin=-0.002	+/-0.091	
forward+skip2	correct= 72%	crossed= 18%	incomplete= 10%	overlap=+0.830
branch-correct=	88% +/-0.052	margin=+0.091	+/-0.089	
bigram	correct= 85%	crossed= 2%	incomplete= 12%	overlap=+0.860
branch-correct=	98% +/-0.025	margin=+0.158	+/-0.066	
multilag	correct= 88%	crossed= 2%	incomplete= 10%	overlap=+0.871
branch-correct=	98% +/-0.025	margin=+0.099	+/-0.043	
coh bigram	correct= 82%	crossed= 12%	incomplete= 5%	overlap=+0.867

```

branch-correct= 88% +/-0.052  margin=+0.098 +/-0.082
coh trigram    correct= 80%  crossed= 8%  incomplete= 12%  overlap=+0.856
branch-correct= 92% +/-0.042  margin=+0.173 +/-0.086
-- gain --
self           correct= 0%  crossed= 0%  incomplete= 100%  overlap= n/a
branch-correct= 48% +/-0.079  margin=+0.003 +/-0.041
forward        correct= 38%  crossed= 62%  incomplete= 0%  overlap=+0.862
branch-correct= 50% +/-0.079  margin=-0.001 +/-0.090
skip-two       correct= 0%  crossed= 0%  incomplete= 100%  overlap= n/a
branch-correct= 100% +/-0.000  margin=+0.135 +/-0.038
self+forward   correct= 38%  crossed= 50%  incomplete= 12%  overlap=+0.885
branch-correct= 52% +/-0.079  margin=-0.001 +/-0.094
forward+skip2  correct= 55%  crossed= 20%  incomplete= 25%  overlap=+0.857
branch-correct= 82% +/-0.060  margin=+0.081 +/-0.085
bigram         correct= 90%  crossed= 0%  incomplete= 10%  overlap=+0.868
branch-correct= 100% +/-0.000  margin=+0.127 +/-0.049
multilag       correct= 75%  crossed= 8%  incomplete= 18%  overlap=+0.872
branch-correct= 92% +/-0.042  margin=+0.130 +/-0.067
coh bigram     correct= 92%  crossed= 0%  incomplete= 8%  overlap=+0.857
branch-correct= 100% +/-0.000  margin=+0.149 +/-0.064
coh trigram    correct= 75%  crossed= 12%  incomplete= 12%  overlap=+0.882
branch-correct= 98% +/-0.025  margin=+0.135 +/-0.070

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=====

```

```

omega = 1 (constant, frequency_std = 0)  2-2-2
-- raw --
self           correct= 0%  crossed= 0%  incomplete= 100%  overlap= n/a
branch-correct= 55% +/-0.079  margin=+0.009 +/-0.054
forward        correct= 35%  crossed= 65%  incomplete= 0%  overlap=+0.838
branch-correct= 42% +/-0.078  margin=-0.013 +/-0.096
skip-two       correct= 0%  crossed= 45%  incomplete= 55%  overlap= n/a
branch-correct= 60% +/-0.077  margin=+0.020 +/-0.108
self+forward   correct= 35%  crossed= 65%  incomplete= 0%  overlap=+0.840
branch-correct= 38% +/-0.077  margin=-0.024 +/-0.106
forward+skip2  correct= 35%  crossed= 60%  incomplete= 5%  overlap=+0.830
branch-correct= 48% +/-0.079  margin=-0.016 +/-0.115
bigram         correct= 50%  crossed= 48%  incomplete= 2%  overlap=+0.840
branch-correct= 52% +/-0.079  margin=+0.006 +/-0.134
multilag       correct= 38%  crossed= 52%  incomplete= 10%  overlap=+0.870
branch-correct= 57% +/-0.078  margin=+0.008 +/-0.067
coh bigram     correct= 48%  crossed= 50%  incomplete= 2%  overlap=+0.846
branch-correct= 55% +/-0.079  margin=+0.010 +/-0.102
coh trigram    correct= 68%  crossed= 25%  incomplete= 8%  overlap=+0.840
branch-correct= 72% +/-0.071  margin=+0.081 +/-0.123
-- gain --
self           correct= 0%  crossed= 0%  incomplete= 100%  overlap= n/a
branch-correct= 55% +/-0.079  margin=+0.009 +/-0.054

```

forward	correct= 35%	crossed= 65%	incomplete= 0%	overlap=+0.838
branch-correct=	42% +/-0.078	margin=-0.013 +/-0.096		
skip-two	correct= 0%	crossed= 45%	incomplete= 55%	overlap= n/a
branch-correct=	60% +/-0.077	margin=+0.020 +/-0.108		
self+forward	correct= 40%	crossed= 55%	incomplete= 5%	overlap=+0.861
branch-correct=	45% +/-0.079	margin=-0.009 +/-0.106		
forward+skip2	correct= 40%	crossed= 50%	incomplete= 10%	overlap=+0.834
branch-correct=	50% +/-0.079	margin=+0.003 +/-0.097		
bigram	correct= 42%	crossed= 50%	incomplete= 8%	overlap=+0.871
branch-correct=	52% +/-0.079	margin=+0.008 +/-0.084		
multilag	correct= 60%	crossed= 35%	incomplete= 5%	overlap=+0.872
branch-correct=	68% +/-0.074	margin=+0.044 +/-0.089		
coh bigram	correct= 40%	crossed= 57%	incomplete= 2%	overlap=+0.855
branch-correct=	45% +/-0.079	margin=-0.009 +/-0.116		
coh trigram	correct= 60%	crossed= 35%	incomplete= 5%	overlap=+0.875
branch-correct=	78% +/-0.066	margin=+0.046 +/-0.064		

```

=====
=====
omega = 1 (constant, frequency_std = 0)    2-3-2
-- raw --
self      correct=  0%  crossed=  0%  incomplete= 100%  overlap= n/a
branch-correct= 55% +/-0.079  margin=-0.002 +/-0.062
forward    correct= 55%  crossed= 38%  incomplete=  8%  overlap=+0.836
branch-correct= 60% +/-0.077  margin=+0.015 +/-0.103
skip-two   correct=  0%  crossed= 52%  incomplete= 48%  overlap= n/a
branch-correct= 45% +/-0.079  margin=-0.007 +/-0.096
self+forward correct= 48%  crossed= 48%  incomplete=  5%  overlap=+0.831
branch-correct= 52% +/-0.079  margin=+0.000 +/-0.107
forward+skip2 correct= 50%  crossed= 42%  incomplete=  8%  overlap=+0.827
branch-correct= 55% +/-0.079  margin=+0.015 +/-0.104
bigram     correct= 42%  crossed= 48%  incomplete= 10%  overlap=+0.828
branch-correct= 57% +/-0.078  margin=+0.003 +/-0.120
multilag   correct= 32%  crossed= 48%  incomplete= 20%  overlap=+0.868
branch-correct= 57% +/-0.078  margin=+0.009 +/-0.072
coh bigram  correct= 42%  crossed= 40%  incomplete= 18%  overlap=+0.833
branch-correct= 60% +/-0.077  margin=+0.009 +/-0.101
coh trigram correct= 30%  crossed= 55%  incomplete= 15%  overlap=+0.809
branch-correct= 48% +/-0.079  margin=-0.008 +/-0.134
-- gain --
self      correct=  0%  crossed=  0%  incomplete= 100%  overlap= n/a
branch-correct= 55% +/-0.079  margin=-0.002 +/-0.062
forward    correct= 55%  crossed= 38%  incomplete=  8%  overlap=+0.836
branch-correct= 60% +/-0.077  margin=+0.015 +/-0.103
skip-two   correct=  0%  crossed= 52%  incomplete= 48%  overlap= n/a
branch-correct= 45% +/-0.079  margin=-0.007 +/-0.096
self+forward correct= 28%  crossed= 35%  incomplete= 38%  overlap=+0.844
branch-correct= 57% +/-0.078  margin=+0.001 +/-0.099

```

forward+skip2	correct= 22%	crossed= 45%	incomplete= 32%	overlap=+0.830
branch-correct=	57% +/-0.078	margin=+0.010 +/-0.093		
bigram	correct= 35%	crossed= 52%	incomplete= 12%	overlap=+0.871
branch-correct=	57% +/-0.078	margin=+0.009 +/-0.096		
multilag	correct= 25%	crossed= 60%	incomplete= 15%	overlap=+0.878
branch-correct=	48% +/-0.079	margin=-0.006 +/-0.100		
coh bigram	correct= 50%	crossed= 42%	incomplete= 8%	overlap=+0.852
branch-correct=	57% +/-0.078	margin=+0.014 +/-0.112		
coh trigram	correct= 18%	crossed= 60%	incomplete= 22%	overlap=+0.883
branch-correct=	50% +/-0.079	margin=-0.001 +/-0.082		

=====

=====

omega = 1 (constant, frequency_std = 0) 1-1-1

-- raw --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	48% +/-0.079	margin=+0.000 +/-0.000		
forward	correct= 40%	crossed= 60%	incomplete= 0%	overlap=+0.967
branch-correct=	40% +/-0.077	margin=-0.065 +/-0.280		
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+1.000
branch-correct=	100% +/-0.000	margin=+0.463 +/-0.054		
self+forward	correct= 35%	crossed= 65%	incomplete= 0%	overlap=+0.908
branch-correct=	38% +/-0.077	margin=-0.042 +/-0.199		
forward+skip2	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+0.997
branch-correct=	98% +/-0.025	margin=+0.422 +/-0.078		
bigram	correct= 18%	crossed= 45%	incomplete= 38%	overlap=+0.868
branch-correct=	100% +/-0.000	margin=+0.036 +/-0.017		
multilag	correct= 88%	crossed= 12%	incomplete= 0%	overlap=+0.978
branch-correct=	40% +/-0.077	margin=-0.010 +/-0.061		
coh bigram	correct= 45%	crossed= 40%	incomplete= 15%	overlap=+0.866
branch-correct=	50% +/-0.079	margin=-0.029 +/-0.093		
coh trigram	correct= 12%	crossed= 2%	incomplete= 85%	overlap=+0.881
branch-correct=	100% +/-0.000	margin=+0.041 +/-0.019		

-- gain --

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	48% +/-0.079	margin=+0.000 +/-0.000		
forward	correct= 40%	crossed= 60%	incomplete= 0%	overlap=+0.967
branch-correct=	40% +/-0.077	margin=-0.065 +/-0.280		
skip-two	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+1.000
branch-correct=	100% +/-0.000	margin=+0.463 +/-0.054		
self+forward	correct= 5%	crossed= 32%	incomplete= 62%	overlap=+0.852
branch-correct=	30% +/-0.072	margin=-0.002 +/-0.005		
forward+skip2	correct= 100%	crossed= 0%	incomplete= 0%	overlap=+1.000
branch-correct=	100% +/-0.000	margin=+0.370 +/-0.036		
bigram	correct= 95%	crossed= 5%	incomplete= 0%	overlap=+0.995
branch-correct=	40% +/-0.077	margin=+0.009 +/-0.049		
multilag	correct= 38%	crossed= 62%	incomplete= 0%	overlap=+0.992
branch-correct=	60% +/-0.077	margin=+0.022 +/-0.088		

```

coh bigram      correct= 100%  crossed=  0%  incomplete=  0%  overlap=+0.965
branch-correct= 52% +/-0.079  margin=+0.033 +/-0.226
coh trigram     correct=  30%  crossed= 70%  incomplete=  0%  overlap=+0.986
branch-correct= 62% +/-0.077  margin=+0.004 +/-0.124

```

```

=====
ALL METRICS AT THE OPERATING POINT  omega = 0.05 constant (frequency_std = 0)
=====

```

```
-- 2-1-2 --
```

rule	cat	correct	crossed	incompl	overlap	branch
margin						
self	raw	0%	0%	100%	n/a	55%
+0.011						
forward	raw	10%	12%	78%	+0.937+/-0.008	52%
+0.003						
skip-two	raw	0%	2%	98%	n/a	98%
+0.161						
self+forward	raw	2%	2%	95%	+0.896+/-0.000	48%
+0.000						
forward+skip2	raw	38%	28%	35%	+0.917+/-0.007	65%
+0.048						
bigram	raw	68%	12%	20%	+0.910+/-0.008	88%
+0.129						
multilag	raw	45%	18%	38%	+0.921+/-0.007	82%
+0.090						
coh bigram	raw	57%	28%	15%	+0.919+/-0.009	75%
+0.085						
coh trigram	raw	70%	8%	22%	+0.899+/-0.008	92%
+0.167						
self	gain	0%	0%	100%	n/a	55%
+0.011						
forward	gain	10%	12%	78%	+0.937+/-0.008	52%
+0.003						
skip-two	gain	0%	2%	98%	n/a	98%
+0.161						
self+forward	gain	2%	2%	95%	+0.895+/-0.000	48%
+0.000						
forward+skip2	gain	38%	28%	35%	+0.913+/-0.007	68%
+0.047						
bigram	gain	68%	20%	12%	+0.920+/-0.008	82%
+0.131						
multilag	gain	48%	12%	40%	+0.915+/-0.007	88%
+0.143						
coh bigram	gain	57%	22%	20%	+0.923+/-0.009	80%
+0.093						
coh trigram	gain	57%	10%	32%	+0.915+/-0.008	90%

+0.174

-- 2-2-2 --

rule	cat	correct	crossed	incompl	overlap	branch
margin						
self	raw	0%	0%	100%	n/a	62%
+0.022						
forward	raw	18%	20%	62%	+0.935+/-0.006	50%
-0.000						
skip-two	raw	0%	50%	50%	n/a	50%
+0.005						
self+forward	raw	0%	0%	100%	n/a	50%
-0.000						
forward+skip2	raw	38%	38%	25%	+0.915+/-0.008	52%
+0.013						
bigram	raw	45%	50%	5%	+0.911+/-0.007	50%
+0.005						
multilag	raw	42%	40%	18%	+0.923+/-0.008	57%
+0.020						
coh bigram	raw	45%	40%	15%	+0.921+/-0.008	55%
+0.015						
coh trigram	raw	62%	28%	10%	+0.904+/-0.007	72%
+0.081						
self	gain	0%	0%	100%	n/a	62%
+0.022						
forward	gain	18%	20%	62%	+0.935+/-0.006	50%
-0.000						
skip-two	gain	0%	50%	50%	n/a	50%
+0.005						
self+forward	gain	0%	0%	100%	n/a	50%
-0.000						
forward+skip2	gain	42%	40%	18%	+0.911+/-0.008	52%
+0.012						
bigram	gain	48%	48%	5%	+0.921+/-0.007	52%
+0.013						
multilag	gain	62%	30%	8%	+0.920+/-0.007	70%
+0.094						
coh bigram	gain	40%	42%	18%	+0.925+/-0.009	50%
+0.008						
coh trigram	gain	68%	22%	10%	+0.921+/-0.008	78%
+0.117						

-- 2-3-2 --

rule	cat	correct	crossed	incompl	overlap	branch
margin						
self	raw	0%	0%	100%	n/a	42%
-0.020						
forward	raw	15%	18%	68%	+0.921+/-0.009	50%

-0.001						
skip-two	raw	0%	57%	42%	n/a	40%
-0.019						
self+forward	raw	0%	0%	100%	n/a	50%
+0.001						
forward+skip2	raw	20%	40%	40%	+0.913+/-0.006	50%
-0.008						
bigram	raw	40%	48%	12%	+0.911+/-0.009	52%
-0.001						
multilag	raw	20%	32%	48%	+0.921+/-0.008	52%
+0.005						
coh bigram	raw	32%	48%	20%	+0.921+/-0.008	48%
-0.005						
coh trigram	raw	32%	50%	18%	+0.903+/-0.009	48%
-0.006						
self	gain	0%	0%	100%	n/a	42%
-0.020						
forward	gain	15%	18%	68%	+0.921+/-0.009	50%
-0.001						
skip-two	gain	0%	57%	42%	n/a	40%
-0.019						
self+forward	gain	0%	0%	100%	n/a	50%
+0.000						
forward+skip2	gain	18%	40%	42%	+0.909+/-0.006	48%
-0.010						
bigram	gain	35%	50%	15%	+0.923+/-0.009	50%
-0.009						
multilag	gain	28%	45%	28%	+0.918+/-0.007	50%
-0.008						
coh bigram	gain	25%	42%	32%	+0.923+/-0.008	48%
-0.007						
coh trigram	gain	32%	48%	20%	+0.919+/-0.008	48%
-0.009						
-- 1-1-1 --						
rule	cat	correct	crossed	incompl	overlap	branch
margin						
self	raw	0%	0%	100%	n/a	57%
+0.011						
forward	raw	0%	0%	100%	n/a	60%
+0.004						
skip-two	raw	100%	0%	0%	+1.000+/-0.000	100%
+0.377						
self+forward	raw	0%	0%	100%	n/a	48%
+0.000						
forward+skip2	raw	42%	0%	57%	+1.000+/-0.000	95%
+0.184						
bigram	raw	0%	0%	100%	n/a	100%

+0.058	multilag	raw	0%	0%	100%	n/a	100%
+0.061	coh bigram	raw	0%	0%	100%	n/a	100%
+0.044	coh trigram	raw	0%	0%	100%	n/a	100%
+0.061	self	gain	0%	0%	100%	n/a	57%
+0.011	forward	gain	0%	0%	100%	n/a	60%
+0.004	skip-two	gain	100%	0%	0%	+1.000+/-0.000	100%
+0.377	self+forward	gain	0%	0%	100%	n/a	48%
+0.000	forward+skip2	gain	48%	0%	52%	+1.000+/-0.000	95%
+0.197	bigram	gain	0%	0%	100%	n/a	100%
+0.063	multilag	gain	0%	0%	100%	n/a	100%
+0.065	coh bigram	gain	0%	0%	100%	n/a	98%
+0.048	coh trigram	gain	0%	0%	100%	n/a	100%
+0.066							

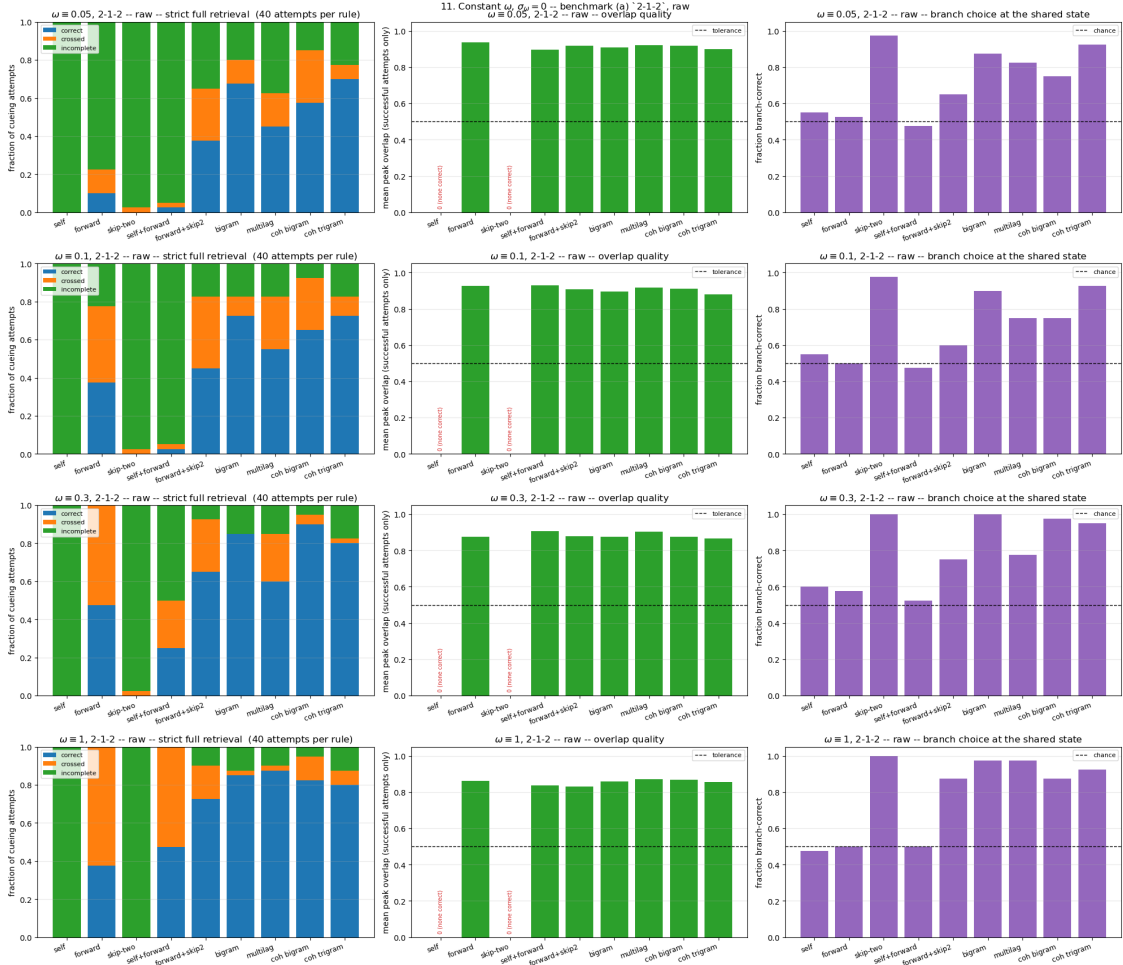
```
[56]: for variant in ("raw", "gain"):
    fig, axes = plt.subplots(len(FIXED_OMEGA), 3, figsize=(20, 4.4 *
    len(FIXED_OMEGA)))
    for row, omega in enumerate(FIXED_OMEGA):
        plot_rule_stats(fixed_omega_stats[(omega, "2-1-2", variant)],
            f"$\\omega \\equiv {omega:g}$, 2-1-2 -- {variant}",
            axes=axes[row])
        fig.suptitle(f"11. Constant $\\omega$, $\\sigma_\\omega = 0$ -- benchmark
        (a) `2-1-2`, {variant}")
        plt.tight_layout(); plt.show()

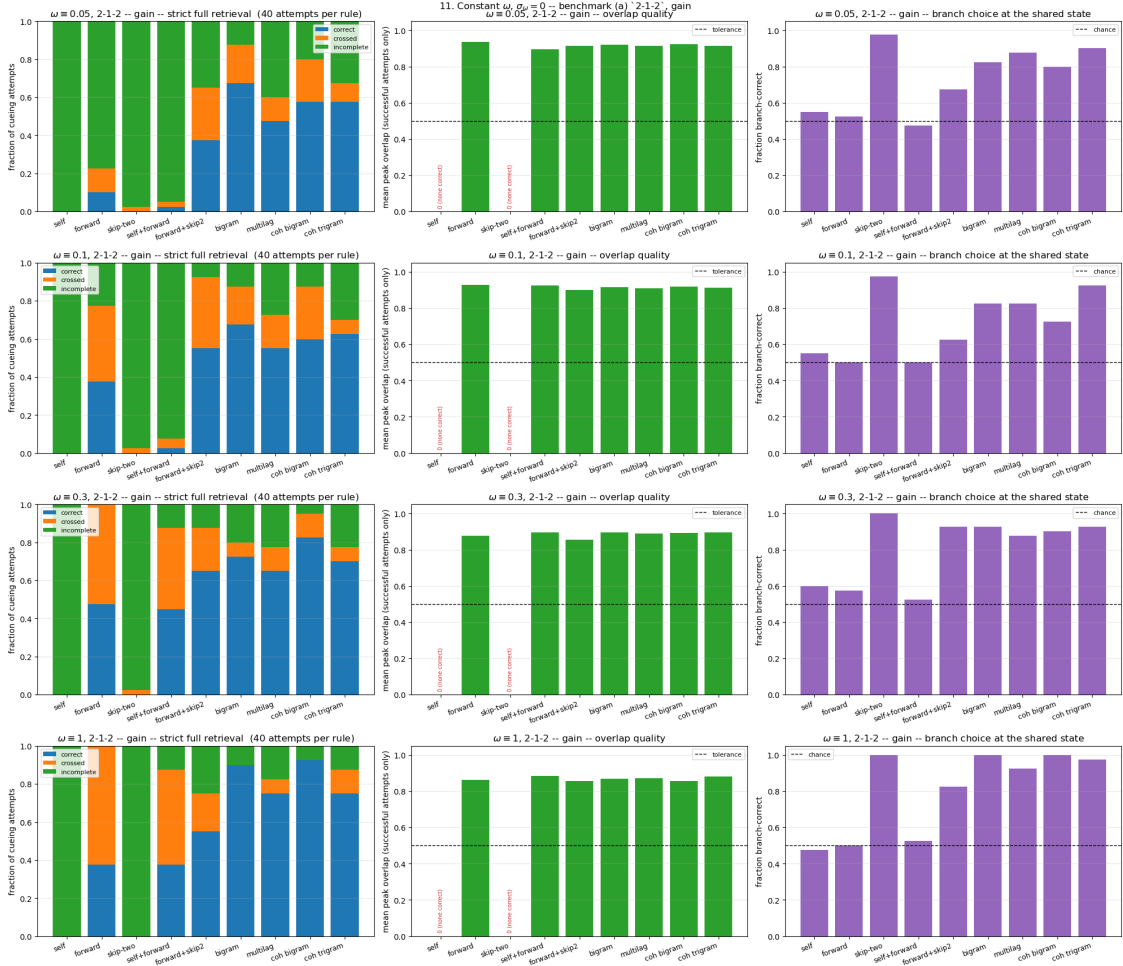
print("strict success / branch-correct, constant omega (frequency_std = 0), raw
rules")
print(f"{'rule':<15}" + "".join(f"{'w={o:g} {t}':>19}" for o in FIXED_OMEGA
for t in FIXED_OMEGA_GEOMETRIES))
for rule in RAW_CATALOGUE:
    row = ""
    for o in FIXED_OMEGA:
        for t in FIXED_OMEGA_GEOMETRIES:
            s = fixed_omega_stats[(o, t, "raw")][rule]
```

```

row += f"{s['counter']['correct'] / s['attempts']:>11.0%} /
↪{s['branch_correct']:>6.0%}"
print(f"{rule:<15}{row}")

```





strict success / branch-correct, constant omega (frequency_std = 0), raw rules									
rule	w=0.05 2-1-2			w=0.05 2-2-2			w=0.05 2-3-2		
w=0.05 1-1-1	w=0.1 2-1-2			w=0.1 2-2-2			w=0.1 2-3-2		
w=0.1 1-1-1	w=0.3 2-1-2			w=0.3 2-2-2			w=0.3 2-3-2		
w=0.3 1-1-1	w=1 2-1-2			w=1 2-2-2			w=1 2-3-2		
w=1 1-1-1									
self	0% / 55%			0% / 62%			0% / 42%		
0% / 57%	0% / 55%			0% / 62%			0% / 42%		
57%	0% /	60%	0% /	57%	0% /	48%	0% /	48%	
0% / 48%	0% / 55%			0% / 55%			0% / 48%		
forward	10% / 52%			18% / 50%			15% / 50%		
0% / 60%	38% / 50%			40% / 48%			35% / 50%		
65%	48% /	57%	42% /	42%	55% /	57%	42% /	48%	
38% / 50%	35% / 42%			55% / 60%			40% / 40%		
skip-two	0% / 98%			0% / 50%			0% / 40%		
100% / 100%	0% / 98%			0% / 48%			0% / 38%		
/ 100%	0% /	100%	0% /	45%	0% /	45%	100% /		

100%	0% / 100%	0% / 60%	0% / 45%	100% / 100%
self+forward	2% / 48%	0% / 50%	0% / 50%	
0% / 48%	2% / 48%	0% / 50%	2% / 50%	0% /
38%	25% / 52%	25% / 48%	18% / 50%	0% / 30%
48% / 50%	35% / 38%	48% / 52%	35% / 38%	
forward+skip2	38% / 65%	38% / 52%	20% / 50%	
42% / 95%	45% / 60%	45% / 52%	30% / 52%	55%
/ 95%	65% / 75%	45% / 52%	42% / 55%	100% /
100%	72% / 88%	35% / 48%	50% / 55%	100% / 98%
bigram	68% / 88%	45% / 50%	40% / 52%	
0% / 100%	72% / 90%	42% / 52%	42% / 55%	0% /
100%	85% / 100%	48% / 60%	28% / 60%	70% / 45%
85% / 98%	50% / 52%	42% / 57%	18% / 100%	
multilag	45% / 82%	42% / 57%	20% / 52%	
0% / 100%	55% / 75%	50% / 57%	32% / 55%	0% /
100%	60% / 78%	52% / 55%	32% / 52%	0% / 100%
88% / 98%	38% / 57%	32% / 57%	88% / 40%	
coh bigram	57% / 75%	45% / 55%	32% / 48%	
0% / 100%	65% / 75%	50% / 55%	38% / 50%	0% /
100%	90% / 98%	38% / 50%	45% / 57%	88% / 82%
82% / 88%	48% / 55%	42% / 60%	45% / 50%	
coh trigram	70% / 92%	62% / 72%	32% / 48%	
0% / 100%	72% / 92%	60% / 75%	30% / 55%	0% /
100%	80% / 95%	62% / 70%	30% / 50%	40% / 62%
80% / 92%	68% / 72%	30% / 48%	12% / 100%	

11.0.1 What to read here

Against §4's disorder runs, check whether the **ordering** survives, not the absolute numbers – removing the disorder changes how the trajectory leaves each pattern, so the rates move.

What should survive if §4's conclusions are about the rules:

- the context rules above chance on branch choice on 2-1-2 and 2-2-2, where their reach meets the shared-run length;
- the whole catalogue at chance on 2-3-2, **at every ω and in both the raw and gain columns**. If branch choice were being decided by ω disorder rather than by the gate, this is the row where it would show up.

A spot check at 10 trials per cell reproduces both: on 2-1-2 the context rules score 80-90% branch-correct at $\omega \in \{0.1, 0.3, 1.0\}$ while **forward** sits at 50%, and on 2-3-2 every rule sits at 50% at every ω .

12 10. Seed sweep – 100 instances at a sampled corruption rate ($\sigma_\omega = 0.3$, then $\omega \equiv 0.3$)

Everything above fixes `corruption_rate` at 0.2. Here a fresh rate is drawn per trial from $U[0.12, 0.28]$, so each rule faces a *range* of difficulties. What is checked is whether the **ordering** holds, not the absolute numbers.

Two things reported here that the per-geometry experiments do not:

- **The variance of metric 2** – high mean with high variance means crisp on easy instances and barely clearing `tolerance` on hard ones. Drawn as ± 1 s.d. on the metric-2 panel.
- **The succession of retrieved peaks** – the cued branch’s peak overlap at each position k , in stored order: metric 2 resolved position by position. Computed over **every** attempt, not just successful ones, so rules that never complete still show *where* they stop.

```
[57]: CORRUPTION_TARGET = CONFIG["corruption_rate"]    # 0.2
CORRUPTION_SPREAD = 0.08                             # per-seed rate ~ U[0.12, 0.28]
SEED_TRIALS = 100

def sample_corruption_rate(target, spread, N, rng):
    """One corruption rate, stochastically rounded to a whole flip count.

    Returns (k + 0.5)/N so that corrupt_phase's int() truncation lands on
    exactly k, where
    k = floor(rN) + Bernoulli(frac(rN)) and so E[k] = rN. """
    rate = rng.uniform(target - spread, target + spread)
    exact = rate * N
    floor = np.floor(exact)
    flips = int(floor + (rng.random() < exact - floor))
    return (flips + 0.5) / N

def seed_sweep(catalogue, seq_len, shared_idx, duration, trials=SEED_TRIALS,
    seed=0,
    config_overrides=None):
    """`rule_sweep` with one corruption rate drawn per trial instead of a fixed
    one.

    Every rule is re-seeded to the same value, so all of them see the identical
    instances at
    the identical rates. Returns rule_sweep's stats, which carry `overlap_var`
    and `peaks`:
    an (attempts, seq_len) array of the cued branch's own per-position peak
    overlaps, in
    stored order, collected on every attempt whether or not it succeeded.
```

```

    config_overrides: passed through to rule_sweep -- used to repeat the sweep
    at constant
    omega (frequency_std = 0) instead of at the configured disorder. """
    sampler = lambda rng: sample_corruption_rate(CORRUPTION_TARGET,
    CORRUPTION_SPREAD, N, rng)
    stats, _ = rule_sweep(catalogue, seq_len, shared_idx, duration,
    trials=trials, seed=seed,
                           rate_sampler=sampler,
    config_overrides=config_overrides)
    rates = stats[next(iter(stats))]["rates"]
    print(f" mean sampled rate over the {trials} seeds: {np.mean(rates):.4f} "
          f"(target {CORRUPTION_TARGET}, range "
          f"[{CORRUPTION_TARGET - CORRUPTION_SPREAD:.2f}, {CORRUPTION_TARGET +
    CORRUPTION_SPREAD:.2f}]))")
    return stats

def plot_peak_profile(stats, tag, shared_idx, axes=None):
    """The succession of retrieved peaks: mean and variance, across attempts,
    of the cued
    branch's own peak overlap at each stored position -- metric 2 resolved
    position by
    position. """
    if axes is None:
        _, axes = plt.subplots(1, 2, figsize=(15, 4.6))
        positions = np.arange(stats[list(stats)[0]]["peaks"].shape[1])
        for rule, s in stats.items():
            axes[0].plot(positions, s["peaks"].mean(axis=0), "-o", lw=1.8,
            label=rule)
            axes[1].plot(positions, s["peaks"].var(axis=0), "-o", lw=1.8,
            label=rule)
        for ax in axes:
            for k in shared_idx:
                ax.axvspan(k - 0.5, k + 0.5, color="tab:grey", alpha=0.15,
                           label="shared" if k == min(shared_idx) else None)
            ax.set_xlabel("position $k$ in the cued sequence (succession of
            retrieved peaks)")
            ax.set_xticks(positions)
            ax.legend(fontsize=7, ncol=2); ax.grid(alpha=0.3)
            axes[0].axhline(CONFIG["tolerance"], color="k", ls="--", lw=1)
            axes[0].set_ylabel(r"mean peak overlap $\langle \max_t \langle \mathbf{cued}, k \rangle \rangle$")
            axes[0].set_ylim(-0.05, 1.05)
            axes[0].set_title(f"{tag} -- mean overlap of successive retrieved peaks")
            axes[1].set_ylabel("variance of peak overlap across attempts")
            axes[1].set_title(f"{tag} -- variance of successive retrieved peaks")

```

```
return axes
```

```
[58]: seed_out = {}
      for tag in SPEC:
          print("=" * 100)
          print(f"{tag} -- {SEED_TRIALS} seeds")
          seed_out[tag] = seed_sweep(RAW_CATALOGUE, **GEOMETRIES[tag])
          print()
```

```
=====
=====
```

2-1-2 -- 100 seeds

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	50% +/-0.035	margin=-0.001 +/-0.057		
forward	correct= 26%	crossed= 38%	incomplete= 36%	overlap=+0.858
branch-correct=	50% +/-0.035	margin=-0.005 +/-0.090		
skip-two	correct= 0%	crossed= 13%	incomplete= 86%	overlap=+0.858
branch-correct=	82% +/-0.027	margin=+0.068 +/-0.081		
self+forward	correct= 20%	crossed= 26%	incomplete= 55%	overlap=+0.864
branch-correct=	49% +/-0.035	margin=-0.003 +/-0.073		
forward+skip2	correct= 54%	crossed= 26%	incomplete= 20%	overlap=+0.858
branch-correct=	70% +/-0.032	margin=+0.055 +/-0.101		
bigram	correct= 45%	crossed= 12%	incomplete= 42%	overlap=+0.795
branch-correct=	86% +/-0.024	margin=+0.107 +/-0.089		
multilag	correct= 52%	crossed= 10%	incomplete= 38%	overlap=+0.876
branch-correct=	78% +/-0.029	margin=+0.097 +/-0.106		
coh bigram	correct= 44%	crossed= 16%	incomplete= 40%	overlap=+0.818
branch-correct=	78% +/-0.029	margin=+0.074 +/-0.090		
coh trigram	correct= 36%	crossed= 10%	incomplete= 54%	overlap=+0.786
branch-correct=	91% +/-0.020	margin=+0.117 +/-0.084		
mean sampled rate over the 100 seeds: 0.2047 (target 0.2, range [0.12, 0.28])				

```
=====
=====
```

2-2-2 -- 100 seeds

self	correct= 0%	crossed= 0%	incomplete= 100%	overlap= n/a
branch-correct=	52% +/-0.035	margin=+0.005 +/-0.068		
forward	correct= 28%	crossed= 38%	incomplete= 34%	overlap=+0.851
branch-correct=	46% +/-0.035	margin=+0.000 +/-0.080		
skip-two	correct= 2%	crossed= 37%	incomplete= 62%	overlap=+0.876
branch-correct=	55% +/-0.035	margin=+0.004 +/-0.077		
self+forward	correct= 22%	crossed= 24%	incomplete= 54%	overlap=+0.855
branch-correct=	50% +/-0.035	margin=-0.001 +/-0.070		
forward+skip2	correct= 34%	crossed= 46%	incomplete= 21%	overlap=+0.857
branch-correct=	50% +/-0.035	margin=+0.001 +/-0.090		
bigram	correct= 22%	crossed= 36%	incomplete= 42%	overlap=+0.799
branch-correct=	51% +/-0.035	margin=+0.002 +/-0.070		
multilag	correct= 35%	crossed= 31%	incomplete= 34%	overlap=+0.880

```

branch-correct= 58% +/-0.035  margin=+0.025 +/-0.102
coh bigram      correct= 26%  crossed= 36%  incomplete= 38%  overlap=+0.819
branch-correct= 48% +/-0.035  margin=+0.002 +/-0.078
coh trigram     correct= 15%  crossed= 22%  incomplete= 63%  overlap=+0.778
branch-correct= 74% +/-0.031  margin=+0.045 +/-0.071
  mean sampled rate over the 100 seeds: 0.2047 (target 0.2, range [0.12, 0.28])

```

```

=====
=====
2-3-2 -- 100 seeds
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 51% +/-0.035  margin=-0.003 +/-0.071
forward       correct= 28%  crossed= 40%  incomplete= 32%  overlap=+0.843
branch-correct= 46% +/-0.035  margin=-0.002 +/-0.079
skip-two      correct=  0%  crossed= 36%  incomplete= 64%  overlap=+0.873
branch-correct= 50% +/-0.035  margin=-0.000 +/-0.074
self+forward  correct= 19%  crossed= 22%  incomplete= 60%  overlap=+0.856
branch-correct= 50% +/-0.035  margin=+0.000 +/-0.070
forward+skip2 correct= 30%  crossed= 46%  incomplete= 24%  overlap=+0.857
branch-correct= 47% +/-0.035  margin=-0.001 +/-0.087
bigram        correct= 18%  crossed= 39%  incomplete= 42%  overlap=+0.799
branch-correct= 44% +/-0.035  margin=-0.003 +/-0.067
multilag      correct= 28%  crossed= 42%  incomplete= 31%  overlap=+0.877
branch-correct= 48% +/-0.035  margin=-0.000 +/-0.096
coh bigram    correct= 22%  crossed= 37%  incomplete= 40%  overlap=+0.816
branch-correct= 45% +/-0.035  margin=-0.004 +/-0.074
coh trigram   correct= 10%  crossed= 30%  incomplete= 60%  overlap=+0.795
branch-correct= 46% +/-0.035  margin=-0.004 +/-0.063
  mean sampled rate over the 100 seeds: 0.2047 (target 0.2, range [0.12, 0.28])

```

```

[59]: fig, axes = plt.subplots(len(seed_out), 3, figsize=(20, 4.4 * len(seed_out)))
      for row, (tag, stats) in enumerate(seed_out.items()):
          plot_rule_stats(stats, f"{SEED_TRIALS} seeds, {tag}", axes=axes[row],
              ↪show_overlap_var=True)
      plt.tight_layout(); plt.show()

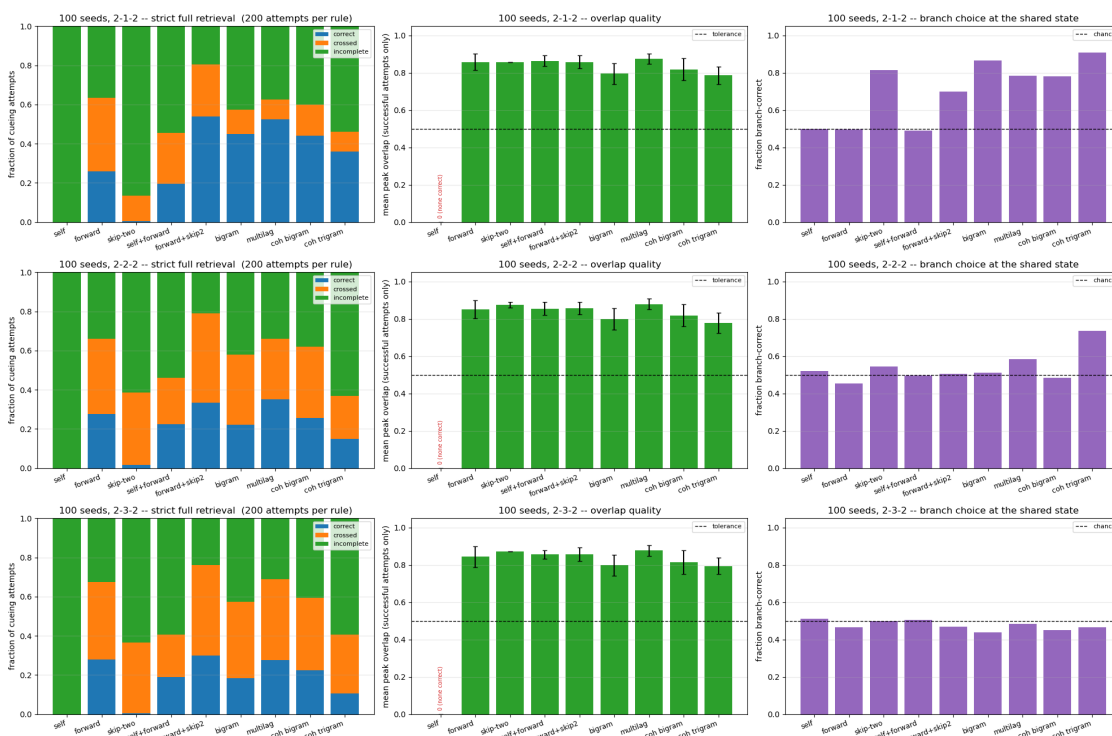
      tags = list(seed_out)
      print(f"{'rule':<15}" + "".join(f"{t:>32}" for t in tags))
      print(f"{'':<15}" + "".join(f"{'correct / overlap (var) / branch':>32}" for t,
          ↪in tags))
      print("-" * (15 + 32 * len(tags)))
      for rule in RAW_CATALOGUE:
          row = ""
          for t in tags:
              s = seed_out[t][rule]

```

```

q = f"{s['mean_overlap']:.2f}" if s["mean_overlap"] ==_
↪s["mean_overlap"] else " n/a"
v = f"{s['overlap_var']:.4f}" if s["overlap_var"] == s["overlap_var"]_
↪else " - "
row += f"{s['counter']['correct'] / s['attempts']:>9.0%} /{q:>6} ({v}) /
↪{s['branch_correct']:>5.0%}"
print(f"{rule:<15}{row}")

```



rule 2-1-2 2-2-2

correct / overlap (var) / branchcorrect / overlap (var) /
branchcorrect / overlap (var) / branch

```

self          0% /  n/a ( - ) /  50%      0% /  n/a ( - ) /
52%          0% /  n/a ( - ) /  51%
forward       26% /  0.86 (0.0021) /  50%    28% /  0.85 (0.0023) /
46%       28% /  0.84 (0.0031) /  46%
skip-two      0% /  0.86 (0.0000) /  82%      2% /  0.88 (0.0002) /
55%          0% /  0.87 (0.0000) /  50%
self+forward  20% /  0.86 (0.0008) /  49%    22% /  0.85 (0.0011) /
50%       19% /  0.86 (0.0005) /  50%
forward+skip2 54% /  0.86 (0.0013) /  70%    34% /  0.86 (0.0012) /
50%       30% /  0.86 (0.0014) /  47%

```

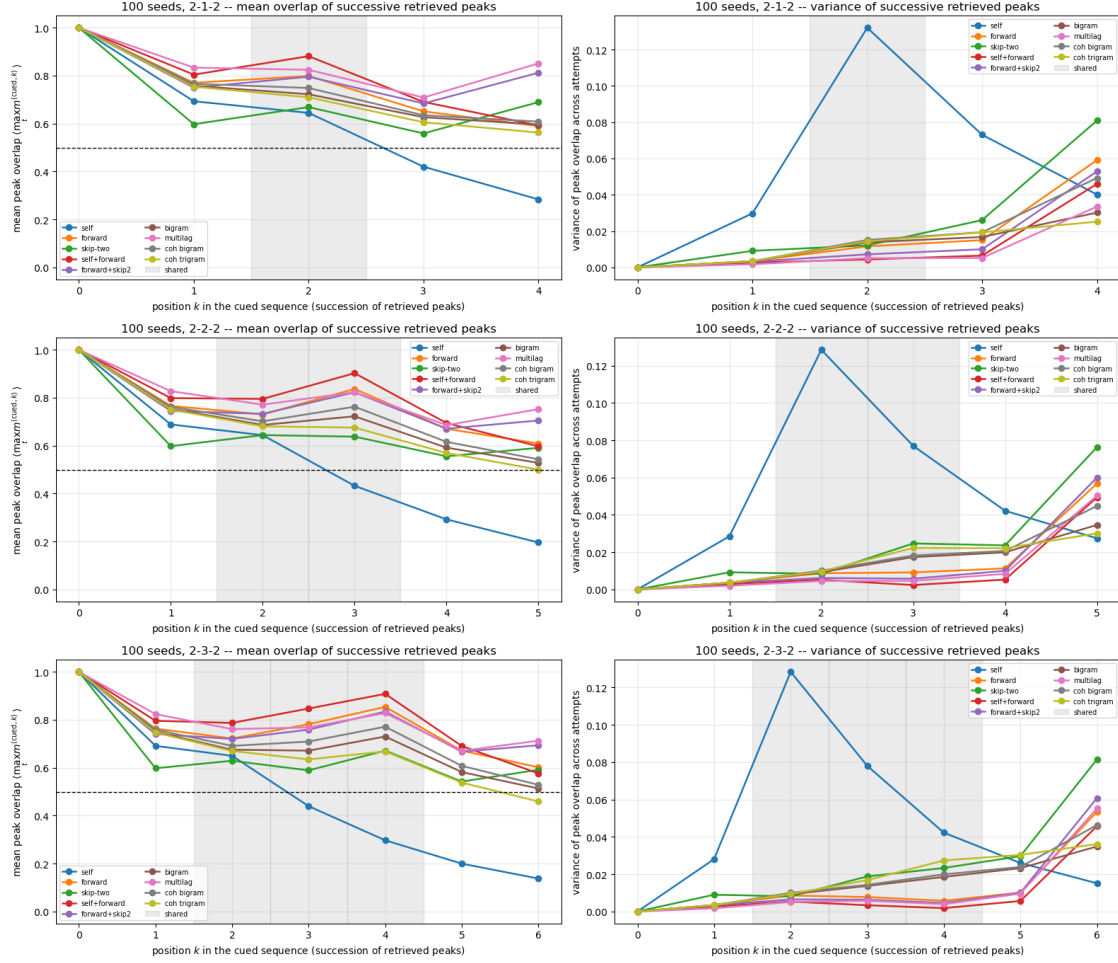
bigram	45% / 0.80 (0.0031) / 86%	22% / 0.80 (0.0035) /
51%	18% / 0.80 (0.0032) / 44%	
multilag	52% / 0.88 (0.0007) / 78%	35% / 0.88 (0.0008) /
58%	28% / 0.88 (0.0009) / 48%	
coh bigram	44% / 0.82 (0.0035) / 78%	26% / 0.82 (0.0035) /
48%	22% / 0.82 (0.0041) / 45%	
coh trigram	36% / 0.79 (0.0023) / 91%	15% / 0.78 (0.0031) /
74%	10% / 0.79 (0.0018) / 46%	

12.0.1 The succession of retrieved peaks

Left: mean peak overlap of the cued branch's pattern at each position, over all attempts. Right: its variance across attempts. Shaded columns are shared positions, where the mean is high for every rule by construction and says nothing about the rule.

How far the mean stays above `tolerance` is the rule's effective retrieval depth; where the variance peaks is where attempts start disagreeing – typically just after the shared run.

```
[60]: fig, axes = plt.subplots(len(seed_out), 2, figsize=(16, 4.6 * len(seed_out)))
      for row, (tag, stats) in enumerate(seed_out.items()):
          plot_peak_profile(stats, f"{SEED_TRIALS} seeds, {tag}",
                             GEOMETRIES[tag]["shared_idx"],
                             axes=axes[row])
      plt.tight_layout(); plt.show()
```



12.0.2 The same sweep at $\omega \equiv 0.3$ (constant, $\sigma_\omega = 0$)

The sweep above runs at the configured disorder, $\omega_i \sim \mathcal{N}(0, 0.3)$. This repeats it with the disorder removed and every oscillator carrying the same constant $\omega = 0.3$: same magnitude, same 100 instances, same sampled corruption rates, same `seed=0`. The only thing that changes is whether the intrinsic frequencies are *spread* or *uniform*.

That makes it the control §9 cannot be on its own. Per the note at the top, the disorder run uses a single fixed ω draw reused across all 100 trials, so any ordering that also holds here is a property of the rules and the corruption range rather than of that one sample. Where the two disagree, the disorder is doing the work.

The final table puts the two side by side, rule by rule and geometry by geometry.

```
[61]: SEED_OMEGA_CONST = {"frequency_mean": 0.3, "frequency_std": 0.0}

seed_out_const = {}
for tag in SPEC:
```

```

print("=" * 100)
print(f"{tag} -- {SEED_TRIALS} seeds, omega = 0.3 constant (frequency_std = 0)")
seed_out_const[tag] = seed_sweep(RAW_CATALOGUE, **GEOMETRIES[tag],
                                config_overrides=SEED_OMEGA_CONST)
print()

fig, axes = plt.subplots(len(seed_out_const), 3, figsize=(20, 4.4 * len(seed_out_const)))
for row, (tag, stats) in enumerate(seed_out_const.items()):
    plot_rule_stats(stats, f"{SEED_TRIALS} seeds, {tag}, $\\omega \\equiv 0.3$",
                    axes=axes[row], show_overlap_var=True)
plt.tight_layout(); plt.show()

fig, axes = plt.subplots(len(seed_out_const), 2, figsize=(16, 4.6 * len(seed_out_const)))
for row, (tag, stats) in enumerate(seed_out_const.items()):
    plot_peak_profile(stats, f"{SEED_TRIALS} seeds, {tag}, $\\omega \\equiv 0.3$",
                      GEOMETRIES[tag]["shared_idx"], axes=axes[row])
plt.tight_layout(); plt.show()

# the two frequency configurations side by side, at equal magnitude
print(f"strict success / branch-correct, {SEED_TRIALS} seeds "
      f"(disorder = sigma_omega 0.3, const = omega 0.3)")
print(f"{'rule':<15}" + "".join(f"{'t' + ' disorder':>22}{'t' + ' const':>22}" for t in SPEC))
for rule in RAW_CATALOGUE:
    row = ""
    for t in SPEC:
        for source in (seed_out, seed_out_const):
            s = source[t][rule]
            row += f"{s['counter']['correct'] / s['attempts']:>14.0%} / {s['branch_correct']:>6.0%}"
    print(f"{'rule':<15}{'row'}")

```

```

=====
=====

```

```

2-1-2 -- 100 seeds, omega = 0.3 constant (frequency_std = 0)
self          correct=  0%  crossed=  0%  incomplete= 100%  overlap=  n/a
branch-correct= 47% +/-0.035  margin=+0.001 +/-0.058
forward       correct= 36%  crossed= 49%  incomplete= 16%  overlap=+0.873
branch-correct= 49% +/-0.035  margin=-0.001 +/-0.076
skip-two      correct=  0%  crossed=  6%  incomplete= 94%  overlap=+0.855
branch-correct= 94% +/-0.017  margin=+0.097 +/-0.062
self+forward   correct= 23%  crossed= 29%  incomplete= 48%  overlap=+0.882
branch-correct= 48% +/-0.035  margin=-0.001 +/-0.078

```

```

forward+skip2  correct= 60%  crossed= 23%  incomplete= 16%  overlap=+0.872
branch-correct= 76% +/-0.030  margin=+0.059 +/-0.097
bigram        correct= 80%  crossed= 12%  incomplete= 8%   overlap=+0.867
branch-correct= 86% +/-0.024  margin=+0.125 +/-0.102
multilag      correct= 54%  crossed= 12%  incomplete= 34%  overlap=+0.889
branch-correct= 82% +/-0.027  margin=+0.098 +/-0.107
coh bigram    correct= 77%  crossed= 13%  incomplete= 10%  overlap=+0.870
branch-correct= 85% +/-0.025  margin=+0.091 +/-0.088
coh trigram   correct= 67%  crossed= 16%  incomplete= 17%  overlap=+0.861
branch-correct= 84% +/-0.026  margin=+0.154 +/-0.123
  mean sampled rate over the 100 seeds: 0.2047 (target 0.2, range [0.12, 0.28])

```

```

=====
=====

```

```

2-2-2 -- 100 seeds, omega = 0.3 constant (frequency_std = 0)
self          correct= 0%  crossed= 0%  incomplete= 100%  overlap= n/a
branch-correct= 50% +/-0.035  margin=+0.003 +/-0.069
forward       correct= 30%  crossed= 46%  incomplete= 23%  overlap=+0.862
branch-correct= 46% +/-0.035  margin=-0.006 +/-0.068
skip-two      correct= 1%  crossed= 52%  incomplete= 47%  overlap=+0.862
branch-correct= 49% +/-0.035  margin=+0.001 +/-0.067
self+forward  correct= 22%  crossed= 29%  incomplete= 48%  overlap=+0.881
branch-correct= 50% +/-0.035  margin=-0.001 +/-0.084
forward+skip2 correct= 36%  crossed= 46%  incomplete= 18%  overlap=+0.871
branch-correct= 50% +/-0.035  margin=-0.002 +/-0.084
bigram        correct= 34%  crossed= 56%  incomplete= 10%  overlap=+0.871
branch-correct= 48% +/-0.035  margin=-0.001 +/-0.082
multilag      correct= 34%  crossed= 33%  incomplete= 33%  overlap=+0.892
branch-correct= 60% +/-0.035  margin=+0.023 +/-0.104
coh bigram    correct= 33%  crossed= 53%  incomplete= 14%  overlap=+0.872
branch-correct= 48% +/-0.035  margin=-0.005 +/-0.071
coh trigram   correct= 53%  crossed= 28%  incomplete= 18%  overlap=+0.857
branch-correct= 72% +/-0.032  margin=+0.068 +/-0.111
  mean sampled rate over the 100 seeds: 0.2047 (target 0.2, range [0.12, 0.28])

```

```

=====
=====

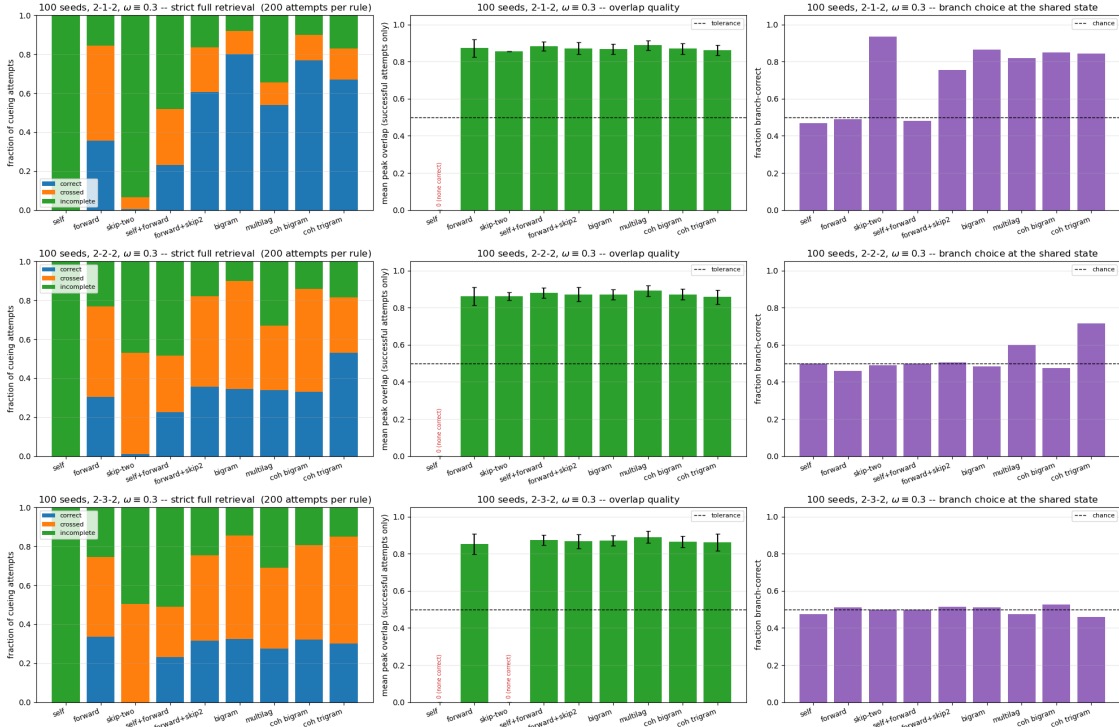
```

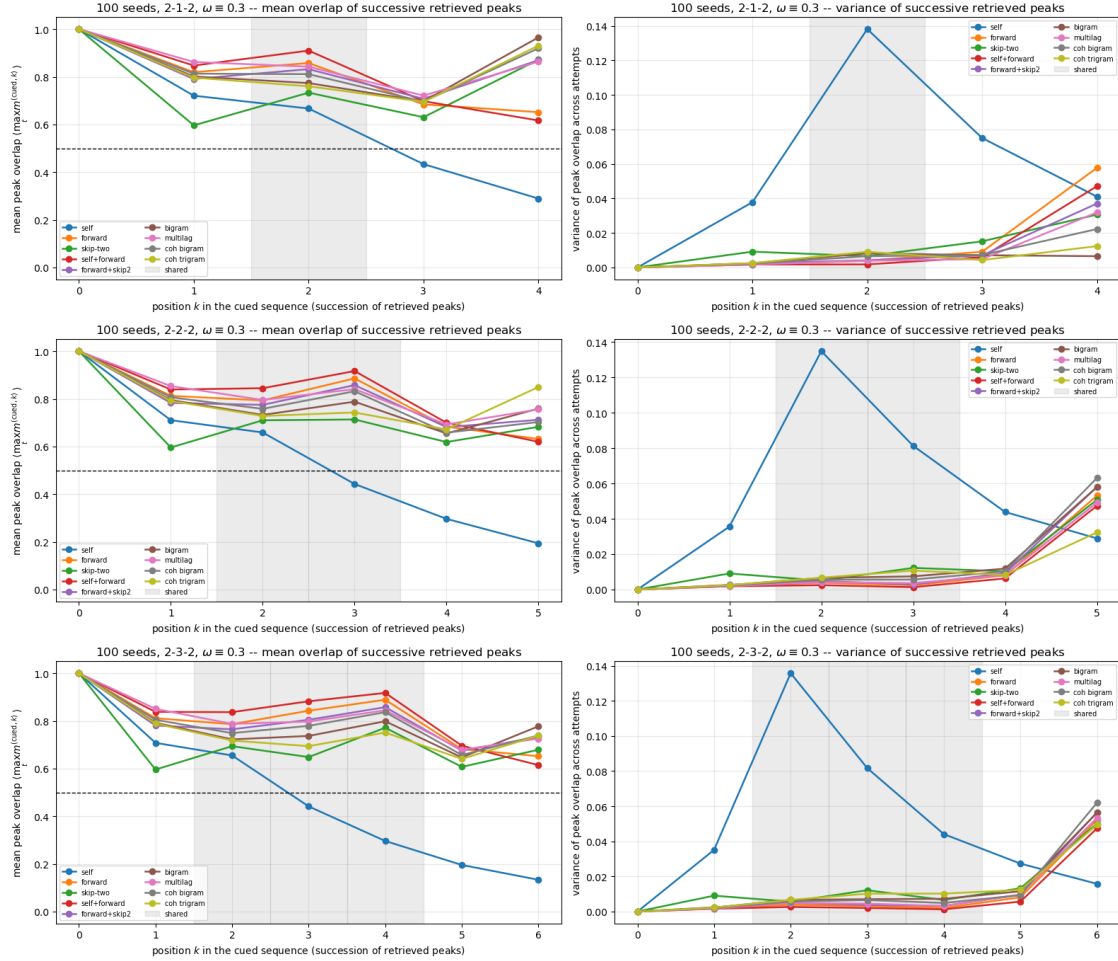
```

2-3-2 -- 100 seeds, omega = 0.3 constant (frequency_std = 0)
self          correct= 0%  crossed= 0%  incomplete= 100%  overlap= n/a
branch-correct= 48% +/-0.035  margin=-0.006 +/-0.074
forward       correct= 34%  crossed= 41%  incomplete= 26%  overlap=+0.854
branch-correct= 51% +/-0.035  margin=-0.001 +/-0.071
skip-two      correct= 0%  crossed= 50%  incomplete= 50%  overlap= n/a
branch-correct= 50% +/-0.035  margin=-0.000 +/-0.063
self+forward  correct= 23%  crossed= 26%  incomplete= 51%  overlap=+0.874
branch-correct= 50% +/-0.035  margin=+0.001 +/-0.079
forward+skip2 correct= 32%  crossed= 44%  incomplete= 24%  overlap=+0.867
branch-correct= 52% +/-0.035  margin=-0.001 +/-0.086

```

bigram correct= 32% crossed= 53% incomplete= 14% overlap=+0.871
 branch-correct= 51% +/-0.035 margin=+0.004 +/-0.082
 multilag correct= 28% crossed= 42% incomplete= 31% overlap=+0.891
 branch-correct= 48% +/-0.035 margin=-0.002 +/-0.101
 coh bigram correct= 32% crossed= 48% incomplete= 20% overlap=+0.865
 branch-correct= 52% +/-0.035 margin=-0.000 +/-0.072
 coh trigram correct= 30% crossed= 55% incomplete= 15% overlap=+0.861
 branch-correct= 46% +/-0.035 margin=-0.003 +/-0.100
 mean sampled rate over the 100 seeds: 0.2047 (target 0.2, range [0.12, 0.28])





strict success / branch-correct, 100 seeds (disorder = sigma_omega 0.3, const = omega 0.3)

rule	2-1-2 disorder		2-1-2 const		2-2-2	
disorder	2-2-2 const		2-3-2 disorder		2-3-2 const	
self	0% / 50%	0% / 51%	0% / 47%	0% / 48%	0% / 48%	0% / 48%
52%	0% / 50%	0% / 51%	0% / 47%	0% / 48%	0% / 48%	0% / 48%
forward	26% / 50%	36% / 49%	36% / 49%	28% / 51%	28% / 51%	28% / 51%
46%	30% / 46%	28% / 46%	34% / 51%	34% / 51%	34% / 51%	34% / 51%
skip-two	0% / 82%	0% / 50%	0% / 94%	2% / 50%	2% / 50%	2% / 50%
55%	1% / 49%	0% / 50%	0% / 50%	0% / 50%	0% / 50%	0% / 50%
self+forward	20% / 49%	23% / 48%	23% / 48%	22% / 50%	22% / 50%	22% / 50%
50%	22% / 50%	19% / 50%	23% / 50%	23% / 50%	23% / 50%	23% / 50%
forward+skip2	54% / 70%	60% / 76%	60% / 76%	34% / 52%	34% / 52%	34% / 52%
50%	36% / 50%	30% / 47%	32% / 52%	32% / 52%	32% / 52%	32% / 52%
bigram	45% / 86%	80% / 86%	80% / 86%	22% / 51%	22% / 51%	22% / 51%
51%	34% / 48%	18% / 44%	32% / 51%	32% / 51%	32% / 51%	32% / 51%
multilag	52% / 78%	54% / 82%	54% / 82%	35% / 48%	35% / 48%	35% / 48%
58%	34% / 60%	28% / 48%	28% / 48%	28% / 48%	28% / 48%	28% / 48%

coh bigram		44% /	78%		77% /	85%	26% /
48%	33% /	48%	22% /	45%	32% /	52%	
coh trigram		36% /	91%		67% /	84%	15% /
74%	53% /	72%	10% /	46%	30% /	46%	

13 11. Branch choice against network size

Branch choice, not full retrieval, is what these benchmarks measure. `correct` bundles it with a completion term – empirically $\text{correct} \approx \text{branch-correct} \times (1 - \text{incomplete})$ – so a rule can decide correctly and still score badly for running out of drive.

This sweep isolates it against N . Error bars are the binomial standard error $\sqrt{p(1-p)/n}$ over cueing attempts.

Read `branch_correct`, not `correct`, across N . `incomplete` rises with N at a fixed duration – a larger network needs longer to traverse and GEOMETRIES is capped at 80 – so `correct` understates the gain. Raise the durations alongside N if you need both.

```
[62]: BRANCH_N = [50, 100, 200, 400]
BRANCH_RULES = ["forward", "bigram", "coh bigram", "coh trigram", "multilag"]
BRANCH_CAT = {k: RAW_CATALOGUE[k] for k in BRANCH_RULES}
BRANCH_TRIALS = 20

_N_KEEP = N
branch_vs_n = {}
for tag in SPEC:
    for n in BRANCH_N:
        N = n                                     # rule_sweep reads the
    ↪module-level N
        stats, _ = rule_sweep(BRANCH_CAT, **GEOMETRIES[tag],
    ↪trials=BRANCH_TRIALS, verbose=False)
        branch_vs_n[(tag, n)] = stats
N = _N_KEEP

fig, axes = plt.subplots(1, len(SPEC), figsize=(6.2 * len(SPEC), 4.6),
    ↪sharey=True)
for ax, tag in zip(np.atleast_1d(axes), SPEC):
    for rule in BRANCH_RULES:
        p = [branch_vs_n[(tag, n)][rule]["branch_correct"] for n in BRANCH_N]
        se = [branch_vs_n[(tag, n)][rule]["branch_se"] for n in BRANCH_N]
        ax.errorbar(BRANCH_N, p, yerr=se, marker="o", capsize=3, lw=1.7,
    ↪label=rule)
        ax.axhline(0.5, color="k", ls="--", lw=1, label="chance")
        ax.set_xscale("log"); ax.set_xticks(BRANCH_N, [str(n) for n in BRANCH_N])
        ax.set_xlabel("N (neurons)"); ax.set_title(tag, fontsize=11)
        ax.set_ylim(0.2, 1.05); ax.grid(alpha=0.3)
```

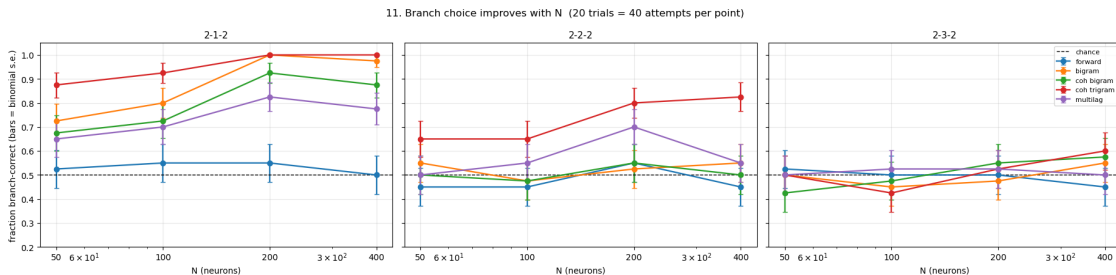
```

np.atleast_1d(axes)[0].set_ylabel("fraction branch-correct (bars = binomial s.e.
↳)")
np.atleast_1d(axes)[-1].legend(fontsize=8)
fig.suptitle(f"11. Branch choice improves with N  ({BRANCH_TRIALS} trials = {2*
↳* BRANCH_TRIALS} "
            f"attempts per point)")
plt.tight_layout(); plt.show()

print(f"branch-correct +/- binomial s.e.  ({2 * BRANCH_TRIALS} attempts per_
↳cell)")
for tag in SPEC:
    print(f"-- {tag} --")
    print(f"{'rule':<14}" + " ".join(f"N={n:<14}" for n in BRANCH_N) + "    N=50_
↳-> 400")
    for rule in BRANCH_RULES:
        row = " ".join(f"{branch_vs_n[(tag, n)][rule]['branch_correct']:>5.0%}"
            f"+/-{branch_vs_n[(tag, n)][rule]['branch_se']:<7.3f}"
↳for n in BRANCH_N)
        d = (branch_vs_n[(tag, BRANCH_N[-1])][rule]["branch_correct"]
            - branch_vs_n[(tag, BRANCH_N[0])][rule]["branch_correct"])
        print(f"{'rule':<14}{row}{d:>+12.0%}")
    print()

print("completion, for contrast -- this is what the duration cap costs at_
↳larger N")
for tag in SPEC:
    print(f"-- {tag} -- incomplete%")
    print(f"{'rule':<14}" + " ".join(f"N={n:<8}" for n in BRANCH_N))
    for rule in BRANCH_RULES:
        print(f"{'rule':<14}" + " ".join(
            f"{branch_vs_n[(tag, n)][rule]['counter']['incomplete'] /_
↳branch_vs_n[(tag, n)][rule]['attempts']:<10.0%}"
            for n in BRANCH_N))

```



```

branch-correct +/- binomial s.e.  (40 attempts per cell)
-- 2-1-2 --

```

rule	N=50	N=100	N=200	N=400
N=50 -> 400				
forward	52%+/-0.079	55%+/-0.079	55%+/-0.079	50%+/-0.079
-3%				
bigram	72%+/-0.071	80%+/-0.063	100%+/-0.000	98%+/-0.025
+25%				
coh bigram	68%+/-0.074	72%+/-0.071	92%+/-0.042	88%+/-0.052
+20%				
coh trigram	88%+/-0.052	92%+/-0.042	100%+/-0.000	100%+/-0.000
+12%				
multilag	65%+/-0.075	70%+/-0.072	82%+/-0.060	78%+/-0.066
+12%				

-- 2-2-2 --

rule	N=50	N=100	N=200	N=400
N=50 -> 400				
forward	45%+/-0.079	45%+/-0.079	55%+/-0.079	45%+/-0.079
+0%				
bigram	55%+/-0.079	48%+/-0.079	52%+/-0.079	55%+/-0.079
+0%				
coh bigram	50%+/-0.079	48%+/-0.079	55%+/-0.079	50%+/-0.079
+0%				
coh trigram	65%+/-0.075	65%+/-0.075	80%+/-0.063	82%+/-0.060
+17%				
multilag	50%+/-0.079	55%+/-0.079	70%+/-0.072	55%+/-0.079
+5%				

-- 2-3-2 --

rule	N=50	N=100	N=200	N=400
N=50 -> 400				
forward	52%+/-0.079	50%+/-0.079	50%+/-0.079	45%+/-0.079
-8%				
bigram	50%+/-0.079	45%+/-0.079	48%+/-0.079	55%+/-0.079
+5%				
coh bigram	42%+/-0.078	48%+/-0.079	55%+/-0.079	57%+/-0.078
+15%				
coh trigram	50%+/-0.079	42%+/-0.078	52%+/-0.079	60%+/-0.077
+10%				
multilag	50%+/-0.079	52%+/-0.079	52%+/-0.079	50%+/-0.079
+0%				

completion, for contrast -- this is what the duration cap costs at larger N

-- 2-1-2 -- incomplete%

rule	N=50	N=100	N=200	N=400
forward	12%	2%	0%	2%
bigram	10%	18%	32%	28%
coh bigram	8%	18%	12%	12%
coh trigram	28%	42%	48%	42%

multilag	20%	22%	12%	12%
-- 2-2-2 -- incomplete%				
rule	N=50	N=100	N=200	N=400
forward	18%	10%	5%	0%
bigram	10%	20%	15%	25%
coh bigram	5%	10%	5%	8%
coh trigram	20%	30%	32%	62%
multilag	18%	25%	2%	8%
-- 2-3-2 -- incomplete%				
rule	N=50	N=100	N=200	N=400
forward	10%	10%	8%	2%
bigram	12%	22%	25%	28%
coh bigram	12%	28%	12%	8%
coh trigram	12%	28%	40%	45%
multilag	22%	22%	18%	8%

14 Findings

2. Gain does not uniformly help – on the smallest geometry it makes several rules dramatically worse. `kc.proportional_gain` removes the $(1 - 2\rho)^{\sum \text{lags}}$ handicap on lag-heavy terms, but raw vs gain is not “weak rule gets a fair boost”:

- **1-1-1** (easiest): actively harmful. **bigram** 55% correct / 5% incomplete raw $\rightarrow$ 18% / 72% gain; **coh trigram** 55%/5% $\rightarrow$ 15%/85%; **multilag** bad either way. Where the raw drive already traversed, amplifying it up to $4.6\times$ destabilizes rather than speeds up.
- **2-1-2** = (a), eq (7): helps the hardest-hit rule – **coh trigram** 45%/42% $\rightarrow$ 57%/22%, **multilag** 45%/22% $\rightarrow$ 52%/22%, **bigram** a wash.
- **2-3-2**: flat-to-worse – **bigram** 57%/0% $\rightarrow$ 48%/20%, **coh trigram** 48%/12% $\rightarrow$ 38%/32%.

Duration is no substitute either: 3x on the hardest geometry gives bit-identical outcome fractions, since the trajectories settle via the stationary early-stop well before the shortest duration tested.

6. A shared run of length L needs a gate reaching back $\geq L$ – necessary, not sufficient.

Below reach L every gate factor reads a pattern that is itself shared, so the rule adds drive but no branch information – the transition matrix’s `nhop` $\geq L + 1$, moved from the drive to the gate. Above it the rules are *still* at chance on the run of three: the gap between the traversed and untraversed predecessor is +0.63 at the cue, +0.02 two patterns later, and still +0.02 at the branch point. The two predecessors are equidistant from the shared run, so their static overlaps are identical and the whole signal is that dynamic residual, which survives about two steps. Reach is cheap to build; the information is what runs out – the constraint Levy and Wu name in the write-up’s context section.

7. The benchmarks are ordered by reach, and the rules fall in that order. (a) 2-1-2 needs reach 1, which every context rule has; (c) 2-2-2 needs reach 2, which only **coh trigram** has; 2-3-2 needs reach 3, which nothing has.

8. 2-3-2 is where the family fails, for the stated reason. It is (c) with the shared segment grown to three patterns, everything else fixed. Every rule reads at most two patterns back, so at the branch point every gate factor is still inside the shared run. Branch choice sits at chance, raw

and gain alike.

9. Success rate and overlap quality rank rules differently. Metric 2 is conditioned on metric 1, so a rule can be rare and crisp, or reliable and sloppy. §10’s variance separates a rule that holds its quality across the corruption range from one propped up by easy instances.

10. The 2-3-2 failure is not an artefact of the frequency disorder. §9 re-runs the whole suite at $\sigma_\omega = 0$ with constant $\omega \in \{0.1, 0.3, 1.0\}$: context rules stay above chance where $\text{reach} \geq L$, and the catalogue stays at chance on 2-3-2, at every ω , raw and gain alike.

14.1 Where to change things

- **Which rules are compared** – RAW_CATALOGUE (and GAIN_CATALOGUE, rebuilt from it automatically). Rules outside the named builders use `kc.term(target, lags, w)`.
- **Drive strength across rules** – `kc.proportional_gain(rule, corruption_rate)`. Both catalogues run in every main experiment; the effect is geometry-dependent and sometimes harmful (Finding 2). `duration` was measured to be no substitute.
- **The benchmark topologies** – GEOMETRIES (`seq_len`, `shared_idx`, `duration`), with SPEC naming the write-up’s benchmarks and FOCUS the subset the knobs run on. The shared run must be contiguous with at least one unshared pattern on each side, and `make_crossover_pair` builds two branches.
- **The reach** – REACH, and `lag=` on `bigram_rule` / `coherent_trigram_rule`.
- **The multilag weights** – GRID_W. **The lifetime measurement** – GAP_TRIALS.
- **Sampled corruption** – CORRUPTION_TARGET, CORRUPTION_SPREAD, SEED_TRIALS.
- **Constant frequency** – OMEGA_CONST / OMEGA_GEOMETRY (§8) and FIXED_OMEGA (§9).
- **Settle vs. loop** – boundary on `net.simulate`; the rules are relative, so they need no re-building for it. Swept in `robustness_sweep.ipynb`.
- **Sharpening / factor count** – `d` in CONFIG *and* in the rule builders; they must match.
- **Statistics** – TRIALS (§4) and SEED_TRIALS (§10).